Refresher Training through Gamified Activities on Maternal and Newborn Care for Frontline Healthcare Workers

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by

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Dedicated to my beloved parents.

Thesis Approval

This thesis entitled **Refresher Training through Gamified Activities for Front-line Healthcare Workers on Maternal and Newborn care** by **Arka Majhi** is approved for the degree of **Doctor of Philosophy**.

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This is to certify that **Arka Majhi** (Roll No. 184350001) was admitted to the candidacy of Ph.D. degree on 12 July 2019, after successfully completing all the courses required for the Ph.D. programme. The details of the course work done are given below.

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2	TD 621	Food Processing and Nutrition Delivery	6
3	DE 668	Instructional Design	4
4	TD 694	Seminar	4
5	TD 610	Contemporary Critical Issues in Technology and Development	AU
6	HS 791	Communication Skills -I	PP
7	TD 792	Communication Skills -II	PP
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Abstract

With the increasing demand for universal healthcare coverage, targeted towards mothers and children under different healthcare programmes in India, there has been a rapid increase in the number of Frontline Workers (FLWs) or Community Healthcare Workers (CHWs), i.e., the Anganwadi Workers (AWWs) and the Accredited Social Health Activists (ASHAs). The job training and the refresher training of these workers leave a lot to be desired. The capacity deficit of field functionaries has been identified as a major factor affecting the effectiveness of state health programmes. Given the massive numbers of field functionaries and the continuously advancing base of knowledge and techniques, conventional training pedagogy is ineffective in building and updating the capacity of the CHWs and their supervisors. With the improvement of technology and the widespread availability of smartphones in cities as well as rural and tribal villages of India, access to digital training systems can potentially benefit the healthcare workers and the healthcare system to a large extent, resulting in an increase in the delivery of maternal and child health services and consequent reduction in undernutrition, morbidity, and child mortality countrywide. The conventional training methods are proving to be inadequate for the task. The gamification of training material for refresher training and the use of digital tools for its dissemination offer one potential solution to this problem, which is the subject matter of the doctoral research. The child and mother care protocol, which includes the immunization schedule, medicine supplements for children, and scheduled visits, check-ups, and medication for antenatal and postnatal mothers, is gamified through various gaming options. Digital versions of these games have been developed, which are playable on Android smartphones. Card-based games turn out to be well-suited for this purpose, as they are easily customizable to accommodate the training content. Besides, they are also easily playable in the physical mode as well, given the familiarity of the field functionaries with traditional card games. The study compares the effectiveness of game-based learning pedagogy and conventional classroom instruction among a group of CHWs, and discusses ways to make it a scalable

and effective refresher training tool. Given its design and scope, we observed that the gamified training is best suited as a scalable training tool for the refresher training of frontline workers to reinforce and update existing skills, rather than as a substitute for the initial, in-person job training that CHWs require for hands-on competence.

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List of Abbreviations

ASHA	Accredited Social Health Activist
AWW	Anganwadi Worker
AWH	Anganwadi Helper
AWC	Anganwadi Centre
CHW	Community Health Worker
NFHS	National Family Health Survey
CNNS	Comprehensive National Nutrition Survey
ICDS	Integrated Child Development Services
IFA	Iron and Folic Acid
MWCD	Ministry of Women and Child Development
LMIC	Low and Middle income Countries
LISA	Local Indicators of Spatial Association
PHC	Primary Healthcare Centre
mHealth	Mobile Health
FDG	Focused Group Discussion
MCPC	Mother and Child Protection Card
IAP	Indian Academics of Paediatrician
ORS	Oral Rehydration Solution
ANC	Antenatal Care
GLO	General Learning Objectives
SLO	Specific Learning Objectives

MCQ	Multiple Choice Question
CINI	Child in Need Institute

Chapter 1

Introduction

Maternal and child health and nutrition are significant indicators of the growth of a nation and the well-being of its population. Healthcare systems worldwide are struggling with a shortage of Community Health Workers (CHWs) or trained health workers, especially in low- and middle-income countries (LMICs), reaching an estimated deficit of 18 million by 2030 (range 16-19) (WHO, 2020). Despite a myriad of interventions and policy actions, these are still chronic problems, including in India, where malnutrition, morbidity, and mortality among children are strong public health concerns. Of the many government schemes addressing these issues, the Integrated Child Development Services (ICDS) scheme under the Ministry of Women and Child Development (MWCD) focuses on reducing malnutrition and overall development of children (below 6 years), as well as improving maternal health of pregnant women and lactating mothers. However, while there has been a noticeable improvement in the rate of child mortality, morbidity, high and persistent child malnutrition levels with their tardy reduction, as seen in successive health surveys, continue to be a matter of concern in India (International Institute for Population Sciences (IIPS) and Ministry of Health and Family Welfare, 2021).

One of the prominent features of the state health programmes is the role played by CHWs or frontline personnel, such as Anganwadi Workers (AWWs) and Accredited Social Health

Activists (ASHAs), who are the interface between the health and nutrition system and the community, where they usually reside. ASHAs are state-government-employed CHWs tasked with delivering last-mile healthcare. They are not salaried employees but considered community volunteers incentivised for their work on a milestone basis. ASHAs' activities involve mobilising and supporting women for antenatal care (ANC) check-ups, ensuring timely postnatal care (PNC) visits, facilitating routine immunisation for children and pregnant women, promoting institutional deliveries, tracking high-risk cases, and guiding households on nutrition, hygiene, and disease prevention.

The state health programme's implementation depends on their ability to converge in community healthcare planning and actions, and deliver essential healthcare services efficiently, which in turn, depends heavily on their training, expertise, and competence (Gujral et al., 1991; Agarwal et al., 2019). Training and supervision have been identified as one of the most ignored intervention-related aspects in systematic reviews of CHW success factors (Kok et al., 2015). However, given the magnitude of the issue and the annual growth in the number of field functionaries, traditional training and capacity-building methods have faced significant limitations. It remains a challenge for LMICs to train or educate CHWs (Sultan et al., 2025), ensuring the most vulnerable population's good health and well-being.

Capacity Building of CHWs

Refresher training for field functionaries, particularly in terms of capacity building, has traditionally been de-emphasized in terms of funding and practice. Traditionally, field workers are trained in a physical classroom environment, following the cascading model of teaching, where the state Master Trainer trains the district trainers, who in turn train the field functionaries. Training program budgets are low, and refresher training, which is essential for retaining and updating knowledge, gets disproportionately impacted by budget cuts. The trainers and learners also complain about inadequate infrastructure for conducting training and assessing the knowledge gains in crowded classrooms (Yadav et al., 2021). These are further compounded by the logistical difficulties of coordinating large-scale, face-to-face training sessions in widely dispersed and geographically distant locations. Conventional training methods have also proven ineffective in motivating field workers and facilitating long-term knowledge retention.

There is a pressing need to study alternative and emerging pedagogies that can bridge such gaps and render attempts at capacity building more affordable, participatory, and efficient. The rapid proliferation of digital technologies and smartphones in India, especially following the COVID-19 pandemic, has opened up new possibilities for leveraging digital tools to enhance the training of field staff. Interestingly, the post-pandemic period has witnessed a considerable surge in digital uptake, and most field functionaries now have basic digital competence and exposure to smartphones. This development has paved the way for building digital training solutions tailored to their specific requirements.

Why haven't the educational paradigms in training materialized?

Over the years, researchers, government programmes, and non-governmental organizations have attempted to revolutionize training approaches for CHWs in India. A wide range of mobile phone-based interventions has been introduced at different points in time, each with varying degrees of effectiveness in promoting learning and knowledge retention (Bassi et al., 2018; Bashingwa and et.al., 2021; Joshi et al., 2023; Thomas et al., 2024). Some of these approaches, however, proved to be less adaptable to field conditions. We briefly review past instances to gain a better understanding of the problem involved.

In the 1930s, proponents of radio broadcasting claimed that it could replace teachers by providing expert instruction on a large scale. Researchers have even recently experimented with community radio like Mobile Vaani, an interactive voice response (IVR)–based mHealth intervention and multimedia content delivered through feature phones and smartphones to train CHWs (Yadav and et.al., 2017; Yadav et al., 2019; Yadav and et.al., 2019). By the 1950s, television broadcasting was considered an advanced educational medium, combining moving images with sound. Lectures were recorded and broadcast simultaneously to multiple classrooms. Research revealed that learners who viewed the broadcast from other rooms achieved the same level of knowledge gain as those who attended face-to-face in-person sessions.

Critics might argue that, though the medium was visual, the incentive was not aligned and motivating enough for people to learn. Researchers implemented an incentive-based training approach for CHWs using basic mobile phones (Shah et al., 2017). The intervention involved preloading the phones with video modules on maternal and child healthcare. Each video was followed by a set of related questions, with their correct answers. CHWs received follow-up phone calls during which they were quizzed on the material. Correct responses were rewarded with cellular talk-time credits, serving as an incentive to reinforce learning and participation. However, evidence suggests that once the incentive is withdrawn, the interest and engagement of CHWs tend to decline. The collection of offline-accessible videos functioned in a manner similar to earlier offline encyclopedic resources distributed by libraries and schools for students—useful as reference material but limited in sustaining long-term motivation without external reinforcement.

Critics have also argued that the lack of interactivity might be an important aspect in such interventions. More recently, in a study conducted in Dharavi, Mumbai (n = 37), instructional illustrations were found to be significantly more effective than the video-based training materials distributed by the Ministry of Women and Child Development (MWCD) for training CHWs (Tulaskar, 2020).

Researchers have also experimented with quizzes and trivia as a teaching method (Majhi et al., 2021). Answering MCQs did not teach CHWs how to create knowledge, but rather how to choose answers. It also encourages guessing, and sometimes they receive rewards for answering something they are unaware of, simply by chance. The multiple answers comprise the wrong ones, too, exposing them to the incorrect content while they read all the options. The MCQs encourage memorizing the questions to beat the system. Quizzes were often part of MOOCs.

Massive Open Online Courses (MOOCs) were also considered transformative for education. Platforms such as Coursera, EdX, IITBombayX, Swawam, and NPTEL have provided access to knowledge and delivered education at scale. However, their effectiveness remains unclear, with studies critiquing high dropout rates (Gupta and Jain, 2014), limited learner engagement (Nieder et al., 2022), and non-consistent learning outcomes across different populations. Spoken Tutorial has experimented with extending this model beyond technical domains on topics such as exclusive breastfeeding, latch and positioning, complementary feeding, and maternal nutrition, designed for self-learning by CHWs and other field health workers (Rupal Dalal, 2019). While learning through MOOCs and other digital technologies generally has delivered factual knowledge effectively, they have yielded mixed results in developing hands-on skills (Mulcahy et al., 2022; Jensen et al., 2023). Scaling up the reach of training through technological solutions might enable wide dissemination and access to expert knowledge. Technology can provide strong lower-level scaffolds (structured, high-quality input at scale), but without deliberate design of interactive, contextual, and mentorship layers around it, the full scaffold needed for complex CHW hands-on skills and judgment remains incomplete.

With recent advancements in artificial intelligence, researchers have experimented with deploying chatbots, voice interfaces, and AI agents to support CHWs (Ramjee et al., 2025; Sivasubramanian and Raval, 2025). These tools serve as external scaffolding, assisting CHWs with daily tasks such as retrieving information for advocacy, estimating child anthropometrics through image processing, classifying images to recognize symptoms, and providing access to

relevant knowledge resources. AI can help offload cognitively demanding processes and provide timely guidance. Excessive dependence on AI might limit the development of CHWs' expertise (UNESCO, 2025). The challenge lies in designing gradual phasing-out strategies, where AI scaffolding initially supports beginners but reduces its assistance as competence grows, allowing CHWs to internalize knowledge, strengthen cognitive processes, and ultimately achieve mastery.

Gamification as an Innovation in Pedagogy

Gamification, the application of game design in non-game contexts, has been recognized as one of the most effective strategies for optimizing participation, motivation, and learning outcomes (Deterding et al., 2011b; Hamari et al., 2014). Field workers, who typically undergo tedious and monotonous training protocols, can benefit from gamification as a means of making learning more enjoyable and engaging. Gamification offers opportunities for competition, cooperation, and reward, which can lead to increased continuity and retention of knowledge (Hamari et al., 2014; Hamari and Koivisto, 2015).

Multiple studies and training documents report that CHWs, including ASHAs in India, often perceive traditional classroom-based training as boring, lecture-heavy, or non-engaging due to its rigid, instructor-focused format lacking interactivity (B and Kumar, 2017; Guleri et al., 2014). It prompted many researchers to explore strategies for making training and learning more enjoyable. While learning through games may not guarantee complete comprehension of a subject, it often serves as a significant motivator. Although the literature on creating enjoyable learning experiences is extensive, there is limited direct evidence connecting fun and enjoyment in game-based learning to improved learning outcomes. However, game-based learning is believed to enhance engagement, which in turn can foster better learning (Li et al., 2023). Various strategies have been employed to achieve this, including the use of diverse media, entertaining instructional methods, and hands-on constructivist approaches. The boundary between enjoyment and education is often blurred. Interestingly, in many cases, learning becomes a byproduct of enjoyable activities, where engagement in fun mechanics, such as challenges, rewards, and narratives, drives knowledge acquisition without explicit instruction (Soyoof et al., 2024).

We therefore looked at gamification and gameful activities as an effective option for exploration. The four studies in this thesis outline these efforts, which involved developing a digital game-based solution for field worker refresher training. Before arriving at the solution, we un-

derwent an iterative design process, where we experimented with various playful activities, as described in the four studies. The details are appended in the appendix. [1]

The objective is to design an Android-centric game as a refresher module that complements the in-person initial core training, alleviating its limitations. This approach leverages the widespread adoption and popularity of smartphones to deliver training content in an efficient and scalable manner.

The game design borrows from traditional card games, which are simple to learn, straightforward to play, and inherently engaging. The physical card game serves as a precursor and introduction to the digital one, offering a seamless transition for those who are upcoming adopters of smartphones and other technologies. The card game format also facilitates the easy incorporation of key elements of gamification, such as rules, procedures, and structured gameplay, making it an ideal choice for this situation.

Health module as Scope of Gamification

During the course of this research, it became evident that gamification requires a well-defined and unambiguous set of rules or protocols for participants to follow. In this regard, the procedures and protocols within the health module, such as immunization schedules and medication supplements for children and mothers, were found to be more clearly codified compared to those related to nutrition. Given this distinction, the gamification efforts were initially confined to the refresher training syllabus for ASHA workers, focusing on health, to ensure clarity and alignment with established health protocols.

ASHA and Anganwadi Workers: Team Building

ASHA workers and Anganwadi workers (AWWs) collaborate to organize Village Health and Nutrition Days (VHNDs), where services such as immunization, antenatal care (ANC), postnatal care (PNC), and nutrition support are provided. They also conduct home visits to ensure that pregnant women and children receive proper care and follow-up. Both workers play a key role in mobilizing the community to participate in health programs and address vaccine hesitancy. Their combined efforts significantly contribute to improving immunization coverage, reducing maternal and child mortality, and raising awareness about maternal and child

[1] Choice of game mechanics, drafting rules of play, and the digital game app are annexed in appendix A.5 and A.6

health, especially in underserved communities. They are essential to the success of national health programs, such as the National Health Mission (NHM) and Poshan Abhiyan (Nutrition Mission). We therefore trained both of these workers, as the game content was relevant to both of them in implementing the MCH programme.

We found that the dynamics of cooperation and competition come into play when AWWs join teams with ASHA workers, either collaborating or competing with each other. This interaction provides valuable insights into inter-cadre teamwork and coordination. It not only increases participation but also helps both cadres better understand each other's roles and responsibilities in community healthcare delivery.

We briefly describe below how the field experiment evolved, building upon the experience of each experiment incrementally. In hindsight, this sequence could not have been anticipated, and in some sense, the flexibility exercised made the research more relevant to the co-creation and improvement of the game's design and the nature of the experiment.

Field Trials and Methodological Considerations

The first field trial [Study 1] (n=100 sampled and 86 of them retained) was conducted in urban areas of Kolkata and rural areas of West Bengal. We conducted play sessions, tests, and interviews in Bengali, which is the local language. We worked in partnership with an NGO named Child in Need Institute (CINI), which also serves as a State Training Centre (STC) for training the ASHAs in the surrounding districts.

The second field trial [Study 2] was conducted at three places. Prior notification of the field testing sessions was given to the ASHA and AWW coordinators, as well as the Health Officers of Burhanpur Urban (Burhanpur, Madhya Pradesh), Joka village (Kolkata, West Bengal), and Mandala village (Slum near BARC, Mankhurd, Mumbai). Out of 400 CHWs selected through convenience sampling, n=368 (184 ASHAs and 184 AWWs) were retained till the end of the study. We conducted play sessions, tests, and interviews in the local languages, i.e., Hindi, Bengali, and Marathi.

The third field trial [Study 3] was conducted at different places in Madhya Pradesh, comprising urban, rural, and tribal areas. Ninety participants (45 ASHAs and 45 AWWs), each from three groups, were retained until the end of the study, totaling 270 participants across the three groups.

The last field trial [Study 4] was conducted at different parts of Burhanpur, Kolkata, and Mumbai. We conducted field testing with n = 198 ASHAs, or 94 from each of two intervention groups, spread across three locations. Here again, we conducted play sessions, tests, and interviews in Hindi, Bengali, and Marathi, the local languages.

Some ASHAs, particularly those with limited formal education, faced challenges reading Hindi numbers on cards and filling out survey questionnaires. To address this, numbers were changed to English while retaining Hindi words. However, ASHAs educated in Madrasas, familiar only with English and Urdu numbers, required English translations to complete forms, but could still play cards using other information. Consequently, all numbers in the cards and the app were updated to English. During questionnaire reviews, discrepancies were identified, including instances where ASHAs skipped Rotavirus-related questions. This was due to the non-operational status of Rotavirus vaccines in Madhya Pradesh, which resulted in their exclusion from training materials. Additionally, confusion arose around the term "booster" for the PCV vaccine, as it was not included in their training schedules, although most ASHAs eventually understood and answered correctly.

Large-scale field trials in multicultural linguistic and geographic settings have guided the development and piloting of the gamified training module. The field trials have provided information about the operational challenges and potentialities entailed in implementing gamification in a field training program. An instance would be how the interrelation between teamwork and collaboration was brought out through observations of ASHAs' and AWWs' interaction when playing the game in cooperation. Additionally, the design process involved iterative prototyping and feedback from field workers to ensure that the game was both culturally appropriate and easy to use.

What emerged during these sessions was the players' intuitive understanding of the card game and their ability to make quick adjustments to new rules or information. The case of Madarsa-educated field workers or those unfamiliar with a new protocol, such as Roravirus, demonstrated that they were able to quickly adjust to the game. This was not anticipated earlier, but it is operationally important in that the game can help them internalize new changes and protocols quickly through a peer-to-peer exchange of notes.

1.1 Dissertation content and structure

As briefly mentioned above, we conducted the experiments in phases of study, each study addressing different yet connected aspects of training and knowledge enhancement for Community Healthcare Workers (CHWs) in India. The first phase was on comparing physical and augmented reality (AR)-based playful activities for refresher training, which was conducted in rural and semi-urban areas of West Bengal. This phase laid the foundation for the effectiveness of game-based training compared to traditional methods. The second phase expanded the scope by integrating digital and physical card games, with a larger cohort of CHWs from various states across India, to test for the retention of knowledge and interest in the long term. Each phase built on the findings of the previous one to create a cohesive narrative that addressed the issues of training, retention, and implementation. The three paper format thesis structure allows for the exploration, analysis, and presentation of each phase separately. This allows for the integration of various methodologies, from qualitative co-design work to quantitative quasi-experimental studies, and provides an overall picture of the effectiveness of game-based training tools. Through the four separate studies, the dissertation explores the multifaceted nature of the problem, contributing to the body of knowledge on gamified training methodologies for CHWs in low-resource settings.

The three studies present a detailed discourse on the implementation of gamified learning, assess its effectiveness, and address the training demands of CHWs. These studies respond to the gaps in the training and performance of CHWs, who are the primary contact points between the health system and rural communities. The studies emphasize the importance of interactive and user-oriented learning methods, stressing the facilitation of engagement and motivation to achieve better learning outcomes. They also emphasize the importance of reinforcement learning and feedback mechanisms in promoting the long-term retention of health information among CHWs. The studies differ in their focus on specific game modalities. While the first study focuses on the comparative effectiveness of physical and computer games, the second and third studies focus on the long-term retention gains of smartphone-based games. This variety of approaches demonstrates the adaptability of gamified learning and its applicability to diverse training needs and settings. The findings of these articles provide valuable insights into the potential of game-based training to enhance knowledge acquisition, retention, and interest among

CHWs, as well as the utility of these instruments in practical applications of data collection and healthcare delivery.

The first study, "Physical and Augmented Reality-based Playful Activities for Refresher Training of ASHA Workers in India", shows the drawbacks of the traditional classroom training model and suggests the concept of playful, interactive training tools. Through a comparison of physical playing cards with an AR-based smartphone game, the study demonstrates that AR-based training leads to significant improvements in knowledge retention and interest among CHWs. The immersive and interactive nature of AR games makes them particularly suitable for visual learners, while physical card games are more suitable for auditory learners who acquire knowledge through verbal communication while playing the games. This study highlights the need to tailor training strategies to the diverse learning styles of CHWs, ensuring that training programs are not only engaging but also effective.

However, field observations identified key barriers to scaling AR training for CHWs in resource-limited environments. First, AR required high-end smartphones and reliable data connectivity, which were often unavailable in rural and tribal areas, leading to frequent disruptions, slow performance, and sometimes complete failure to load content. Second, CHWs encountered usability problems with AR on small screens, which distracted them from the core learning material. Third, sustaining AR long-term proved difficult due to high costs for device repairs, software updates, and technical support, which we struggled to provide at scale.

In comparison, physical playing cards offered greater practicality for these settings. They needed no power or internet, were simple to print and share, and encouraged natural group discussion and collaborative play, making them well-suited to CHWs' capacity levels and local constraints.

Following these findings, we further focused on developing the card game for both digital and physical play and tested it with CHWs. The development of the physical card game was not an easy process in itself. Segmentation of the topics of maternal and child care into four 'suites' was the first. The unequal number of cards in each suit was a strong psychological block that took time to overcome. The issue of numbering the cards, particularly when we had multiple actions, e.g., 3 doses of DPT, was also debated with the workers, and we finally decided to give sub-numbers to each sub-activity to emphasize the bundled nature of the activity. These are

described in some detail in the appendix [2]. The digital version subsequently presented technical challenges, albeit not a significant one, with different game modes posing their own algorithmic challenges. While we have focused on a traditional team game, 'Satti', the game can also be played in solitaire and rummy style.

The second study, "Refresher Training through Digital and Physical, Card-Based Game for Accredited Social Health Activists (ASHAs) and Anganwadi Workers (AWWs) in India", further explores the relative effectiveness of physical and digital card games. The study finds that while digital games yield higher short-term knowledge gains, long-term knowledge retention is equally high, with no significant difference between digital and physical play modes. This suggests that both modalities are viable options for refresher training, with digital games having the added advantage of scalability and accessibility. The study also highlights the necessity for ongoing reinforcement in an attempt to preserve knowledge retention, as the control group, without training, performed significantly worse than the game-based groups. These results highlight the importance of ongoing training and support for CHWs in delivering high-quality healthcare services.

Following these findings from the second study, we attempted to deploy and test both modalities, physical and digital card games, in different locations to evaluate the product and determine if the results are generalizable. The third study, "Replay, Revise, and Refresh: Smartphone-Based Refresher Training for Community Healthcare Workers in India", examines the long-term effect of smartphone-based refresher training. The study compares digital and physical card games with conventional classroom teaching and finds that both game-based methods outperform conventional training in knowledge retention. While the digital game shows higher short-term knowledge gain, long-term knowledge retention is similar for both digital and physical gameplay modes. The qualitative answers provided by CHWs indicate a strong preference for digital games due to their convenience and interactivity, yet highlight the need for physical card decks for purposes of learning and reference when undertaking home visits, where individual mothers can't be expected to be familiar or experienced with the digital mode. This study emphasizes the importance of incorporating game-based training into existing programmes, while addressing the practical needs and interests of CHWs.

In light of the findings from these articles, several key themes emerge. Firstly, game-

[2]The development of the game content is detailed in A.5 and A.6

based training methods, with digital and AR capabilities, are better than classroom training in facilitating learning and knowledge retention among CHWs. Their interactive and immersive nature makes them highly engaging, responsive to different learning styles, and makes training fun and effective. Second, while digital games offer access and scalability, physical card games are also an appropriate choice in resource-constrained contexts, or/where mobile phone access is restricted. The choice between them on which mode to deploy must be tailored to the specific requirements and contexts of the targeted population. These studies focus on knowledge gain and knowledge retention over time through game-based training. While these tools may have the ability to maximize short-term knowledge acquisition, long-term effectiveness is contingent upon regular use and reinforcement.

Overall, the evidence from the following articles provides strong support for game-based training methods' ability to improve the knowledge, and engagement of CHWs in India. These game-based trainings are an effective alternative to traditional training methods, counteracting the challenge of limited literacy, resource constraints, and poor retention. They are, however, dependent upon careful planning, ongoing assistance, and alignment within existing healthcare structures.

Study-1 Physical and Augmented Reality based Playful Activities for Refresher Training of ASHA Workers in India			
Aim	**Methods**	**Results**	**Conclusions**
To compare the effectiveness of physical and augmented reality (AR) card games for improving knowledge gain and retention on child immunization among ASHA workers.	Focus group discussions (FGDs) and participatory design exercises with CHWs. Field trials were conducted to evaluate the effectiveness of both physical and AR-based games.	AR-based games demonstrated higher engagement, knowledge gain, and retention compared to physical card games, mainly due to their interactive and intuitive nature.	AR-based games were found to be more effective in learning enhancement and knowledge retention, suggesting potential as a complementary training tool for ASHA workers.
Study-2 Refresher Training through Digital and Physical, Card-Based Game for Accredited Social Health Activists (ASHAs) and Anganwadi Workers (AWWs) in India			
Aim	**Methods**	**Results**	**Conclusions**
To investigate the effectiveness of digital and physical card-based games in enhancing knowledge of child immunization among community healthcare workers (CHWs) in India.	A quasi-experimental study with 368 participants (n=92 for each group: digital, physical, traditional classroom, and control). Quantitative gameplay data and qualitative feedback were collected through interviews.	Both digital and physical card games showed significant knowledge improvements compared to traditional classroom and control groups. Digital gameplay demonstrated slightly higher knowledge retention.	Game-based refresher training can be a scalable, cost-effective, and engaging tool for improving CHWs' knowledge and practices in resource-constrained settings.
Study-3 Replay, Revise, and Refresh: Smartphone-Based Refresher Training for Community Healthcare Workers in India			
Aim	**Methods**	**Results**	**Conclusions**
To assess the differences in knowledge gain and retention among digital, physical, and traditional classroom training methods for CHWs.	270 CHWs (n=90 per group) were divided into three groups: digital game, physical card game, and traditional classroom training. Pre- and post-intervention tests and retention tests after six months were conducted.	All groups demonstrated significant knowledge improvement, with digital gameplay yielding higher post-intervention scores and superior long-term knowledge retention.	Smartphone-based training outperformed both physical and classroom methods, indicating its potential for improving long-term knowledge retention through frequent, interactive learning sessions.

Table 1.1: Summary of Aims, Methods, Results, and Conclusions from Three Studies

1.2 Research Questions

This research addresses several key questions across the four studies, focusing on the effectiveness of game-based training, knowledge retention, user engagement, and data collection efficiency. These questions aim to explore the potential of content gamification and digital tools in improving healthcare training contributing to the growing body of evidence supporting innovative pedagogical approaches in resource-constrained settings. These research questions are as follows:

1. **Effectiveness of Game-Based Training**

   RQ1 To what extent are game-based training methods more effective than traditional classroom-based refresher training for community healthcare workers (CHWs) in terms of knowledge acquisition and retention?

   RQ2 How do digital and physical gameplay modalities compare in improving CHWs' knowledge and performance?

2. **Knowledge Retention**

   RQ3 Are there significant differences in long-term knowledge retention between physical and digital gameplay modes for CHWs?

   RQ4 How does knowledge retention from game-based training compare to that achieved through traditional training methods after an extended period?

3. **Engagement and User Experience**

   RQ5 How do CHWs perceive and engage with digital and physical game-based training tools in terms of ease of use, motivation, and satisfaction?

Chapter 2

Review of Literature

2.1 ANC, PNC and Child Immunization

Antenatal Care (ANC) and Postnatal Care (PNC) are essential parts of maternal and newborn health services and form an important component of the game's learning focus. Early registration of pregnancy, preferably within the first trimester, allows timely identification of health risks and access to maternal services. A minimum of four ANC visits is recommended to ensure complete care, including monitoring of blood pressure and weight, hemoglobin testing, tetanus vaccination, nutrition advice, and birth preparedness. PNC services focus on the health of both the mother and the newborn after delivery, emphasizing early initiation of breastfeeding, detection of post-delivery complications, counseling on family planning, and linking with immunization services.

Immunization saves 2-3 million lives each year from diseases that vaccines could prevent (Andre et al., 2008). Across the world, about 14.5 million children under 12 months missed the necessary vaccination, and 1.6 million of them were Indian (WHO, 2024). A study on NFHS-4 data found that the immunization coverage of children in 163 districts in India was less than

half (Panda et al., 2020). 47% of children from the lowest wealth quantile households in India are not fully immunized, and any health insurance does not cover 71% of families (International Institute for Population Sciences (IIPS) and Ministry of Health and Family Welfare, 2016). As a result, they end up spending significant amounts on healthcare, pushing almost 32-39 million people annually to dip below the poverty line (van Doorslaer et al., 2006; Garg and Karan, 2009).

A study (Panda et al., 2020) using LISA and a spatial regression model with NFHS-4 data revealed that mothers who received ante- and post-natal care had a better chance of fully immunizing their children. It also found a strong association between full immunization of children and empowerment indicators among women, such as mothers' literacy and women being the family's authority. However, despite free and locally available vaccination services in Government Hospitals and Primary Healthcare Centres (PHCs) in India, and the launch of Mission Indradhanush (MoHFW, 2016), full immunisation remains a distant goal (International Institute for Population Sciences (IIPS) and Ministry of Health and Family Welfare, 2021). The study also found that ignorance and unawareness among family members regarding vaccination increases the likelihood of undernutrition in children and results in child mortality and morbidity. The study also suggests the need to sensitize the population, which would improve access to community healthcare. Pockets of the population with limited access to healthcare require more targeted steps to bridge the immunization gap. mHealth technologies could help improve health outcomes and decrease health inequalities (Peek, 2017).

2.2 Knowledge Gap of healthcare workers and mothers

We conducted a focus group discussion with mothers in a rural village in the South 24-Parganas district of West Bengal. We found that, due to a knowledge gap among healthcare workers, mothers and children in the community usually remain unaware of medical care, healthcare delivery, check-up services, and family planning. Studies conducted in other parts of India also show similar results. A study (Mani et al., 2020) found that mothers have low awareness of tetanus, immunization, iron, and folic acid (IFA) tablets. More than half of the pregnant women were unaware of availing pregnancy care services at ANC visits, such as blood pressure and

weight measurement, and abdominal examination. Knowledge regarding high-risk conditions in pregnancy and danger signs in pregnancy was also low. Few mothers were aware of the various components of institutional delivery and maternity benefit schemes. The designed game aims to bridge the knowledge gap of community healthcare workers in the aforementioned topics.

2.3 Access to mobile phones and internet services

According to a press release (Telecom Regulatory Authority of India, 2021), as of June 30, 2024, there were 1,205.64 million telephone subscribers in India, of which 44.66% (538.51 million) were rural telephone subscribers. Out of them, 99.44% (535.53 million) were rural wireless telephone subscribers, making 45.75% share of the total wireless subscribers. Based on the NSS 75th Round, Household Social Consumption Education Survey conducted between July 2017 and June 2018 (MoSPI, 2015), it was found that 35.8% of households (24.8% rural and 51.3% urban) had access to the internet. The rise in smartphone ownership among healthcare workers in India opens up an opportunity to utilize mHealth approaches to bridge the gap between healthcare access and engagement with the beneficiaries of the target population (Ganapathy and Ravindra, 2008; Bassi et al., 2018; Madanian et al., 2019).

2.4 Introductin to mHealth

Mobile Health (mHealth) is defined as "medical and public health practice supported by mobile devices" (WHO, 2011), thus having the potential to overcome the limitations of traditional methods of receiving child and maternal healthcare, such as job cards or printed paper brochures. mHealth is a means to disseminate healthcare information cost-effectively and in a personalized manner, tailored to socioeconomic and cultural needs, thereby fitting the context.

Researchers from developing countries have provided suggestions for designing effective mHealth interventions (Kumar and Anderson, 2015). These include communicating with the community through radio and text messages (both automatic and human-aided). Studies have been conducted to motivate health workers and increase their knowledge by deploying short

videos on mobile phones (Ramachandran et al., 2010). Another study (Kumar and Anderson, 2015) from India found that watching short films on mother and child care empowered Health Workers and mothers to manage pregnancies proactively. Shah. et al. (Shah et al., 2017) from India did a similar study on an incentive-based approach for training AWWs through mobile-based videos. The AWWs' feature phones were loaded with videos related to mother and child healthcare. A few questions with their corresponding answers related to the video's content were appended to the video. After watching these videos, AWWs were supposed to call a toll-free number, where they would be quizzed on the same questions as shown in a particular topic video. AWWs were awarded cellular talk time as an incentive for answering correctly. Another study (Pérez et al., 2020) from India found that 'Tika Vaani', an IVR-based mHealth solution designed to improve knowledge on immunization, has shown significant results in enhancing healthcare knowledge when the content is customized to meet the needs of less-literate users.

2.5 mHealth interventions through e-Learning, Serious Games and Gamification

e-Learning is a teaching and learning approach that utilizes an educational model to improve access to training, communication, and interaction by leveraging electronic media and devices, thereby promoting novel understanding and enhancing learning (Sangrà et al., 2012). 'Blended Learning' happens when e-learning and face-to-face teaching are combined. 'Educational game' pedagogy requires learners' participation while performing competitive activities following previously set game rules. A Cochrane review (Akl et al., 2010) examined two physical games for educating healthcare professionals and confirmed that games are an effective strategy for teaching. The application of games with a 'serious intent' focus has been observed in various domains, including health, training, defense, education, military, aviation, city planning, politics, and ecology, with multiple approaches. Researchers have used similar terms like Game-Based-Learning, Games-for-Good, Alternative-Purpose-Games, (Sawyer and Smith, 2008), and Edugaming (Angarita et al., 2005) while describing similar interventions.

Clark Abt first outlined the idea of 'Serious Games' in his book, published in 1970, named 'Serious Games.' He described 'Serious Games' as the games that "have an explicit and care-

fully thought-out educational purpose and are not intended to be played primarily for amusement" (Abt, 1970). Serious Games combines learning theory and empirical outcomes to improve skill learning and game design principles. Interventions through serious game design can enhance cognitive, social, and health-related skills beyond the game context (Ritterfeld et al., 2009; Giunti et al., 2015). The number of articles and systematic reviews on serious games in healthcare education is growing (Graafland et al., 2014; de Ribaupierre et al., 2014; Laamarti et al., 2014; De Wit-Zuurendonk and Oei, 2011).

The games created with the serious purpose of providing healthcare education delivered through digital devices are Serious Games. Serious Games provide a gaming experience through rules, engines, and mechanics. However, gamification tries to create a similar experience through applying game mechanics and involving fun while performing mundane activities, while developing motivational and cognitive abilities. Deterding et al. defined gamification as the "use of game design elements in non-game contexts" (Deterding et al., 2011b,a; Deterding, 2011, 2012). Gamification potentially allows higher involvement of the player in setting up the objectives and outcomes, thus personalizing the intervention and making it cost-effective (Gentry et al., 2019). The increase in inequality of access to resources for healthcare, lack of adherence to treatment (Vicente et al., 2014), and increased healthcare cost (Lenihan, 2012) generated the need for applying gamification in digital healthcare services. Self-determination theory suggests using intrinsic and extrinsic motivations in the context of game design (Ryan and Deci, 2000). Extrinsic motivators are game elements such as badges, points, progress bar, and leaderboards (Ryan and Deci, 2000). At the same time, intrinsic motivators are the feelings of 'mastery' or accomplishment towards the goal, 'autonomy' or the freedom of a sense of play, and 'relatedness' or connecting players to the goal and passion (Deterding et al., 2011b).

2.6 Educational Theories and Serious Gaming

In his book (Knowles, 1980), Malcolm stresses the factors supporting adult learning. Serious games and gamification techniques can also help achieve those goals.

- With every experience faced, there is a tendency to move from dependency to autonomy. Games facilitate this by providing an active learning environment through independence

in moves or choices made in games.

- Past learning is used in dealing with our current problems. Games challenge us to deal with different situations as per our prior knowledge.
- With age, people tend to be more focused on the task goal. Games train us to be goal-oriented by targeting small tasks and completing levels.
- With age people tends to learn from tasks, based on practical problems and less on educational content. Games facilitate learning through design problems, which are inspired by practical problems.

Learning through games supports the 'cognitive theory of multimedia learning' (Mayer, 1997), which says that learning happens better when both visual and auditory senses are involved. Siemens proposes that learning is not a linear process, but meandering through changing environment (Siemens, 2005). Serious Games also follow similar mechanisms. Researchers (Mey, 2005; Gee, 2012) showed that games are compelling as learning happens within meaningful contexts.

Training through Serious Games has resulted in more extended training engagement and improvement in learning gains. Experiential Learning Theory (Kolb, 1984) suggests that after gaining knowledge, every learner observes, reflects, and analyzes it, creating abstract concepts in mind and actively experimenting with them by applying their learnings to see the results. Serious Games can target every stage of this process and support the creation of knowledge through the experience of gameplay. For combining learning and gaming, Ritterfeld suggests three ways (Ritterfeld and Weber, 2006).

- Reinforcement model: Learning rewarded with entertainment
- Motivation model: Entertainment encourages learner's interest and attention
- Blended model: Process and result of learning itself becomes exciting

The blended approach seems to be most effective because it provides intrinsic satisfaction of learning (Breuer and Bente, 2010) and provides fun while learning (Koster, 2010).

To achieve predefined learning objectives in mind, a Serious Game in its design phase needs to follow a grounded design approach (Hannafin et al., 1997) and align with educational

theories. Choosing behaviorism or constructivism while designing the game can influence the playful activity's learning theories and instructional strategies (Gunter et al., 2008). The learning approach defines the characteristics of the training environment. It also controls the pace and frequency of interactions intended for critical learning through the Serious Game targeted toward the learning outcomes of the learner.

2.7 Theoretical Framework: Why aren't CHWs Learning?

The challenges in CHW training are often attributed to institutional inertia, limited commitment from state health systems, or overhyped technological solutions. Another perspective is offered by the framework of fast and slow thinking by Daniel Kahneman (Kahneman, 2011). According to dual-process theories of cognition, humans operate through two interacting systems of thought.

System 1 (S1) thinking is fast, intuitive, and automatic. It processes information rapidly, recognizing patterns based on prior experiences, and draws on long-term memory without conscious effort. System 2 (S2) thinking, in contrast, is slow, deliberate, and analytical; it monitors for errors, walks through procedures step by step, and allows for meta-cognition—thinking about thinking. While S2 is essential for novel or complex problems, much of real-world problem-solving relies on the well-developed capabilities of S1.

Expertise emerges from repeated engagement, practice, and reflection, which gradually consolidates knowledge into long-term memory. Over time, S2-mediated processes help structure experiences, allowing S1 to recognize patterns and make decisions automatically. In effect, S1 functions as a highly efficient, largely unconscious problem-solving system; most decisions and actions are executed seamlessly, without deliberate analytical effort.

For CHWs, this framework underscores the importance of experiential, practice-based, and reflective learning. Simply providing information is insufficient. Training must be structured to repeatedly engage CHWs in meaningful tasks, allowing them to internalize procedures and knowledge in long-term memory. This enables S1-driven recognition and decision-making, making fieldwork both effective and adaptive to complex, real-world situations.

2.8 Designing Implications: Training Tools for CHWs

Effective learning design requires a careful balance of cognitive load. Extraneous load should be minimized through clear and intuitive materials, such as well-structured interfaces, simple visuals, and minimal distractions. At the same time, intrinsic load must be matched to the learners' prior knowledge. This means introducing concepts gradually, revisiting familiar ground at the start, and avoiding too much new material at once. Germane load should be encouraged by presenting moderately challenging content that prompts learners to think more deeply and connect ideas.

Learning happens through effortful practice. Learners need repeated opportunities to attempt tasks, make mistakes, and receive timely feedback. This cycle helps strengthen memory, gradually shifting performance from deliberate reasoning to more automatic responses. Confidence is built through iteration. Small improvements over time, reinforced with feedback, allow learners to see their progress. This steady reinforcement motivates continued engagement and helps fragile skills become more stable and reliable in practice. Effective design must allow space for productive struggle. If tasks are over-automated or solutions are given too quickly, learners may miss the chance to develop resilience and adaptive strategies. Allowing them to work through difficulties, even when the process feels slow, is key to creating learning that lasts.

Constructivism views learners as active builders of their own knowledge, but discovery learning does not always work as intended. The challenge lies in how scaffolding is implemented. If support is removed too quickly, learners may feel lost and unable to progress. Consider the analogy of arriving in a new city without GPS. You could rely on S2 to carefully follow road signs, but this is effortful and slow. With GPS, you can offload that cognitive effort, but if you always rely on it, you never truly learn the city. The best approach is a balance: guidance at first, followed by gradual phasing out so learners can eventually navigate on their own.

This highlights the limits of working memory. S2 has fewer resources and cannot handle too many complex tasks at once. The goal of instruction is to support learners until the knowledge becomes internalized and shifts into S1. Experts often forget what it feels like to be a novice because their own S1 knowledge is automatic and effortless. Learners struggle with basics, and if these are not mastered through repeated, effortful practice, moving on to

advanced topics becomes overwhelming. Effective learning design therefore requires balancing support, practice, and the gradual transition from effortful reasoning to automatic mastery. CHWs should understand information and extract meaning from the game's content. Further, they might construct knowledge by applying the learning to overcome the game's situations, ideally cascading later in their daily practices.

2.9 Learning: A Social Activity

Much of the technology-driven hype in education stems from the assumption that the primary barrier to learning is limited access to information. While this may hold true in resource-constrained contexts, in most cases, access alone is insufficient. Effective learning requires skilled instructors and a community of peers with whom learners can engage and collaborate. Social Learning Theory (Bandura, 1977) states that learning is a social activity where we learn by observing, imitating, and modelling others. For community health workers (CHWs), this social dimension is particularly important: while their role emphasizes connecting with the community, they must also engage with peers to foster a collaborative and socially supported learning environment.

Chapter 3

Methods

3.1 Ethics Declaration

The study procedure was approved by the IIT Bombay Institutional Review Board (IRB) (Proposal Number: IITB-IRB/2022/051), and the IRB approval is attached in the Appendix (Figure: C.1). All participants provided informed consent prior to data collection. We conducted the studies by relevant guidelines and regulations set by the Institute Ethics Committee, which strictly adheres to the Declaration of Helsinki (WMA, 2013) developed by the World Medical Association.

Participation in the playtesting activity was voluntary, with an option to quit anytime during the research. We tried to compensate CHWs for the valuable time and expertise they brought in through knowledge gain and non-monetary collaboration. Their contribution to our research was immeasurable materialistically. We served tea, snacks, and sweets as a token of gratitude. However, we have not given the participants any monetary or materialistic compensation.

Aparajita Mondal's contributions to research are not a part of her academic pursuits.

3.2 Evaluation method

This research study assessed knowledge gain, retention, and satisfaction after playing the designed game with the intervention group. We compared it with a control group using the traditional mode of the classroom refresher course. The evaluation of the gameplay intervention program generally follows the Kirk Patrick Model of Evaluation (Kirkpatrick, 1950). This model considers evaluation across four levels. We limited the scope of this research study to conducting level 1 and level 2 evaluations only, as levels 3 and 4 evaluate the prolonged effects of the intervention program.

We conducted a baseline knowledge evaluation before the intervention. Then, we conducted 4-5 physical and digital card play rounds. It was followed by a post-test using the questionnaire survey. We provided a deck of cards to every CHW group who lived nearby and installed an Android app for other groups. They were encouraged to play when they met for about an hour every day for a week. Master trainers trained the control groups through regular classroom training. After a few weeks, we tested the knowledge levels through a questionnaire survey.

1. Kirk Patrick Level 1 evaluation (Reaction): CHWs' reaction toward the game, gameplay experience, enjoyment, and satisfaction after playing a few game rounds will be recorded through feedback forms. The standard SUS questionnaire for evaluating usability covers most Level 1 questions. The responses would provide inputs for further modification of the game. We translated the GEQ or Game Experience Questionnaire (Ijsselsteijn et al., 2013) into Indian languages. We started with Marathi, Hindi, and Bengali, as the respondents were majorly into these three languages.

2. Kirk Patrick Level 2 evaluation (Learning): A pre-and post-questionnaire on both intervention and control groups will assess the delta learning by measuring the change in the CHWs' knowledge of child immunization and pregnancy care. To compare if the delta learning is significantly different in one of the cases, we conducted a paired t-test (considering normal distribution). We further compared the post-test scores to delayed post-test scores conducted a few weeks later to compare knowledge retention. We used t-tests to find significant differences.

3.3 Instrumentation

3.3.1 Knowledge Questionnaire

The instrument used in this study had three sections. All participants consented to this research by reading the first section of the questionnaire and signing it before every survey/test. The second section was the demographic data sheet to identify potential group variances. The demographic data sheet consisted of a checklist and gap-fill questions, such as age, highest educational grade/degree, and years of experience as an ASHA. The third section was 40 Multiple Choice Questions (MCQ) on all four silos, Children below 1 year, above 1 year, ANC and PNC. Questions related to infants and children were on immunization and medicine supplements. This test evaluates the participants' knowledge at the baseline, gained through intervention, and retention after a week of practicing the playful activity. A reference to the 3-page questionnaire is in the Appendix section: (Figure: B.1 , B.2, and B.3). This questionnaire was initially prepared in English and then auto-translated to Indian languages by Google translation services, followed by human corrections. Most of the respondents' mother tongues are comprised of Hindi, Marathi, and Bengali, so we translated the questionnaire into these languages.

3.3.1.1 Game Experience Questionnaire

We tried to conduct the game experience survey through the GEQ (Game Experience Questionnaire) (Ijsselsteijn et al., 2013). The questionnaire contains 33 questions assessing gaming experience on seven components: Immersion, Flow, Competence, Positive and Negative Affect, Tension, and Challenge. There are approximately five questions per item to make it a robust experience measure. The questions generally capture the feelings that they experienced while gaming. The answers are on 5 5-point Likert scale, which ranges from "not at all" at the low end to "extremely" at the high end. The GEQ is available in global Western languages like English, Dutch, Finnish, and German (Ijsselsteijn et al., 2008). In this research, we made an attempt to translate it into Indian languages. We started with Marathi and Hindi as the respondents were mainly into these two languages. The translation process of GEQ was similar to the translation process of the main evaluation questionnaire. Firstly, we translated the English

texts into Marathi with the help of the Google Translate service and the IndiaTyping service to get two linguistic variations. Then, a human was involved in correcting the translation of the questions individually. As Indian languages are cultural and gender-sensitive, we considered that the questions were asked to an elderly lady formally or respectfully. We chose the vocabulary appropriate for the rural respondent, thus avoiding urban vocabulary. For simplicity of understanding, we used English words in Marathi texts, which local people commonly use. A reference to the GEQ and the translation process (Figure: B.4). Later, the GEQ is now in the process of translation into 16 Indian languages: Bengali, Gujarati, Hindi, Kannada, Malayalam, Marathi, Tamil, Telugu, Odia, Punjabi, Urdu, Assamese, Nepali, Bhojpuri, Konkani, and Maithili. The translated questionnaires are going through rounds of human correction phases. We decided to exclude the social presence module (player's social engagement with characters) and the Post-game module (fatigue after a long gap in playing) from this study as they were beyond the scope of this research.

3.4 Experiments in Three Studies

The course of this experimental research has not been linear but instead characterized by discontinuities and leaps. The author has been lucky enough to have the effort come full circle in reputable journals. Due to the evolving and iterative work, we chose the 'three-publications' format for this doctoral thesis to present flexibly. Each chapter provides an introduction, methodology, results, and conclusion. The three summaries of publications are not chronological in order. However, it illustrates how the gamification efforts developed while focusing on different aspects. Towards the end of the dissertation, a summary is provided that links all the studies through reflections.

The papers presented in this thesis collectively explore innovative, gamified, and technology-enabled strategies to enhance the knowledge, engagement, and data collection efficiency of Community Healthcare Workers (CHWs) in India. While each study is rooted in a different facet of public health intervention—refresher training, immunization schedules, geospatial mapping, and anthropometric measurement—the overarching narrative revolves around leveraging Human-Computer Interaction for Development (HCI4D) to address systemic inefficien-

cies in CHW training and performance evaluation.

The first publication-'Physical and Augmented Reality based Playful Activities for Refresher Training of ASHA Workers in India' [Publication 1 - (Chapter 4)] compares a physical card-based game to Augmented Reality game for training ASHA workers. 'Refresher Training through Digital and Physical, Card-Based Game' [Publication 2 - (Chapter 5)] and 'Replay, Revise, and Refresh: Smartphone-Based Refresher Training for Community Healthcare Workers in India' [Publication 3 - (Chapter 6)] demonstrate the efficacy of gamified interventions in improving knowledge acquisition and retention among CHWs.

Publication 2 compares traditional classroom training with physical and digital card-based gameplay, showing significant short-term and long-term knowledge improvements, particularly in the digital group. Publication 3 further supports these findings by revealing that digital interventions offer a significantly higher learning curve and immediate knowledge gain. However, both card-based formats outperformed traditional training in long-term retention.

Chapter 4

Study 1: Comparing Physical and Augmented Reality-based Playful Activities for Refresher Training of ASHA Workers

This study seeks to design and evaluate playful refresher training activities for CHWs on immunization schedules and maternal-child healthcare practices. It compares a physical card game and an Augmented Reality (AR)-enabled smartphone application. Hypothesis (**H1**): ASHAs playing the AR-based digital version would show a higher improvement in knowledge of immunization compared to the group using the physical card-based game.

This study aims to empirically validate the pedagogical potential of immersive technologies like AR in public health training, address gaps in CHW knowledge through scalable learning tools, and contribute a novel perspective to digital education research in low-resource settings. The study proposes an inclusive training methodology easily adapted across diverse public health topics and geographies by making learning interactive, language-independent, and

play-based.

4.1 Study-1 Details

Parts of this section are from a published research Study [1]

1. **Introduction**

   Research conducted on training CHWs using technology through means of texts, voice messages, videos, and quizzes has shown mixed results (Section 2.4). While discussing with CHWs, we found that most of them prefer to learn when they are doing something tangible with their hands or receive instant feedback. Augmented Reality can address this gap (Billinghurst and Dünser, 2012; Dunleavy et al., 2009; Wu et al., 2013) by layering a virtual world over the physical world through a smartphone screen, which enables learning by doing (Azuma, 1997). With the increasing use of mobile devices, the adoption of interactive and gaming-style learning strategies (Peek, 2017) presents new opportunities for training in AR (Santos et al., 2014; Sarkar et al., 2020). AR provides an opportunity to transform an average room into an interactive learning lab, where child profiles and vaccine cards come to life, and choices made by the player have consequences within that world.

[1] **Majhi, A.**, Agnihotri, S. B., Mondal, A., 2022. Physical and Augmented Reality-based Playful Activities for Refresher Training of ASHA Workers in India ", *CHI 2022: Asian CHI Symposium 2022* (Best Paper and Best Presentation award in Long Paper category) https://dl.acm.org/doi/10.1145/3516492.3558788

2. **Methods**

We conducted the study in multiple stages, i.e., conceptualization of a training game, content validation, game design, development, field testing, and comparative assessment.

(a) **Focused Group Discussion with CHWs**

To complement the quantitative findings, a qualitative component was incorporated into the study through a Focus Group Discussion (FGD). While the quantitative data assessed measurable outcomes related to immunization training, the FGD was conducted to gain a deeper understanding of Accredited Social Health Activists' (ASHAs) lived experiences with existing training practices, the challenges they encountered, and their perceptions of the effectiveness of current training approaches. These qualitative insights provided contextual explanations for the quantitative findings and informed the interpretation of the results and the design recommendations presented in this thesis.

A FGD was conducted with (n=14) ASHAs who were available and willing to participate at the time of the study. Participants were recruited using convenience sampling from the study area. The discussion was facilitated using a semi-structured discussion guide consisting of open-ended questions related to participants' experiences with immunization training, perceived barriers to learning, confidence in performing immunization-related tasks, and suggestions for improving future training programmes. The discussion encouraged participants to share their experiences freely while allowing the facilitator to probe emerging topics where appropriate.

As audio recording was not feasible during the session, detailed written field notes were taken throughout the discussion to document participants' responses, examples, interactions, and key observations. Immediately after the session, these field notes were expanded to preserve contextual details and improve the completeness of the qualitative dataset.

The qualitative data were analyzed manually using reflexive thematic analysis following the approach proposed by Braun and Clarke (Braun and Clarke, 2006, 2019). The thematic analysis was conducted on the detailed field notes. The analysis began with repeated reading of the field notes to achieve familiarity with the data. Meaningful segments of text were then coded inductively, allowing codes to emerge from

participants' responses rather than being predetermined. Related codes were iteratively compared and grouped into broader candidate themes. These themes were subsequently reviewed and refined to ensure they accurately represented recurring patterns across participants' experiences. Throughout the analysis, principles of grounded theory, particularly inductive coding and constant comparison, informed the development and refinement of themes (Charmaz, 2014). The qualitative analysis was performed manually, and no qualitative data analysis software was used.

The themes identified from the FGD were used to contextualize and interpret the quantitative findings reported in this thesis. Specifically, the qualitative findings helped explain participants' experiences underlying the observed quantitative results, identified practical challenges encountered during immunization training, and informed the discussion and design implications for future game-based training interventions for ASHAs.

An important theme that emerged is the knowledge gap among participants, particularly in recalling key aspects of the immunization schedule. Many ASHAs reported difficulty remembering the names of vaccines, the sequence in which they are administered, and the specific timing of each dose. This gap was often described in relation to their day-to-day work, where accurate recall is essential for effective service delivery.

Participants also reflected on their prior training experiences, noting that these sessions were predominantly lecture-based and, in many cases, delivered within limited timeframes. As a result, the training was perceived as rushed and difficult to comprehend, with limited opportunities for interaction, clarification, or reinforcement of key concepts. Several participants indicated that this format made it challenging to retain information over time.

These observations suggest that the issue extends beyond the complexity of the content itself and is closely tied to how the information is presented and reinforced. The lack of engaging, accessible, and contextually relevant learning formats appears to contribute to reduced retention. These insights informed our subsequent design direction, motivating us to explore more visual, interactive, and practice-oriented approaches aimed at improving both understanding and long-term retention among ASHAs.

(b) **Co-Design with CHWs**

We chose to design along with the ASHAs. We hosted small co-design sessions where ASHAs reviewed sample illustrations, suggested modifications, and made them draw. Their inputs directly influenced the choice of imagery, terminology, and the flow of the physical and AR-based games. This collaborative approach ensured that the final tools were intuitive and familiar. It also helped build a sense of ownership among the ASHAs, making them more invested in learning.

(c) **Gamified Activity - Physical Card Game**

We developed a deck of 20 cards, covering the administration of age-specific vaccines, which was used across both the physical and AR-based interventions to maintain consistency and comparability of content. The detailed process of designing the card deck is discussed in Section 7.2.2 and 7.4.2. Players received a random mix of cards in the Physical Card Game version. The task in the game is to simulate the experience of examining child profiles and administering age-appropriate vaccines. Players receive points for correct matches and can challenge each other's combinations. The activity encouraged peer validation, discussion, and friendly competition. Each round took about 15 minutes and was played by 2–5 CHWs.

(d) **Gamified Activity - AR-based Digital Game**

The AR-based game or digital version added an interactive twist. CHWs placed printed stickers on the walls representing children and mothers. When scanned with a smartphone using the AR app, a 3D model appeared, showing age and other key details. Players then scanned the appropriate vaccine cards to administer them virtually. The game rewarded correct matches with laughter and positive reinforcement. Mistakes led to crying sounds and point deductions. This instant feedback made the experience memorable and emotionally engaging. Players can play the AR game solo or in teams. In multiplayer settings, we observed role-playing dynamics, with some participants taking the lead in scanning while others handled cards or assisted with strategy.

(e) **Evaluation**

To test both tools, we collaborated with an NGO involved in ASHA training. We aligned our interventions with their existing schedule and conducted five sessions. Each group of participants went through either the physical or digital version of the

training. ASHAs were informed of the purpose of the study and agreed to it through SMS or WhatsApp. The participation was not mandatory, and they were free to exit anytime. Of the initial 100 ASHAs recruited for the study, 86 went through all phases. Most were between 30 and 50 years old and had a secondary school level of education. They were all proficient in Bengali, although some had difficulty reading or writing it. The demographic data is tabulated in Table 4.1.

We used a questionnaire consisting of 10 questions focused on the immunization schedule to measure knowledge gains. We conducted the questionnaire survey both before and after the training. Questions covered topics such as the timing of vaccines, matching vaccines to ages, and interpreting information from the MCP card. We divided the participants into two groups: one experienced the card game (n=42), and the other used the AR app (n=44). We analyzed the change in their scores using paired t-tests within groups and two-sample t-tests between groups (Dimitrov and Rumrill, 2003).

We divided the participants into two groups: one used the physical card game, and the other used the AR app. Before starting, we asked each ASHA to complete a short quiz (pre-test) covering key points of the immunization schedule. We repeated the same quiz right after the play session (post-test).

Table 4.1: Study-1 Participants Demographics

Parameter	Number of ASHAs (%)
Age	
<30	4 (4.65%)
30–40	41 (47.67%)
40–50	37 (43.02%)
No information	4 (4.65%)
Education (Grade)	
Below 8th	8 (9.30%)
8th–10th	12 (13.95%)
10th–12th	56 (65.12%)
Graduation and above	8 (9.30%)
No information	2 (2.33%)
Experience (Years)	
0–5	4 (4.65%)
5–10	32 (37.21%)
10–15	34 (39.53%)
Above 15	11 (12.79%)
No information	5 (5.81%)

3. **Results**

On average, ASHAs who played the physical card game improved their scores by about 3.4 points. Users of the AR app improved by over 6.5 points. We observed a statistical difference between the two groups. The results showed that the AR-based and physical games significantly increased the ASHA workers' knowledge of child immunization schedules. The intuitive and interactive nature of the interface of the AR game enhanced its appeal among the participants, making learning more effective. We also found that AR-based gaming was effective for visual learners. [2] However, physical card-based gameplay was liked by auditory learners or those who support verbal cues during playtime.

[2] "Visual learners" prefer to understand and remember information they can see, such as images, diagrams, or spatial layouts, whereas "auditory learners" tend to learn more effectively from spoken explanations, discussions, and other sound-based inputs.

Intervention	Pre-test score	Post-test score	Difference in Score
IG-1 (n=44) (AR)	2.7 (SD=1.1)	9.2 (SD=2.1)	6.5
IG-2 (n=42) (Physical Cards)	2.8 (SD=1.4)	6.2 (SD=1.8)	3.4
Difference of difference			Difference in Score
(IG1post - IG1pre) - (IG2post - IG2pre)			3.1

Table 4.2: Study 1: Summary of means of pre-test and post-test scores and their difference. Learning through Mobile App (Intervention Group 1 or IG 1). Learning through Physical Card (Intervention Group 2 or IG 2)

4. **Discussion**

(a) **Findings in Playful Learning**

Both modalities showed overall effectiveness, but the study shows a few counterintuitive findings.

i. **Short-term vs Long-term knowledge retention**

The AR-based game showed effective, immediate knowledge gains compared to the physical card group. This reflects Kahneman's dual-process theory (Kahneman, 2011). AR-based game engaged System-1 through fast, intuitive cues like the crying or laughing baby, signalling a right or wrong move. Whereas playing with physical playing cards demanded carefully weighing arguments and perspectives to make a thoughtful decision through taking turns, discussion, and rule-based negotiations. System-1 supported immediacy and memorability, but System-2 supported deeper understanding and codifying the content. An ASHA who had been in the physical card play group stayed silent through the first play round, later started correcting her peers mid-game, saying, "No, not at six months, the vaccine comes later—let me explain properly," suggesting that peer interaction and slow reasoning supported knowledge retention.

ii. **Low literate CHWs performed better with AR than physical cards**

Another counterintuitive finding was that participants with lower literacy or limited smartphone exposure often showed higher comfort with AR-based gameplay than with physical cards. The conjecture is that in the digital version, the

multimodal cues reduced the cognitive burden of textual recall. This is consistent with embodied cognition (Wilson, 2002). During playtesting, an ASHA burst out laughing at the crying baby sound and remarked, "Now I will never forget this mistake—the baby scolded me," while making a sarcastical facial expression. It shows that sensory feedback can act as a strong anchor for memory. However, literate participants often preferred the tactile and visual-symbolic reasoning of the cards, where they could compare options.

iii. **Constructive and Social Learning pathways**

While the AR-based game offered a learning loop through trial and error, the physical card game offered social play, elements of collaboration, competition, and correction. This exhibits the factor of relatedness in Self-Determination Theory (SDT) (Miller et al., 1985). ASHAs frequently joked and teased one another and took the game in good spirits. One of them said, "In this game, mistakes are fun, not shameful." This environment of safe social space aligns with Vygotskian ideas (Vygotsky, 1978) where scaffolding from colearners turns mistakes into a shared learning opportunity. Facilitators of the experiment observed that the physical card gameplay group had more laughter and intense debates, as compared to the digital group. This suggests that a technologically complex method or solution does not necessarily translate linearly into engagement or effectiveness. Instead, each form of gameplay was supported by different motivational and cognitive pathways. AR-based gameplay supported competence through feedback, whereas the physical card version supported relatedness through collaboration, and both were complementary to each other.

(b) **Knowledge Graph**

By about 10-15 rounds, the novelty and challenge dipped since the scenarios were stereotypical. It suggests incorporating more realistic, diversified scenarios and adaptive gameplay for long-term interest. Peer collaboration in both versions also allowed for discussion, dispelling misconceptions, and learning together, pointing out the social facet of learning through play.

(c) **Co-Design**

The intervention became contextually grounded and language-independent by involving ASHAs in the participatory co-design process, by selecting illustrations and

developing rules.

(d) **Institutional Support**

The co-design and field-testing were supported by the Child in Need Institute (CINI), which also serves as the State Training Centre for ASHAs in West Bengal. CINI's long presence in communities and the trust it has built through its social work enabled us to reach remote areas and work closely with people who might otherwise have been difficult to engage. This experience shows how important institutional support is for enabling access and ensuring interventions are welcomed in community settings.

5. **Conclusion** This study began with the conjecture that training can be more effective and enjoyable if it is more game-like. Based on our observations and measurements, we found that both tools, the physical card game and the AR app, made learning more interactive, memorable, and enjoyable for ASHAs. The AR app, in particular, showed greater improvement in test scores. These findings matter because ASHAs often have limited time and resources for refresher training. If we can make those brief windows more impactful using tools they enjoy, we can improve their knowledge and empower them in their vital roles.

This study demonstrated the potential of AR-supported learning tools to enhance healthcare training for CHWs by providing interactive, immersive, and visually rich learning experiences. The findings suggest that AR-supported learning tools can overcome the limitations of traditional training methods. Future efforts should prioritize refining AR game design for low-literacy users and scaling up this intervention across various healthcare settings. The current game's content is limited to a fixed number of scenarios. However, in reality, there can be infinite possibilities. Adding more complex narratives might make the game's content more comprehensive, but it also makes the gameplay more challenging. The long-term impact is the actual test of any training tool. With further refinement and support, tools like these can become part of mainstream ASHA training across states, translated into different languages, and used in both online and offline formats.

4.2 Study-1 Reflections

This study reaffirms our conjecture that experiential, game-based refresher learning tools can improve capacity-building efforts. By shifting the mode of training from passive to active engagement, we can create an effective learning system for frontline health workers.

Critics have argued that the use of Augmented Reality (AR), a relatively new technology, may have induced a novelty effect that influenced the learning outcomes in this study. It leaves us wondering if the observed improvements in knowledge were due to the pedagogical value of the intervention or the new technology.

One of the challenges was balancing complexity and learning effectiveness. Realistic scenarios and advanced skill-building tasks demand more proficiency from players. The game must remain accessible to CHWs with varying levels of technical and smartphone literacy, as well as technical proficiency. Simplified instructions, visual cues, and intuitive interfaces can help make the game experience more inclusive and user-friendly.

Therefore, it was decided to drop the AR route, given the limitations in scalability, the need for relevant infrastructure, and the difficulty in adopting it faced by less educated and more elderly workers. The Card game was more compatible with these constraints. The subsequent part of the study proceeds with Card games, examining their physical and digital variety.

This study also lacked a placebo or no-intervention control condition, limiting the ability to control for the sole effect of the intervention. This limits the generalization of the findings. In the next study, we added a placebo group to enable a more controlled comparative design and better isolate the effect of the intervention from usual practices.

Chapter 5

Study-2: Refresher Training through Digital and Physical, Card-Based Game for Accredited Social Health Activists (ASHAs) and Anganwadi Workers (AWWs) in India

This study was built on the findings of the previous research by expanding the comparison to include both digital and physical versions of the same card game, following the same rules of play for refresher training in immunization knowledge. The objective was to evaluate the comparative effectiveness of these two game modalities in improving knowledge retention and engagement over time.

A significant methodological limitation of the previous study is its lack of an appropriate control group. Educational research demonstrates that most instructional activities, regardless of their novelty, tend to yield positive learning gains. Assessing a new teaching practice against

a condition of "no teaching" only provides a weak basis for causal inference. A more rigorous approach would evaluate the intervention in relation to traditional or established teaching methods, thereby providing a more valid and interpretable measure of its effectiveness (Hattie, 2008).

Building upon the learnings from the previous experiment in Study 1, this experiment is four-armed and compares the impact of four training groups: (i) A smartphone-based digital card game, (ii) A physical card-based game mimicking similar mechanics, (iii) Traditional classroom teaching, (iv) A control group receiving no training.

The intervention is designed around the ANC, PNC, and mother and child immunization schedule and medications, integrating age-appropriate vaccination logic into the game mechanics. We categorized vaccines into children's age groups: Below 1 year, above 1 year, and mothers' phases: Antenatal care, and Postnatal Care. The research questions are as follows: **RQ1:** Does knowledge gain and retention differ significantly among CHWs trained using digital games, physical card games, and traditional classroom methods? **RQ2:** Is there evidence of a novelty effect that fades over time?

5.1 Study-2 Details

Parts of this section are from a published research Study [1]

1. **Introduction** The prevailing concerns regarding training CHWs have already been discussed in the previous Chapter. This study examines the effectiveness of card-based games—both digital and physical—as refresher training tools for CHWs. We also wanted to find whether these games helped participants retain information over time and how they compared with conventional classroom instruction. Throughout the design process, we involved CHWs to ensure the games reflected their learning needs and constraints. We prototyped and piloted a card game-based refresher training solution for CHWs, particularly focusing on child immunization. This game tried to make learning more interactive

[1] **Majhi, A.**, Agnihotri, S. B., Mondal, A., 2024. Refresher Training through Digital and Physical, Card-Based Games for Accredited Social Health Activists (ASHAs) and Anganwadi Workers (AWWs) in India. CHI PLAY Companion '24, October 14–17, 2024, Tampere, Finland (https://dl.acm.org/doi/10.1145/3665463.3678819)

and engaging, allowing CHWs to practice decision-making in a social and dynamic setting.

2. **Methods**

We contacted the CHW supervisors and recruited about 400 CHWs (almost equal number of ASHAs and AWWs) through convenient sampling to account for possible dropouts. By the end, 368 from those who retained,were selected. They paired up as ASHA AWW groups as per their areas where they live and work, forming 181 groups. Then they were randomly alloted into four experimental groups of 92 each. Group 1 (IG-1) (45 pairs of ASHAs and AWWs): Trained with the digital card game on smartphones, Group 2 (IG-2) (45 pairs of ASHAs and AWWs): Trained with physical cards, Group 3 (IG-3) (45 pairs of ASHAs and AWWs): Received a traditional classroom session with the same content, and Group 4 (CG) (46 pairs of ASHAs and AWWs): Control group with no training during the study period. The demographic details of the CHWs in this study are tabulated in Table 5.1 on the next page. We tried our best to balance age groups, educational levels, and years of experience among CHWs.

All groups underwent a baseline test. The three intervention groups received their respective training formats. A post-test followed. After three weeks, we conducted a final delayed retention test (DRT) to measure how much they remembered. We trained the control group only after completing all evaluations to ensure fairness. Before participating, we gave all CHWs verbal and written explanations about the study's purpose and process. We designed digital and physical card games to help CHWs memorize the sequence and dosage of immunizations and supplements required for children and mothers. We used questionnaire surveys to check knowledge gain and retention. Focus group discussions with CHWs captured engagement levels, user satisfaction, and feedback on the training experience. Data analysis included paired t-tests for knowledge score comparisons and thematic analysis for qualitative data.

We built the training game using 60 cards. The process of designing the card deck is discussed in Section 7.2.2 and 7.4.2. We printed the playing cards on card paper for the physical version, and the digital game mimicked the same format. The rules of play are discussed in Appendix A.6. The players played the game in teams of four. Each player received 15 shuffled cards. The goal was to place cards in the correct chronological order

	Intervention Group-1 Digital Card Game		Intervention Group-2 Physical Card Game		Intervention Group-3 Classroom training		Control Group No training	
Parameters	ASHAs(%)	AWWs(%)	ASHAs(%)	AWWs(%))	ASHAs(%)	AWWs(%)	ASHAs(%)	AWWs(%)
Age								
Less than 30	2(4.35%)	1(2.17%)	2(4.35%)	2(4.35%)	1(2.17%)	1(2.17%)	2(4.35%)	1(2.17%)
30-40	24(52.17%)	22(47.83%)	25(54.35%)	20(43.48%)	24(52.17%)	21(45.65%)	24(52.17%)	20(43.48%)
40-50	19(41.3%)	22(47.83%)	18(39.13%)	23(50%)	20(43.48%)	23(50%)	18(39.13%)	24(52.17%)
No information	1(2.17%)	1(2.17%)	1(2.17%)	1(2.17%)	1(2.17%)	1(2.17%)	2(4.35%)	1(2.17%)
Education (grade)								
$Below 8^{th}$	2(4.35%)	2(4.35%)	2(4.35%)	2(4.35%)	2(4.35%)	3(6.52%)	2(4.35%)	2(4.35%)
$8^{th} - 10^{th}$	10(21.74%)	15(32.61%)	10(21.74%)	12(26.09%)	9(19.57%)	11(23.91%)	8(17.39%)	13(28.26%)
$10^{th} - 12^{th}$	30(65.22%)	24(52.17%)	29(63.04%)	28(60.87%)	30(65.22%)	28(60.87%)	31(67.39%)	27(58.7%)
Graduate and above	3(6.52%)	4(8.7%)	4(8.7%)	3(6.52%)	4(8.7%)	3(6.52%)	4(8.7%)	3(6.52%)
No information	1(2.17%)	1(2.17%)	1(2.17%)	1(2.17%)	1(2.17%)	1(2.17%)	1(2.17%)	1(2.17%)
Experience as CHWs (years)								
0-5	5(10.87%)	4(8.7%)	4(8.7%)	4(8.7%)	4(8.7%)	3(6.52%)	4(8.7%)	4(8.7%)
5-10	19(41.3%)	16(34.78%)	20(43.48%)	16(34.78%)	19(41.3%)	16(34.78%)	17(36.96%)	17(36.96%)
10-15	17(36.96%)	19(41.3%)	16(34.78%)	18(39.13%)	16(34.78%)	19(41.3%)	19(41.3%)	18(39.13%)
Above 15	4(8.7%)	5(10.87%)	5(10.87%)	7(15.22%)	5(10.87%)	7(15.22%)	5(10.87%)	6(13.04%)
No information	1(2.17%)	2(4.35%)	1(2.17%)	1(2.17%)	2(4.35%)	1(2.17%)	1(2.17%)	1(2.17%)

Table 5.1: Study-2 Participants' Demography

based on the child's or mother's age and stage of care. Players took turns placing cards, and correct moves awarded bonus turns. The game ends when one player finishes their cards in their hand. The digital game automated the process of shuffling, distribution, and ensuring rules are followed. This reduces the cognitive load on the players. Instant feedback gives them the opportunity to learn and reflect.

We used a pen-and-paper questionnaire (Appendix B) comprising 40 multiple-choice questions covering the MCP card's contents. Each question had a single correct answer. Public health experts reviewed the questions and pilot-tested them with a small CHW group. Each survey phase—pre-test, post-test, and DRT—used the same question pool but in different orders to reduce memory bias. We translated the questionnaire into local languages to include the diversity of participants across India in the evaluation. We conducted interviews and informal discussions with CHWs to obtain qualitative feedback on their gameplay experience.

3. **Results** A mixed-design ANOVA was conducted with training condition (digital card game on smartphones, physical cards, traditional classroom session, and control with no training) as a between-subjects factor and stage (Pre, Post, Long Post) as a within-subjects factor.

   Figure 5.1 visualizes the dispersion of the scores of each group through a series of box plots.

   The analysis revealed a significant main effect of stage, $F(2, 648) = 85.58$, $p < .001$, $\eta^2 = .174$, indicating significant changes in scores over time. A significant main effect of training condition was also observed, $F(3, 648) = 37.16$, $p < .001$, $\eta^2 = .113$, suggesting overall differences between instructional approaches. A significant interaction between training condition and stage was found, $F(6, 648) = 8.75$, $p < .001$, $\eta^2 = .053$, indicating that learning trajectories differed across conditions.

   Post-hoc analyses showed that participants trained using the digital card game on smartphones exhibited the largest improvement from Pre to Post ($d = 1.83$), followed by those trained with physical cards ($d = 1.66$). The traditional classroom condition showed a smaller improvement ($d = 0.65$), while the control group exhibited only modest gains ($d = 0.65$), likely reflecting practice effects.

   At Long Post, both card-based training conditions retained significant improvements rel-

	Intervention Group-1 (IG-1) Digital Card Game		Intervention Group-2 (IG-2) Physical Card Game	
Test phase	Mean (SD)	Median (min-max)	Mean (SD)	Median (min-max)
Pre-test	25.80 (4.81)	26 (15-36)	24.35 (5.02)	24 (16-35)
Post-test	33.83 (4.03)	34 (25-40)	32.12 (4.36)	32 (21-40)
Late Post-test (After 3 weeks)	30.55 (4.04)	31 (23-39)	30.43 (4.20)	30 (23-39)
	Intervention Group-3 (IG-3) Digital Card Game		Control Group (CG) Physical Card Game	
Test phase	Mean (SD)	Median (min-max)	Mean (SD)	Median (min-max)
Pre-test	25.35 (4.30)	25 (15-34)	24.61 (5.65)	24 (10-35)
Post-test	28.05 (3.73)	28 (19-36)	27.64 (3.69)	28 (19-36)
Late Post-test (After 3 weeks)	26.25 (3.32)	26 (19-33)	24.67 (4.10)	25 (15-32)

Table 5.2: Study-2: Test Scores

Table 5.3: Mixed ANOVA Results

Effect	df	F	p	η^2
Group	3	37.16	<.001	.113
Stage	2	85.58	<.001	.174
Group × Stage	6	8.75	<.001	.053

ative to baseline (both $p < .001$), although scores declined slightly from Post. In contrast, participants in the traditional classroom condition did not show significant retention ($p = .267$), performing similarly to baseline. The control group likewise showed no sustained improvement ($p = .939$).

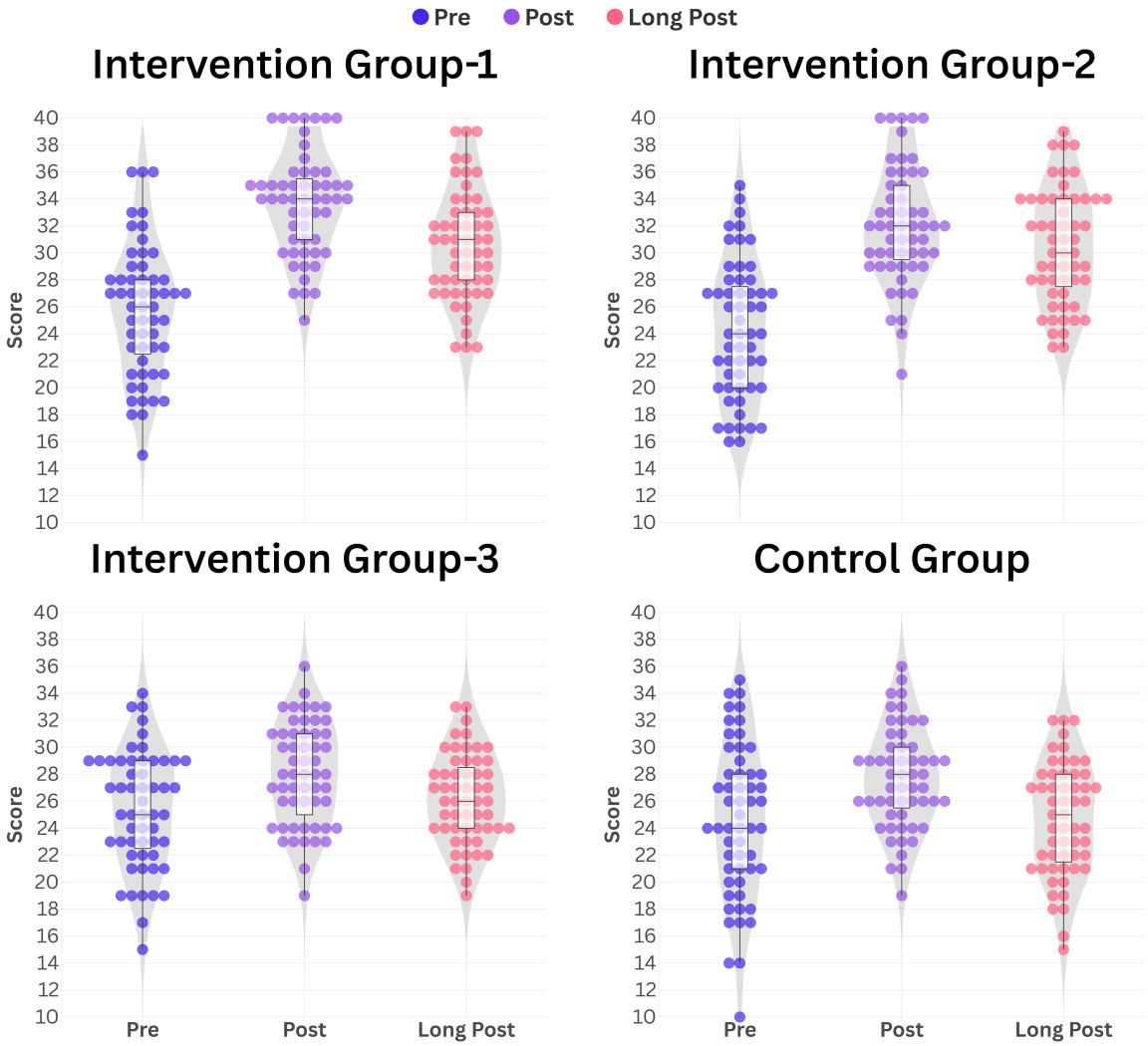


Figure 5.1: Study 2 : Results from comparing the scores of IG1 (Digital Card Games group), IG2 (Physical Card Game group), IG3 (Classroom group) and CG (Control Group)

Between-condition comparisons confirmed that, while all groups were equivalent at baseline, both the digital card game and the physical card conditions significantly outperformed the control group at both Post and Long Post. The digital card game condition consistently outperformed the traditional classroom condition, and the latter did not differ significantly from the control group at Long Post.

These findings indicate that card-based interactive training, particularly when delivered digitally, produces stronger and more durable learning outcomes than traditional classroom instruction.

4. **Discussion**

Table 5.4: Within-Group Comparisons Across Training Conditions

Condition	Comparison	*t*	*p*	*d*	95% CI
Digital card game (smartphone)	Pre–Post	-9.57	<.001	-1.83	[-2.25, -1.41]
	Post–Long	4.17	<.001	0.79	[0.40, 1.18]
	Pre–Long	-5.66	<.001	-1.08	[-1.47, -0.69]
Physical cards	Pre–Post	-8.72	<.001	-1.66	[-2.05, -1.27]
	Post–Long	2.03	.044	0.39	[0.01, 0.77]
	Pre–Long	-6.91	<.001	-1.32	[-1.70, -0.94]
Traditional classroom	Pre–Post	-3.42	<.001	-0.65	[-1.02, -0.28]
	Post–Long	2.70	.008	0.51	[0.14, 0.88]
	Pre–Long	-1.12	.267	-0.21	[-0.57, 0.15]
Control (no training)	Pre–Post	-3.39	<.001	-0.65	[-1.01, -0.28]
	Post–Long	4.03	<.001	0.77	[0.40, 1.14]
	Pre–Long	-0.08	.939	-0.01	[-0.36, 0.34]

Table 5.5: Between-Condition Comparisons (Tukey HSD)

Stage	Comparison	Result
Pre	All conditions	Not Significant
Post	Digital game > Control	***
	Physical cards > Control	***
	Digital game > Classroom	***
	Physical cards > Classroom	*
Long Post	Digital game > Control	***
	Physical cards > Control	***
	Classroom vs Control	Not Significant
	Digital game > Classroom	**

The results support both H1 and H2. Consistent with H1, knowledge gain and retention differed significantly across training modalities, with both the digital card game on

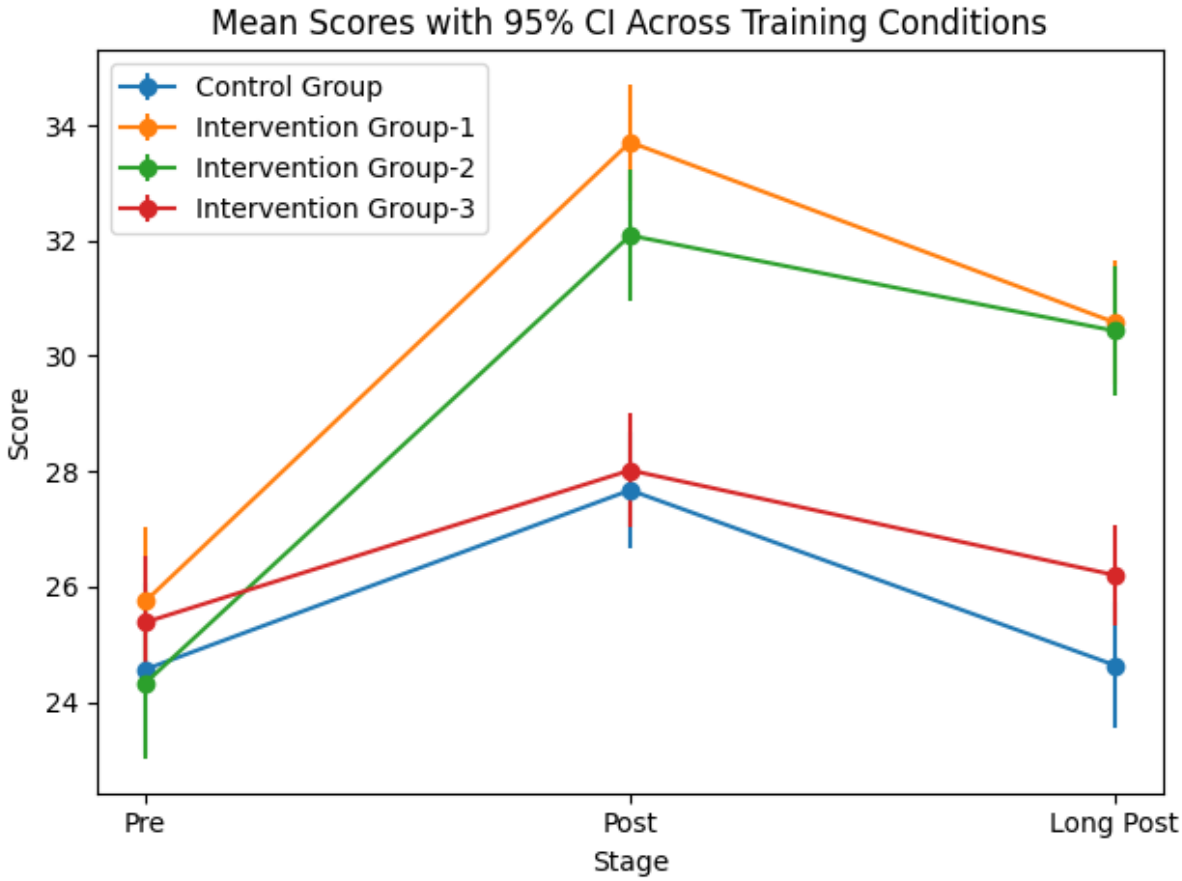


Figure 5.2: Mean scores with 95% confidence intervals across training conditions and time. Participants trained using the digital card game on smartphones showed the largest improvement, followed by those trained with physical cards.

smartphones and the physical card game producing substantially greater improvements than traditional classroom instruction and the control condition. These gains persisted at the delayed assessment, indicating sustained learning. Supporting H2, although some decline from Post to Long Post was observed, the continued advantage of the card-based conditions over baseline suggests that improvements cannot be attributed solely to a novelty effect. While the digital intervention showed the largest immediate gains—likely driven by engagement—the physical card condition demonstrated comparable retention, suggesting that interactive, game-based approaches enable deeper, more durable learning than traditional methods.

Findings in Playful Learning

Although digital and physical card games both significantly improved knowledge, several observations challenged conventional assumptions about game-based learning for CHWs.

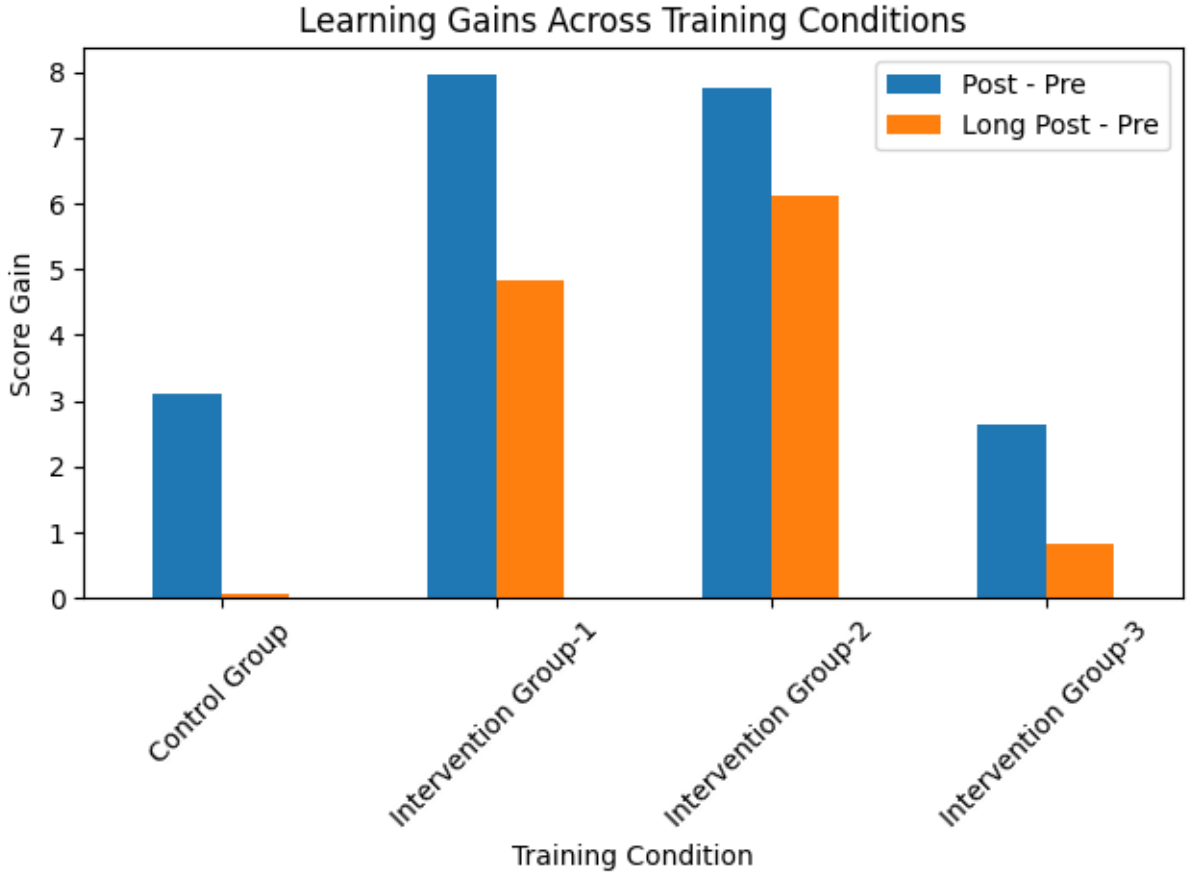


Figure 5.3: Learning gains relative to baseline. Both card-based interventions demonstrate substantially larger gains than the traditional classroom and control conditions. While some decay is observed at Long Post, card-based training retains meaningful gains, whereas classroom and control conditions show minimal sustained improvement.

(a) **Familiarity with Technology reduced Cognitive Engagement**

Some CHWs with prior experience with smartphones performed poorly in the digital game compared to expectations. Discussions with the participants later revealed that they relied on rapid chances and mindless tapping of cards, rather than carefully evaluating the sequence of the immunization schedule. A CHW remarked, “I thought I knew the order because I use phones every day, but I kept placing cards wrong because I was rushing.” This is in line with "Cognitive Load Theory" [2] (Sweller, 2022), which suggests that familiarity with technology often leads to

[2]Cognitive Load Theory (Sweller, 1988) : Technology familiarity creates overconfidence, leading to surface-level engagement where interface cues overload working memory instead of supporting deliberate content processing.

surface-level engagement and overreliance on interface cues. However, CHWs with minimal digital experience approached the game more purposefully, taking extra time to reason through each card placement, which improved knowledge retention. This is supported by dual-process theory (System-2 thinking) [3] (Kahneman, 2011), highlighting the benefits of conscious or thoughtful learning.

(b) **Highly competitive groups bargain Accuracy for Speed**

Another counterintuitive finding was that highly competitive teams of CHWs sometimes showed lower overall accuracy during early rounds of digital-based card play. While competition increased engagement and motivation, it occasionally triggered speed-over-accuracy strategies. CHWs prioritized finishing first over correctly sequencing cards. Meanwhile, we observed that CHWs playing with physical cards were less competitively driven and more collaborative in their approach. They achieved higher correctness by playing at a slower pace. This is in line with the principles of Self-Determination Theory [4] (Miller et al., 1985).

Learning - A Social Experience

The results suggest that when CHWs engage in game-based learning, they are not just having fun but learning in a way that retains them. The interactive and hands-on nature of the digital and tangible card games increased the interactivity and engagement of the training. It aligns with the Social Learning Theory [5] (Bandura, 1977).

Number Literacy and Content Standardization

Several issues came up during testing. A few CHWs struggled with numeric literacy, especially those who had studied in madrasas, where English numerals were more familiar than Hindi. We responded by changing card numbers to English digits. We also found that some vaccines mentioned in our materials were not administered in all districts, which confused participants during the evaluation. A handful of participants attempted to

[3] Kahneman's dual-process theory: System 1-driven quick moves in the game can result in immediate gains through game scores, but shallow knowledge retention without slow reflecting and reasoning driven by System 2. (Explained in detail in Theoretical Framework In the chapter Background and Literature Review)

[4] Self-Determination Theory : Intrinsic motivation and social relatedness can override extrinsic incentives like competition in supporting effective learning.

[5] Social Learning Theory: Learning is a social activity where we learn by observing, imitating, and modeling. We learn better when peers are actively engaged in the learning process.

refer to printed job aids during testing. We removed these cases from the final analysis. To prevent bias, we randomized the order of questions in later survey rounds.

Learning Experience through Games

During informal feedback sessions, most of the CHWs indicated that they enjoyed playing the game. They found it better than classroom sessions and more relevant to their work. Some teams were very competitive, which helped reinforce memorization of the order of the cards. The findings point to a broader potential shift in how we might train CHWs. Physical card games were easy to adopt as they did not require high-tech hardware or advanced literacy. They can be designed locally for local health agendas and scaled up with little technology. They offer a practical, scalable solution to this problem, even at their simplest.

5. **Conclusion**

   This study shows that learning through games with smartphones or paper cards can be a robust and feasible method for training community health workers. Both formats helped CHWs learn and retain key health information better than classroom sessions. With rising smartphone access, digital tools offer slightly better outcomes and added flexibility, which has the advantages of scalability and long-term retention. This strategy can help bridge long-standing training gaps and improve public health outcomes across India by scaling the solution. Physical card-based games can be an alternative at places where access to digital infrastructure is scarce. The findings also suggest that training approaches combining physical and digital game-based learning can be more effective.

5.2 Study-2 Reflections

The findings of this study show that both digital and physical card-based game interventions significantly outperformed traditional classroom training and the control group in terms of immediate knowledge gain. The digital version of the game demonstrated better outcomes in retention, although the statistical significance between digital and physical modes remained marginal.

Competitive and collaborative elements also improved peer learning and social reinforce-

ment, which are the core components of community work. Gamification also improved intrinsic motivation and increased participants' confidence, an element often lacking in traditional training programs. The initial rounds of digital play show the novelty effect. Players showed increased interest due to the introduction of new technology and instant feedback. However, as the players adapted to the given limited set of scenarios, interest started to level off. This insight suggests the need for repeated revisions to the content and the introduction of more varied, context-dependent challenges, enabling CHWs to experience long-term interest and learning benefits.

Critique of these two experimental studies highlighted that the sample populations selected were unlikely to be representative of the broader demographic and contextual variability of Community Healthcare Workers (CHWs) in India. In response to this limitation, the experimental design of the subsequent experiment sought to expand the geographic and contextual reach by implementing the intervention in several demographically varied sites.

There were practical challenges in executing multi-arm study designs, particularly in geographically constrained or community-based settings. There were risks of intervention spillover and sample contamination. To mitigate these risks and enhance the simplicity of implementation, the study design of the subsequent experiment was condensed to three arms, excluding the placebo or no-intervention group. This adaptation allowed for better comparisons without interfering with the intervention and data collection process.

Chapter 6

Study-3: Replay, Revise, and Refresh: Smartphone-Based Refresher Training for Community Healthcare Workers in India

This study is similar to the previous study. However, in this study, the experiment was conducted in various locations in Madhya Pradesh, improving the representativeness of the population. We chose rural, tribal, and urban areas of Madhya Pradesh and Maharashtra. Mostly Hindi versions of digital and physical cards and questionnaires were used; however, in some areas bordering Maharashtra, we also used the Marathi version of the questionnaire.

6.1 Study-3 Details

Parts of this section are from a published research Study [1]

1. **Introduction**

   The prevailing concerns on training CHWs are already discussed in the chapter 1.

   This study investigates the comparative effectiveness of three refresher training approaches: standard classroom training, a physical card game, and a smartphone digital game.

   This study particularly wants to address the following research questions:

   - RQ1: Is gaming an effective replacement for standard classroom-based refresher training in enhancing CHW knowledge?
   - RQ2: Do digital and physical gameplay modalities differ significantly in terms of the immediate gains in knowledge?
   - RQ3: Do the two modalities differ significantly regarding long-term knowledge retention?

2. **Methods**

   We built the training game using 60 cards (Chapter 7.4.2). The process of designing the card deck is discussed in Appendix A.3. We printed the playing cards on card paper for the physical version, and the digital game mimicked the same format and rules of play, discussed in Appendix A.6. We developed a digital version of the card game as a smartphone application with the same content and rules of play as the physical game. We calculated the sample size using G*Power software (d = 0.5, 1-β = 0.95, α = 0.05), leading to a required sample of 88 participants per group. Considering attrition, we initially recruited 95 participants per group. 90 participants per group retained, totaling 270 participants. Each group had an equal distribution of ASHAs and AWWs (45 each).

[1] **Majhi, A.**, Agnihotri, S. B., Mondal, A., 2024. Replay, Revise and Refresh: Smartphone-Based Refresher Training for Community Healthcare Workers in India ", *HCI International 2024, Washington DC, USA* https://link.springer.com/chapter/10.1007/978-3-031-61966-3_34)

A quasi-experimental design was conducted. We divided the participants into three groups:

- IG1: Digital card game (smartphone-based)
- IG2: Physical card game
- IG3: Traditional classroom training (control group)

We conducted three assessments for each group: a baseline pre-test, a post-intervention test, and a delayed post-test after three weeks. All tests included 45 multiple-choice questions (Appendix B) focused on immunization and healthcare scheduling, shuffled across instances to reduce recall bias. We performed quantitative analysis to evaluate intra- and inter-group differences.

3. **Results**

Immediate knowledge gain was significantly greater in the digital card game group, while there was no significant difference in knowledge retention between physical and digital card game-based groups. Traditional classroom training showed significantly low retention of knowledge. The qualitative feedback indicated that CHWs liked the digital version of the card game due to its convenience and interactivity, but had reservations about distractions and the requirement of printed versions for reference.

A. Participants' Demography						
	Intervention Group-1 Digital Card Game (n=90)		**Intervention Group-2 Physical Card Game** (n=90)		**Intervention Group-3 Classroom In-Person training** (n=90)	
Parameters	ASHAs(%) (n=45)	AWWs(%) (n=45)	ASHAs(%) (n=45)	AWWs(%)) (n=45)	ASHAs(%) (n=45)	AWWs(%) (n=45)
Age group (years)						
Less than 30	1 (2.22%)	0 (0%)	1 (2.22%)	1 (2.22%)	2 (4.44%)	2 (4.44%)
30-40	22 (48.89%)	19 (42.22%)	22 (48.89%)	20 (44.44%)	23 (51.11%)	19 (42.22%)
40-50	21 (46.67%)	25 (55.56%)	20 (44.44%)	24 (53.33%)	20 (44.44%)	24 (53.33%)
No information	1 (2.22%)	1 (2.22%)	2 (4.44%)	0 (0%)	0 (0%)	0 (0%)
Education (schooling grade)						
Below 8^{th}	3 (6.67%)	2 (4.44%)	1 (2.27%)	2 (4.44%)	1 (2.22%)	2 (4.44%)
$8^{th} - 10^{th}$	9 (20%)	11 (24.44%)	9 (20.45%)	14 (31.11%)	10 (22.22%)	12 (26.67%)
$10^{th} - 12^{th}$	29 (64.44%)	28 (62.22%)	29 (65.91%)	23 (51.11%)	29 (64.44%)	27 (60%)
Graduate and above	4 (8.89%)	4 (8.89%)	4 (9.09%)	5 (11.11%)	5 (11.11%)	4 (8.89%)
No information	0 (0%)	0 (0%)	1 (2.27%)	1 (2.22%)	0 (0%)	0 (0%)
Experience as CHWs (years)						
0-5	2 (4.44%)	2 (4.44%)	4 (8.89%)	3 (6.67%)	4 (8.89%)	2 (4.35%)
5-10	18 (40%)	17 (37.78%)	16 (35.56%)	16 (35.56%)	19 (42.22%)	17 (36.96%)
10-15	18 (40%)	18 (40%)	18 (40%)	19 (42.22%)	17 (37.78%)	18 (39.13%)
Above 15	7 (15.56%)	8 (17.78%)	7 (15.56%)	6 (13.33%)	4 (8.89%)	8 (17.39%)
No information	0 (0%)	0 (0%)	0 (0%)	1 (2.22%)	1 (2.22%)	1 (2.17%)

Table 6.1: Study-3: ParticipantsDemography

Experimental Study Results	**Intervention Group-1 Test Scores**		**Intervention Group-2 Test Scores**		**Intervention Group-3 Test Scores**	
Test phase	Mean (SD)	Median (min-max)	Mean (SD)	Median (min-max)	Mean (SD)	Median (min-max)
Pre-test scores	24.85 (3.34)	24.16 (18-33)	24.28 (2.88)	24.13 (19-30)	25.02 (3.19)	24.69 (19-32)
Post-test scores	35.62 (3.57)	35.56 (28-42)	31.86 (4.1)	31.48 (24-42)	30.01 (4.32)	29.71 (19-39)
Delayed Post-test scores (After 3 weeks)	31.75 (4.69)	32.23 (22-41)	30.62 (4.1)	30.82 (21-40)	26.46 (3.54)	26.75 (19-34)

Table 6.2: Study-3: Study Results

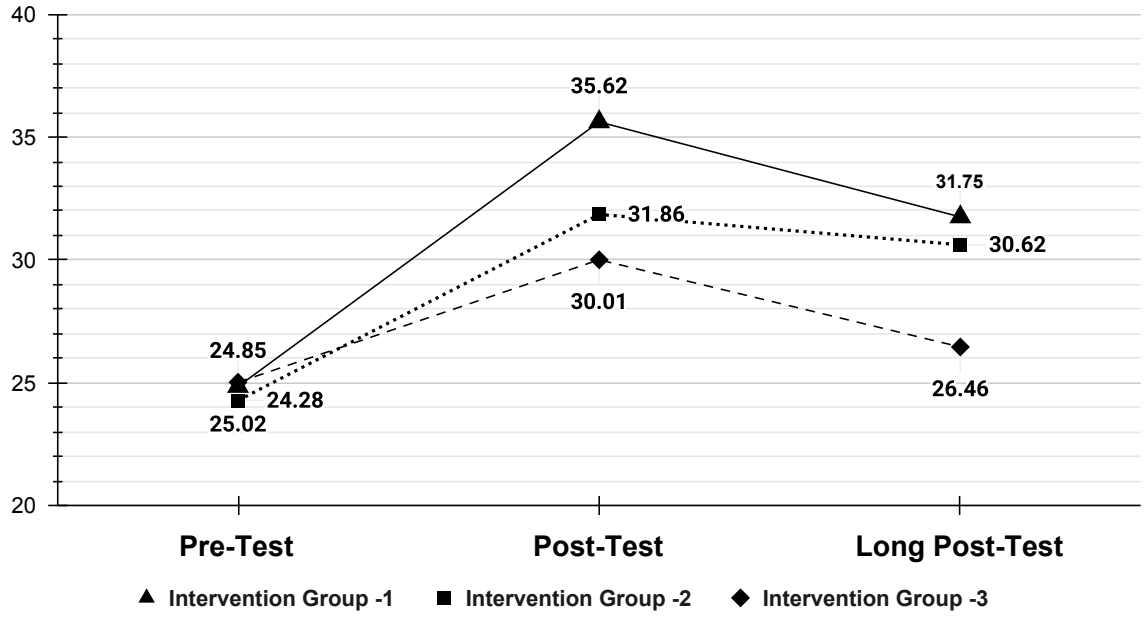


Figure 6.1: Study 3: Line chart showing trends of change of mean value in the pre-test, post-test, and delayed post-test

Pre-test scores did not differ significantly across the three groups ($F = 0.84$, $p = 0.44$), testifying to an equal baseline. Post-intervention scores differed significantly ($F = 22.96$, $p < 0.00001$). Tukey's HSD test revealed that IG1 performed significantly better than both IG2 ($F = 6.36$, $p = 0.00004$) and IG3 ($F = 9.39$, $p < 0.000001$). We did not observe

a statistically significant difference between IG2 and IG3 ($F = 3.03$, $p = 0.085$).

After three weeks, delayed post-test scores varied significantly ($F = 20.62$, $p < 0.00001$). IG1 and IG2 scored significantly higher than IG3 ($p < 0.05$). No significant difference existed between IG1 and IG2 ($F = 1.84$, $p = 0.4$), indicating equivalent retention. There were significant gains from the pre-test to the post-test for all groups. IG1 gained the most in scores, followed by IG2, then IG3. While IG1 gained the most immediately, its delayed retention dropped more significantly than IG2, which retained more steadily. IG3 consistently had the lowest scores at every phase.

4. Discussion

This study validates the hypothesis that gamified training methods, particularly smartphone-based interventions, can outperform traditional classroom training in immediate knowledge acquisition. However, the performance of digital and physical card games converged when we checked knowledge retention over three weeks. This insight underscores the potential of both gamified methods as viable tools for CHW refresher training.

Findings in Playful Learning

Although digital and physical card games both significantly improved knowledge, several observations challenged conventional assumptions and revealed nuanced interactions between digital literacy, local context, and learning behavior.

(a) **Technical Constraints hinders Cognitive Processing**

CHWS from rural and tribal areas who were in the intervention group, which used the digital card game, sometimes performed more slowly than those who were intervened with the physical card game, despite having prior experience with smartphones. Interviews from the feedback session and careful introspection revealed that limited network connectivity and small smartphone screens caused "cognitive friction", reducing engagement. This behavior is in line with Cognitive Load Theory (Sweller, 2022). [2]

One ASHA from a tribal block remarked, “I know how to use a phone, but tapping through this many screens was confusing. I liked spreading the cards on the table better.” Physical cards enabled better reasoning and collaborative discussions, which led to a reduction in extrinsic cognitive load and supported "distributed cognition" [3] (Hutchins, 1995) through visible sequencing and scaffolding from coparticipants.

(b) **Switching to Physical Cards**

There were instances where CHWs in the intervention group, who had access to the digital card game, voluntarily opted to switch to physical cards for peer learning or practicing at home. One CHW commented, “I liked the app, but with the real cards, I could show my friends and explain everything. It stays in my memory better.”

[2] Cognitive Load Theory (Sweller, 1988): Learning suffers when extraneous demands (e.g., poor interfaces) overload limited working memory alongside essential content.

[3] Distributed Cognition (Hutchins, 1995): Intelligence emerges across people/tools; physical cards externalize thought processes for group reasoning.

This behavior is in line with the principles of "Social Constructivism" [4] (Vygotsky, 1978). Collaborative dialogue between co-learners and tangible artifacts facilitates the co-creation of knowledge. These findings have important implications for public health training programs. Gamified learning can offer a scalable, interactive, and efficient approach to training. However, hybrid models that combine digital and printed materials may more appropriately address the varied needs and constraints of CHWs.

(c) **Drop in Retention of Knowledge among Digital Play Group**

CHWS in the intervention group who played the digital card game sometimes showed a sharper drop in retention compared to those in the intervention group who played the physical card game. The novelty and interactivity of the app generated intrinsic motivation (Miller et al., 1985), but in some cases also promoted overconfidence and choosing quick strategies, reducing careful thinking. A participant reflected, "I wanted to finish first, but then I realized I had misplaced some cards—I remembered better after playing slowly with my friends." This gels with Kahneman's dual-process theory [5] (Kahneman, 2011).

(d) **Context-Dependent Perception of Games**

We observed that urban CHWs sometimes treated the digital game as a leisure activity. However, rural and tribal CHWs engaged with it as a formal learning tool. One rural CHW explained, "Playing on the phone with my community CHWs felt serious here, but when I go for training and play it with the city CHWs, it seemed like a game to pass the time." This underscores that learning outcomes are co-determined by social, cultural, and environmental factors, rather than being solely determined by the modality itself. This is in line with Situated Learning Theory [6] (Lave and Wenger, 1991).

[4] Social Constructivism: Knowledge builds through social interaction within the zone of proximal development, where peers scaffold understanding via shared tools and dialogue.

[5] Kahneman's dual-process theory: System 1-driven quick moves in the game can result in immediate gains through game scores, but shallow knowledge retention without slow reflecting and reasoning driven by System 2. (Explained in detail in Theoretical Framework In the chapter Background and Literature Review)

[6] Situated Learning Theory (Lave & Wenger, 1991): Knowledge emerges through participation in "communities of practice," where context shapes meaning-making. Rural CHWs' serious community setting fostered legitimate peripheral participation; urban mixed groups diluted it into casual play.

5. **Conclusion**

This study shows that training using smartphones could be an effective method for refresher training, by enhancing the knowledge retention of CHWs. Though the digital game showed a higher short-term gain in knowledge, long-term retention was almost similar in both digital and physical card games. Evidence also shows that game-based training can complement conventional training programs in low-resource contexts. Future research can investigate long-term effects, motivational determinants of the use of games, and integration of gamified learning into routine CHW workflows.

6.2 Study-3 Reflections

The experiment results show that game-based learning interventions—both physical and digital—are significantly more effective than traditional classroom-based methods in enhancing knowledge acquisition among CHWs. Post-test results showed the highest gain in the digital gameplay group (IG1), followed by the physical card group (IG2), with the classroom training group (IG3) showing the least improvement. These findings validate our first hypothesis that interactive and engaging formats, such as games, can be superior alternatives for refresher training, especially when aiming for immediate knowledge enhancement.

In terms of long-term retention, measured three weeks after the intervention, the digital and physical groups performed significantly better than the classroom group. The difference between digital and physical modes was not statistically significant. We conducted a focused group discussion with CHWs, during which written notes were taken and subsequently analyzed manually using a thematic analysis approach (Braun and Clarke, 2006, 2019) informed by grounded theory guidelines Charmaz (2014). The analysis involved repeated reading of the notes, initial coding of recurring ideas, and the identification of broader themes that captured shared patterns across participants' responses. The themes that emerged from participants' feedback supported major sociocultural determinants of learning. While some CHWs enjoyed the interactivity and independence offered by the smartphone game, others used physical cards because they were tangible, easy to use, and could serve as counseling tools during field visits. Smartphone distraction, especially when shared among the CHWs, also constituted a significant

challenge.

These findings validate the imperative for the creation of educational interventions for effectiveness as well as contextual appropriateness. Physical card play, while less technologically advanced, is more accessible and can be more easily integrated into the everyday routines of CHWs. Digital smartphone-based games, however, incorporate automation of feedback, progression tracking, and scalable opportunities to disseminate, especially when presented by public health platforms or app stores.

At a policy level, the research suggests that investment in low-cost, culturally targeted, gamified training interventions can improve the quality and consistency of health services delivered by CHWs. These interventions can act as reinforcement strategies following in-class training, filling the knowledge gap and building confidence, hence driving better health outcomes at the community level. Looking to the future, the results endorse a hybrid training paradigm that leverages the complementary strengths of physical and digital interventions.

6.3 From Knowledge Acquisition to Field Performance

Building upon the insights gained from the three preceding experimental studies conducted across diverse locations in India, this research sought to investigate the practical implications of gamified training interventions. While earlier studies focused primarily on knowledge acquisition and retention, our objective in the subsequent phase was to explore how such interventions influence the day-to-day practices and performance metrics of Community Healthcare Workers (CHWs). Specifically, we aimed to assess the extent to which game-based training modalities, particularly those with location-based mechanisms, translate into measurable improvements in CHWs' routine field activities. We addressed this inquiry in some limited detail in the fourth experimental study, presented in the next study.

6.4 Small Experimental Study - 1 : Mapping Child Malnutrition and Measuring Efficiency of Community Healthcare Workers through Location-Based Games in India

Most CHWs are poorly trained and lack digital tools and skills for adequate and sustained data collection. CHWs' access to smartphones opens up new opportunities to leverage digital tools for improved health surveillance. Healthcare planning and intervention depend upon timely and accurate data collection. Conventional data entry computer software is often incapable of facilitating extended CHW participation due to usability and a lack of contextual appropriateness. Previous resistance to digital tools like the Poshan Tracker points to the need for user-centered design principles that emphasize simplicity, training, and user participation.

This study examines the use of geolocation-based games to enhance the quality and engagement of CHWs in collecting child anthropometric measurements. Recognizing the value of CHWs in providing community-level health interventions, this study examines how gamified data collection tools can improve the frequency and quality of anthropometric measurements in difficult-to-reach populations.

This study answers the following research questions:

RQ1: Does implementing a location-based game improve the measurement efficiency of CHWs more than a conventional data entry application?

RQ2: Is the efficiency gained through the game-based intervention maintained in the long term?

The Measurement Efficiency Index (MEI) is a composite score ranging from 1 to 100 that measures the effectiveness of Community Health Workers (CHWs) in performing anthropometric data collection tasks, such as measuring height, weight, and mid-upper arm circumference (MUAC) of newborn children during household visits. It combines three main parts, each scored from 0 to 100, with equal weight: MEI = (Accuracy + Coverage + Timeliness)/3. Accuracy checks the precision of measurements by comparing CHW readings to re-measurements performed by a local supervisor who shadows the CHW and verifies random cases on site (for

example, full points are awarded if weight is within measurement tolerance, with partial scores given for minor errors. Coverage measures how completely CHWs visit assigned households, using GPS data from the game to calculate the percentage of eligible children reached (with extra points for hard-to-reach rural or tribal areas). Timeliness scores are based on app timestamps, including the average time per child (target under 5 minutes) and visits completed within due dates, benchmarked against the group average. This index offers a clear and objective way to compare game-based training with traditional methods.

Data collection happens through the Android geolocation game app. CHWs receive a list of tasks to perform daily. They use GPS to navigate to homes, pin locations, scan child IDs, enter measurements with photos of tools, and get instant feedback. The app syncs data to a central dashboard when online, building community heatmaps for stunting, wasting, and underweight rates while giving game points for streaks and exploration. Supervisors use a linked app to log re-checks anonymously.

we recruited around 200 CHWs across multiple Indian states following convenience sampling, and 94 participants per group completed the study. We conducted a quasi-experiment with 188 CHWs in two groups: one using the game-based app and the other using a typical non-game data entry app. There were pre- and post-intervention tests and a delayed retention test (DRT) three months after the intervention to assess long-term engagement and data collection efficiency.

The intervention group (geolocation game app) demonstrated statistically significant improvements in measurement efficiency compared to the control group (regular data collection app) ($p = 0.00004$; Cohen's $d = 1.6$). While both groups showed score reduction in the delayed post-test, the IG maintained a significantly higher performance level than the CG, supporting both RQ1 and RQ2. Mean scores for the IG improved from 49.04 at baseline to 73.9 post-test and slightly declined to 69.14 at DRT. In contrast, the CG scores moved from 51.46 to 54.84 post-test and returned to 52.58 in DRT.

The findings indicated that the game-based application significantly improved the efficiency of data measurement among CHWs compared to the traditional non-game app. The game-based model showed better engagement and retention. CHWs indicated a higher motivation level for gathering data in untapped regions. The study established that the initial motivation towards the game-based application gradually reduced, indicating the need for sustained

reinforcement and support to achieve long-term engagement.

We were also further interested in exploring the effectiveness of these tools with nursing students, who share functional and educational overlaps with CHWs, particularly in frontline healthcare delivery, community engagement, and training needs.

6.5 Small Experimental Study – 2: Pilot Testing Game-Based Training with Pre-Service Health Trainees

The aim of this study was to check the applicability of the intervention across different levels of formal healthcare training. The inclusion of nursing students as study participants was intended to provide insights into how such game-based interventions could be integrated into early-stage training.

The first-year ANM batch (n=39, Average Age=19 years, SD=2.28) had recently joined the course in the last 2 months. They didn't have any prior knowledge about immunization and mostly relied on guesswork while filling out the pre-test questionnaire. Conducting t-tests, we found no significant difference between the pre- and post-test scores of the physical card group [p=0.197, average improvement in scores = 4.6], but a significant difference was observed in the digital group [p=0.0009 ($p < 0.001$), average improvement in scores = 16.5]. As the intent of this research is to study the effectiveness of the tool as a refresher training, this cohort does not align well with the study requirements due to the absence of prior knowledge. However, the digital group shows promising improvements.

The second-year ANM batch (n=29, Average Age=20.25 years, SD=2.25), who had prior introductory coursework in immunization, showed significant differences in both the physical card group [n=19, p=0.003, average improvement in scores = 3.63, SD=4.7] and the digital app group [n=10, p=0.002, average improvement in scores = 8.5, SD=6.4]. The digital app group showed higher gains, but with larger variation. A t-test comparing the differences between the card and digital app groups showed no statistically significant difference (p=0.051), indicating a marginal trend but insufficient evidence to establish superiority of one intervention over the other.

The first- and second-year B.Sc. Nursing batches were excluded as they had not yet been taught immunization in details, and fourth-year students were unavailable due to clinical placements. Therefore, the study focused on the third-year B.Sc. Nursing batch (n=35, Average Age=21 years, SD=1.14), who had prior exposure to immunization coursework. Significant differences between pre- and post-tests were observed in both the physical card group [n=25, $p=0.0000007$ ($p < 0.001$), average improvement in scores = 9.16, SD=6.88] and the digital app group [n=10, $p=0.00005$ ($p < 0.001$), average improvement in scores = 12.3, SD=5.38]. Although the digital app group showed higher average gains, a t-test comparing the differences between groups indicated no statistically significant difference ($p=0.17$). Thus, while both interventions are effective as refresher training tools, no significant difference between them can be concluded.

Figure 6.2: Gameplay session with nursing students

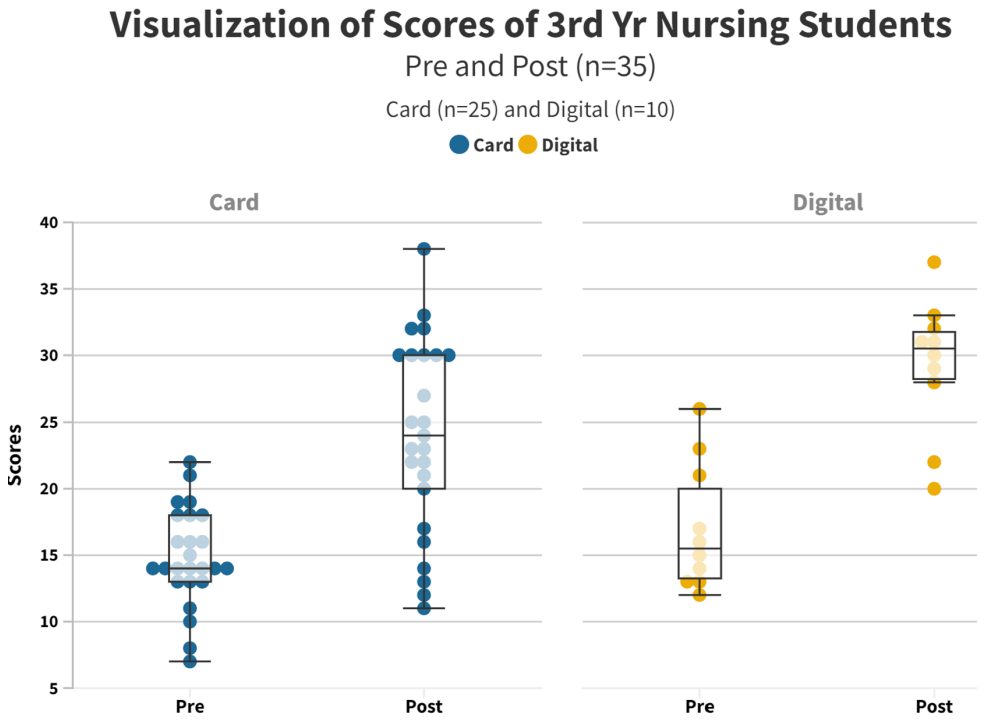
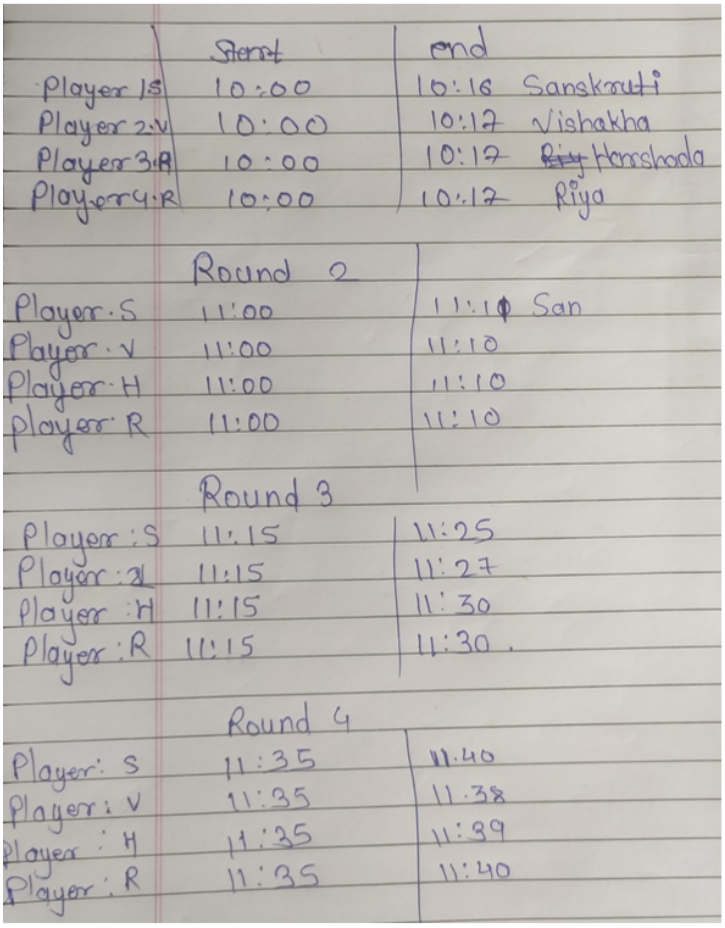


Figure 6.3: Results from comparing the scores of nursing students



	Start	end	
Player 1:S	10:00	10:16	Sanskruti
Player 2:V	10:00	10:17	Vishakha
Player 3:H	10:00	10:17	~~Riya~~ Harshada
Player 4:R	10:00	10:17	Riya

Round 2		
Player:S	11:00	11:16 San
Player:V	11:00	11:10
Player:H	11:00	11:10
player:R	11:00	11:10

Round 3		
Player:S	11:15	11:25
Player:V	11:15	11:27
Player:H	11:15	11:30
Player:R	11:15	11:30

Round 4		
Player:S	11:35	11:40
Player:V	11:35	11:38
Player:H	11:35	11:39
Player:R	11:35	11:40

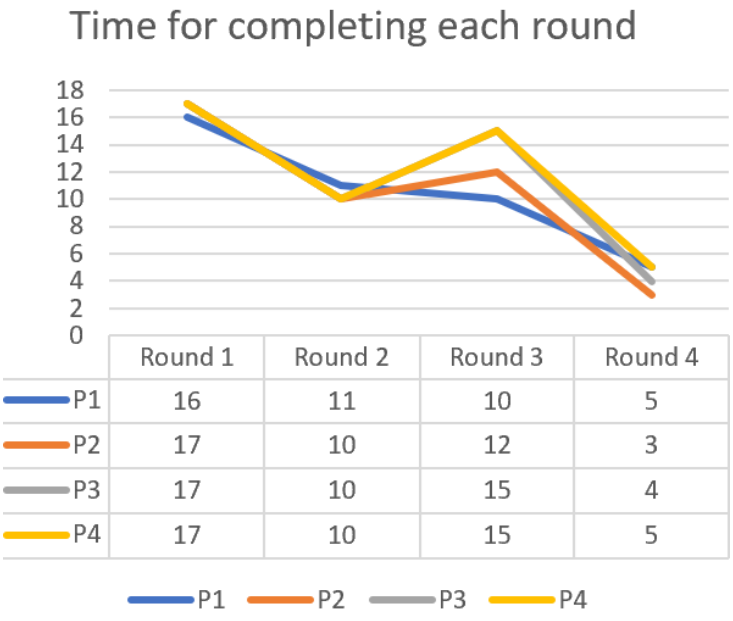


Figure 6.4: Decrease in time taken to complete a round showing mastery

Chapter 7

Development of the Game

7.1 From Guidelines to Game Concept

Initially, we began with content focusing on the health and nutrition of both the mother and child. Abstract nutrition advice in Indian guidelines, such as "balanced diet" or "gradual complementary feeding", inherently resists binary game outcomes (win/lose), as real counseling involves cultural variability and no fixed "correct" meal. Nutrition guidelines in India, such as those in ASHA Modules 6 and 7, often emphasize broad dietary mixes (e.g., cereals, pulses, greens) without precise quantities or unambiguous rules, which complicates the derivation of game rules. In contrast, ANC, PNC, and immunization follow prescriptive schedules (e.g., 4+ ANC visits, specific postnatal checks on days 3, 7, 14, 28, 42), which suit the incremental scaffolding of content and decision-making mechanics. The game-based learning literature notes similar issues, where ambiguous health content could lead to superficial mechanics unless simplified, thereby risking an ambiguous knowledge transfer to CHWs.

The primary challenge was converting detailed healthcare guidelines on child immunization, ANC, and PNC into playable game mechanics without compromising clinical fidelity.

The raw content was in the form of tables, checklists, and training slides, which do not map directly to meaningful card actions or win conditions, a difficulty also noted in other educational card-game projects and research. The design task was to identify "gameable" patterns, such as age windows, sequences, and common disruptions, and decide what could become card types, suits, or event triggers. Early drafts attempted to represent every nuance of the guidelines, resulting in a complex list of card types and conditions. The playable structure was missing. There was no clear sense of what a "round" was, what players were trying to optimize, or how a session would end.

We resolved this issue by revising the core metaphor of the game. Play rounds represent months (the time period during which the vaccine or care is administered), and the win condition is completing the correct care sequences for children and mothers with minimal missed opportunities. Guideline details that did not translate well to simple actions were shifted to facilitator notes and debrief discussions, rather than being forced into the game rules. All the information was categorized into four silos: children below 1 year, children above 1 year, ANC, and PNC, but with an important difference.

7.2 Development of the Augmented Reality Game

7.2.1 Idea of game

CHWs can play in single-player or multiplayer mode (based on group size), focusing on modules such as pregnancy (ANC) and child immunization. Before starting, players stick 10-20 printed child and mother cards from the kit onto room walls at a height easy to scan with smartphones, while carrying bags with physical vaccine and medicine cards.

In single-player mode, the goal is to immunize all children and care for mothers correctly and efficiently according to the schedule. Players scan a wall card (eg. 0-month child) to display a 3D model with age details, then scan matching cards (BCG, HepB, OPV, colostrum) for points and smiling sounds. Wrong cards result in crying sounds and point loss. Missed cards also have negative points from all players or teams. Play runs in 4 sequential rounds where children age (e.g., $1\frac{1}{2}$ to $2\frac{1}{2}$ months), with scores shown on a leaderboard.

In multiplayer mode, 4 players form teams of 2 with randomized card bags. Teams trade cards within or between groups to finish rounds; hoarding slows others, but cooperation leads to higher team scores and wins. Players choose roles freely, such as one scanning while the other trades cards.

7.2.2 Design of the card deck

Unity served as the primary game development platform, with Vuforia integrated for augmented reality (AR) functionality. Card images were uploaded to the Vuforia server as image markers, requiring unique visual designs to enable distinct detection by the vision model. Text modifications alone proved insufficient for reliable recognition, necessitating graphic variations for each vaccine or information card to ensure accurate AR tracking.

Cards were designed to integrate textual statements with illustrations for effective information delivery, featuring graphics, titles, and brief descriptions. Images proved more effective

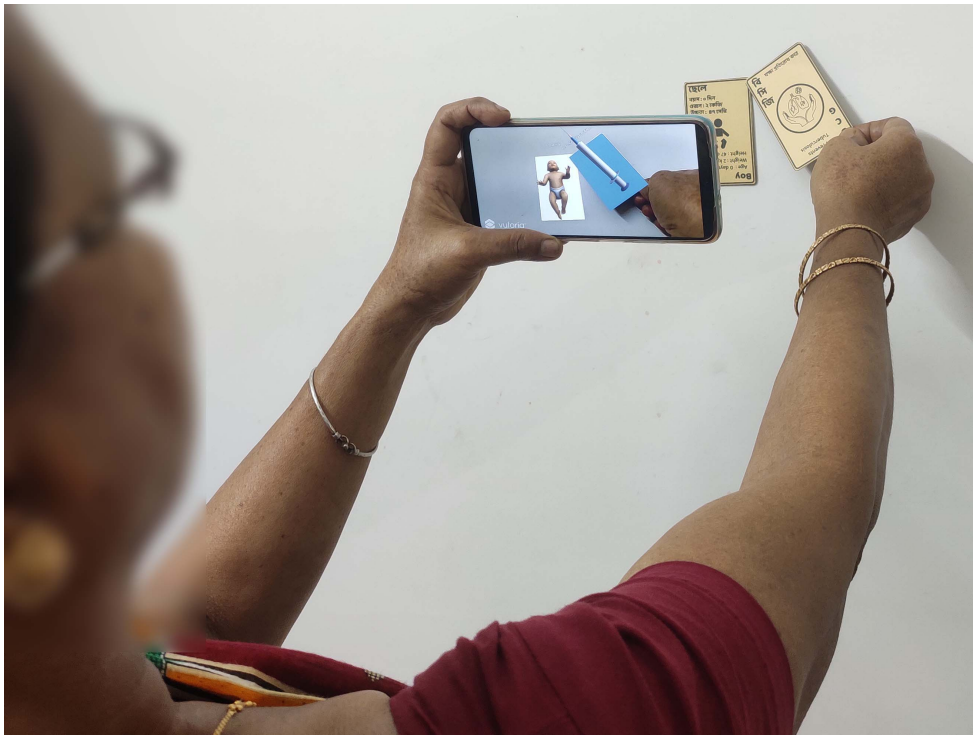


Figure 7.1: ASHAs play-testing the Tikakaran-AR play app

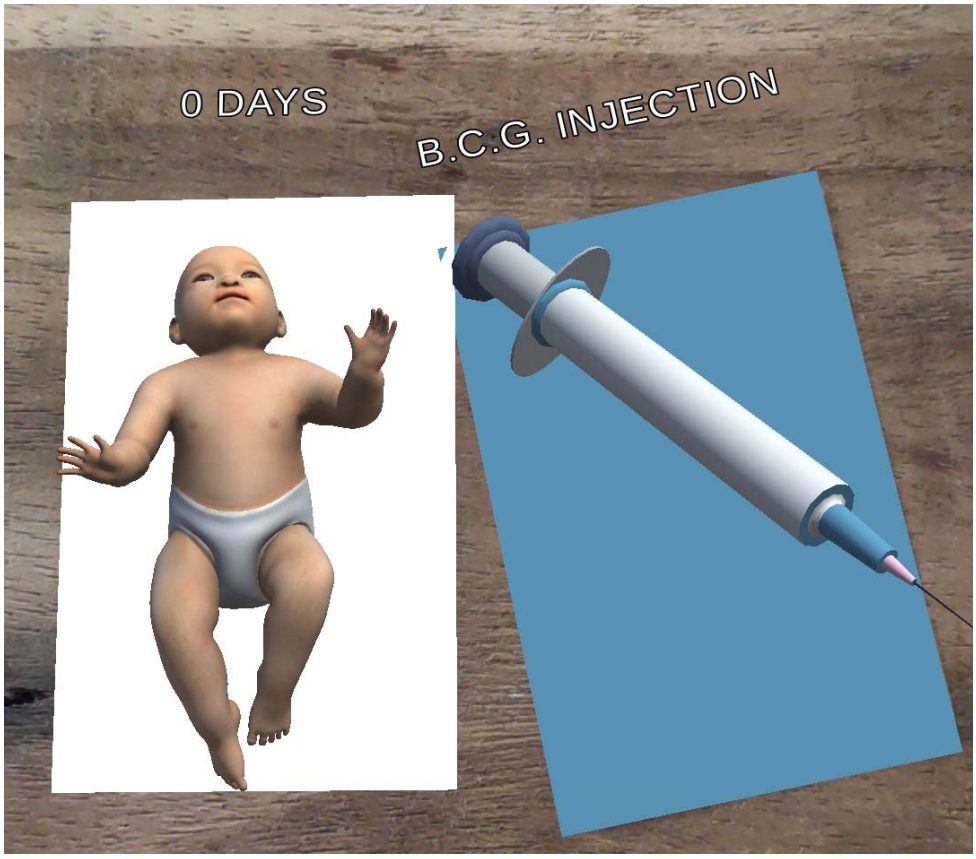


Figure 7.2: Child 3D model and Vaccine representation in Augmented Reality

Figure 7.3: CHWs playing in teams of two - one holds the smartphone while the other shows the card

Figure 7.4: CHWs playing in teams of two with physical playing cards

than text alone for straightforward communication, enhancing recall, reducing language barriers, and increasing self-explanatory value. In focus group discussions (FGDs), participatory design exercises involved Community Health Workers (CHWs) selecting royalty-free illustrations resembling specific vaccines, medicines, or conditions. The majority's preferences determined final choices, supplemented by CHW sketches for contextual relevance.

Bengali was selected as the primary language for card text, targeting field studies in rural and suburban West Bengal. The deck comprised 20 unique cards: 10 on immunization, 5 on child medicines/supplements, 3 on maternal medications/immunization, and 2 sets depicting boys/girls across ages, heights, and weights. Cards measured 5.5 × 8.5 cm (standard playing card or credit card size), printed with graphics and information.

Figure 7.5: Card design for AR play and physical card play

7.3 Development of the Sequence Game

Insights from the Augmented Reality game prototype indicated that matching age-appropriate vaccines should be retained across various game formats. However, the games must remain simple for use by novice smartphone users and elderly women.

The sequencing game presents bins representing different age groups with vaccines in randomized order; players drag vaccines into the correct bins. On dragging and dropping the items, the list automatically rearranges, and correctly positioned items change to green to provide immediate visual feedback to the learner. A five-minute countdown timer is displayed to create a sense of urgency and encourage the timely completion of the task.

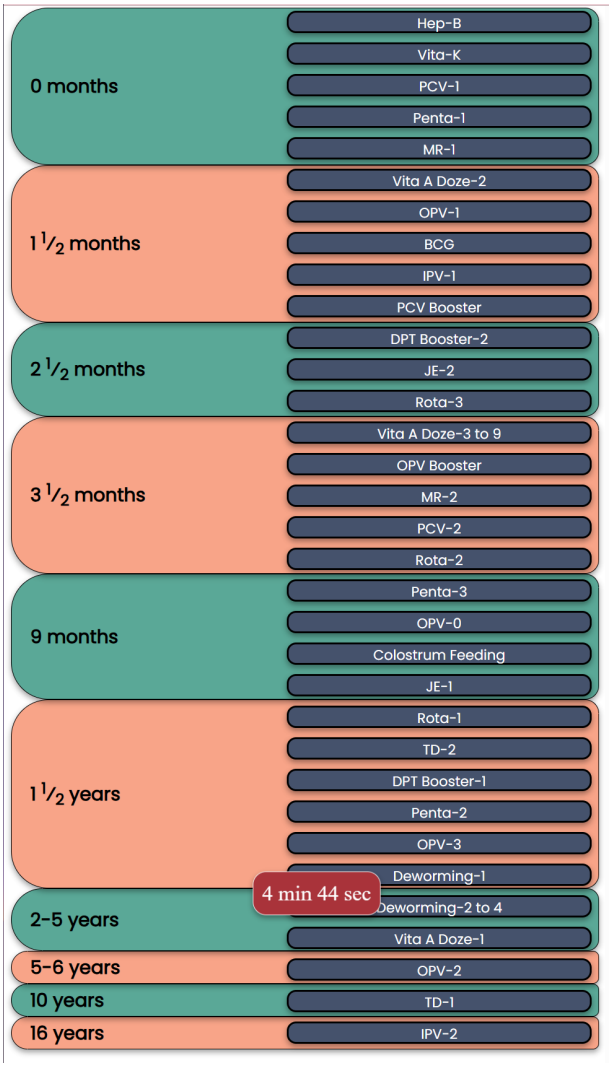


Figure 7.6: Gamestart - Screenshot of the Sequencing Game

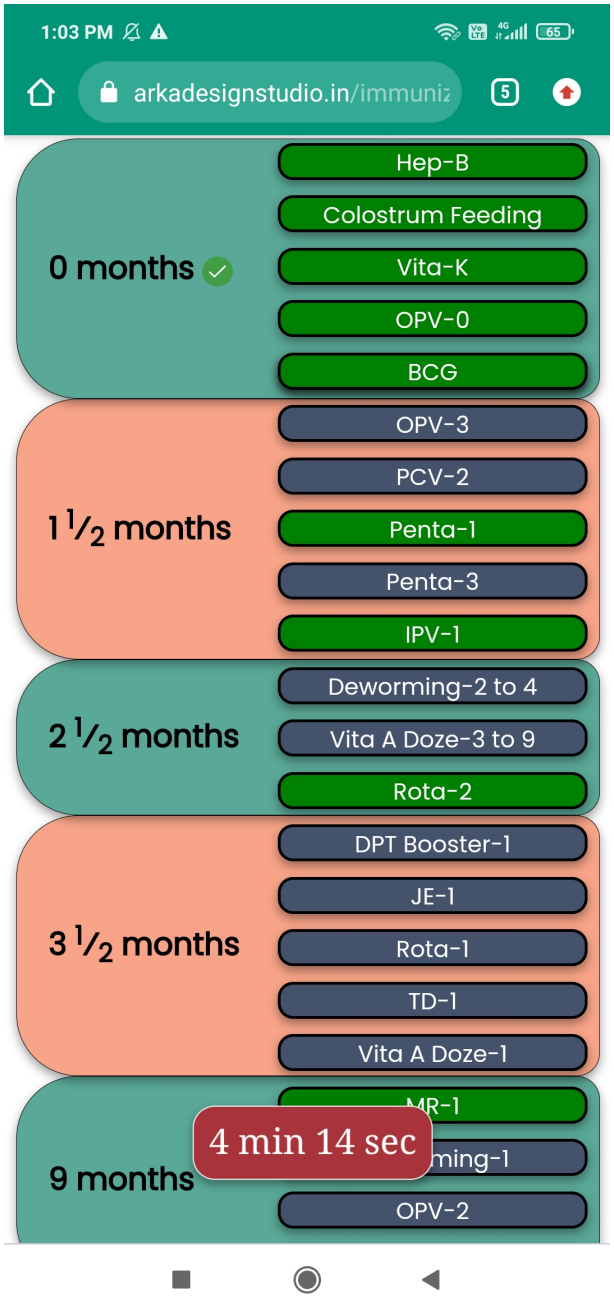


Figure 7.7: Upon sequencing the first bin of birth age (0 months) - Screenshot of the Sequencing Game

Out of the full code, we will discuss a few important sections that make the major decisions in the game design. The link to the GitHub repository and the Playable Live link to the game are in the footnote [1] [2].

A large interactive container was designed to hold smaller draggable items representing different vaccines. These items can be repositioned and arranged according to the immunization schedule. The schedule consists of ten stages of interventions, and each stage may include multiple vaccines, denoted as A, B, C, D, and E.

```
<div id="draggableContainer">

    <div id = "1A" draggable="true">OPV-0</div>
    <div id = "1B" draggable="true">Hep-B</div>
    <div id = "1C" draggable="true">BCG</div>
    <div id = "1D" draggable="true">Vita-K</div>
    <div id = "1E" draggable="true">Colostrum Feeding</div>

    <div id = "2A" draggable="true">OPV-1</div>
    <div id = "2B" draggable="true">Penta-1</div>
    <div id = "2C" draggable="true">Rota-1</div>
    <div id = "2D" draggable="true">PCV-1</div>
    <div id = "2E" draggable="true">IPV-1</div>

    <div id = "3A" draggable="true">OPV-2</div>
    <div id = "3B" draggable="true">Penta-2</div>
    <div id = "3C" draggable="true">Rota-2</div>

    <div id = "4A" draggable="true">OPV-3</div>
    <div id = "4B" draggable="true">Penta-3</div>
    <div id = "4C" draggable="true">Rota-3</div>
    <div id = "4D" draggable="true">PCV-2</div>
    <div id = "4E" draggable="true">IPV-2</div>

    <div id = "5A" draggable="true">MR-1</div>
    <div id = "5B" draggable="true">JE-1</div>
    <div id = "5C" draggable="true">PCV Booster</div>
    <div id = "5D" draggable="true">Vita A Doze-1</div>
```

[1] Link to the repository: https://github.com/arkamajhi/1000days/tree/main/Child_Imm_Lvl2

[2] Link to the Live Url: http://1000days.in/Child_Imm_Lvl2/index.html

```
29
30      <div id = "6A" draggable="true">MR-2</div>
31      <div id = "6B" draggable="true">JE-2</div>
32      <div id = "6C" draggable="true">OPV Booster</div>
33      <div id = "6D" draggable="true">DPT Booster-1</div>
34      <div id = "6E" draggable="true">Vita A Doze-2</div>
35      <div id = "6F" draggable="true">Deworming-1</div>
36
37      <div id = "7A" draggable="true">Vita A Doze-3 to 9</div>
38      <div id = "7B" draggable="true">Deworming-2 to 4</div>
39
40      <div id = "8A" draggable="true">DPT Booster-2</div>
41
42      <div id = "9A" draggable="true">TD-1</div>
43
44      <div id = "10A" draggable="true">TD-2</div>
45
46  </div>
```

Listing 7.1: Draggable Items Initialization

7.4 Development of the Card Game

7.4.1 Idea of game

Insights from the sequencing game prototype confirmed the need to retain age-appropriate vaccine matching. However, presenting information modules in a list format was less intuitive, leading to the exploration of card-based interfaces, which are more familiar to users. Cards were tested to determine if sequencing them proves more user-friendly while maintaining similar learning outcomes.

The design process followed an iterative, design-based research journey with recurring bottlenecks around content translation, rule complexity, materials, and institutional fit, each addressed through cycles of simplification, re-framing, and co-design with CHWs and trainers.

7.4.1.1 Unequal "Suits" from Endogenous Content

In classic card game, the standard deck consists of four suits with exactly 13 ranks each, which underpins core mechanics like balanced distribution, complete sequences, and symmetry between suits. In this game, the four "suits" (silos/foundations) emerged directly from health content: Child under 1 year, Child above 1 year, ANC, and PNC.

Because the immunization schedule and care protocols do not distribute evenly across these four categories, each suit ended up with a different number of cards. Some age bands and care periods are critical, characterized by a high density of interventions, while others have relatively few. Some "suits" (e.g., under-1 vaccines) had long, dense sequences, while others (e.g., PNC) had relatively few cards, upsetting expectations of symmetry. This pattern is also observed in other endogenous educational game designs that derive their suits from content structures rather than abstract symmetry. Respecting domain fidelity means giving up some of the mathematical neatness of traditional card decks. As a result, standard 4×13 assumptions behind Solitaire-like or Badam Satti-like rules could not simply be imported.

The first instinct was to treat these four content-based suits like the four suits in a regular 52-card deck and apply familiar patterns from Solitaire and Badam Satti. In Badam Satti,

for example, balanced suits allow sequences to grow outward from the 7 in both directions, with reasonably similar possibilities across suits. Using unmodified Badam Satti rules could make certain suits structurally advantaged or disadvantaged, because they offered more or fewer playable extension points. Concerns arose that the imbalance might break game flow or make the game feel "rigged" or incomplete to players familiar with standard decks. In Solitaire, equal suit lengths support the predictable completion of foundations.

Through playtesting, it became clear that strict numerical symmetry was not necessary for the game to support Solitaire-like or Badam Satti-like modes. What mattered more was preserving the relational structure. Other card-game design work similarly notes that non-standard decks can still support familiar patterns, as long as the ordering and adjacency relations within suits remain coherent (Palmer and Denelsbeck, 2024; Yan et al., 2005). In our case, the realization was that unequal suit lengths did not prevent:

- A Badam Satti mode, where any intervention can act as an anchor (analogous to 7s), and players extend care sequences up/down in time within each category, even if some sequences are shorter or longer.
- A Solitaire mode, where players build foundations of correct sequences for each category (under-1, above-1, ANC, PNC) and manage a "stock" of remaining interventions, with completion defined as finishing all required sequences rather than filling 13 slots per suit.

Thus, instead of retrofitting content into a 4×13 grid, we preserved endogenous suit sizes and reinterpreted the victory conditions and extension rules. Because suits are unequal, there is no expectation that every suit will reach the same length. The completion condition occurs when players exhaust all the cards in their hands or a team empties its hand. This approach aligns with broader serious-game design guidance, which suggests that mechanics can be adapted from existing games, but they must be flexibly re-engineered to fit the logic and scope of the educational content (Stefany et al., 2024; Arnab et al., 2015).

This psychological transition from the standard 4×13 card format to variable numbers of cards in each suit, though straightforward in hindsight, took considerable time. One probable reason could be that I was not an avid card player in my childhood, when one typically develops an intuitive understanding of card games and of the process of modifying the "standard" rules to suit a group's convenience.

7.4.1.2 Low-Fidelity Prototyping and Rule Explosion

Once a basic game concept was drafted, we created low-fidelity paper prototypes by designing handwritten cards, sketched icons, and a one-page rules sheet. This stage mirrored pen-and-paper prototyping practices in card-game design, where the emphasis is on testing ideas quickly rather than refining graphics. Early playtests with colleagues and a small number of CHWs exposed the problem of rule explosion.

Rules were too complex to hold in working memory. Initial rules included a range of possible actions (draw, play, swap, reserve, "counsel," "refer," etc.), multiple scoring categories, and various exceptions. Players spent cognitive effort on remembering what they were allowed to do, rather than thinking about the immunization schedule, ANC, and PNC sequences. Structural complexity might overwhelm the intended learning focus. Thus, simplifications of rules were made. The action set per turn was reduced to a small, fixed repetitive moves (e.g., draw 1, play or discard 1, optionally support a teammate), following serious games design guidance that early prototypes should test the minimal rules necessary for making meaningful decisions.

Rules were also reorganized into a progression according to difficulty levels for first-time players. Introductory rounds used only child–vaccine sequences. Only after players became fluent were ANC and PNC event cards introduced, echoing phased complexity and scaffolding recommendations in educational game design, a strategy that came in handy even in the digital version of the game.

7.4.1.3 Translating Tactile Play to Touch Interaction

In the physical game, CHWs spread cards on a table, point, rearrange, and jointly sequence, which naturally supports discussion and distributed cognition. Moving this to Android required redefining what "play a card," "show a card," or "build a sequence" looks like in a single-screen touch interface. Similar issues are reported in other research on translating physical card games to digital platforms (Van Rozen et al., 2023).

The major bottleneck is the loss of spatial flexibility. A phone screen cannot show the same dense layout of multiple cases and long sequences without becoming cramped. Designing responsive layouts that keep cards legible on small, low-resolution devices while still showing

enough context for reasoning became a core UI challenge. We resolved this issue by utilizing stacked views and "zoom" or "detail" screens, rather than attempting to display the entire table at once, an approach commonly employed by other digitalized card-game platforms.

The next issue was regarding gesture mapping. deciding whether players tap, drag-and-drop, or use long-press menus, and ensuring these gestures are easily discoverable for CHWs with varying smartphone familiarity. We resolved this by choosing a small, consistent set of gestures (tap to select, drag to place) and adding subtle animations and highlights to show valid targets. Drag-and-drop functionality for playing or reordering cards was implemented with robust hit detection, collision targets, and snap zones that are tolerant of finger imprecision, especially for CHWs who are unused to fine-grained touch gestures. Handling conflicting gestures (scroll vs. drag, tap vs. long press) required clear priority rules and visual feedback to prevent accidental moves or frustration.

Social cues, such as physical pointing and eye contact, are replaced by on-screen indicators, which can weaken the sense of collaborative play. We resolved this by adding explicit indicators of turn, hints, chat, and prompt mechanisms to support social signals.

7.4.1.4 Encoding Rules, Turn Logic, and State

Physical play relies on human judgment. Players know whose turn it is, whether a move is legal, and when a sequence is complete. In the physical game, players enforce rules. In digital play, the app must track the state and verify the legality of every move, automatically enforcing rules in real-time. Encoding all constraints (correct sequence, no skipping required care, suit compatibility) into a rule engine was both technically and conceptually demanding. Multiple suits (under-1, above-1, ANC, PNC) and their asymmetric sequences were transformed into data structures that can be checked efficiently. We implemented "adjacency check" (correct next vaccine or visit) as code, rather than simple numeric adjacency like 6–7–8, which many standard card engines assume. We created explicit rule engines or lookup tables that codify allowed transitions between interventions, similar to logic-model-driven health apps that embed behaviour and content rules.

Handling incomplete cases, such as skipped visits, catch-up schedules, or deliberate "wrong" moves for learning, required the system to distinguish between illegal moves (blocked) and ped-

agogically allowed but incorrect moves (allowed but flagged with feedback). This distinction complicates validation logic and UI prompts (warnings vs. hard blocks vs. auto-correction of moves). Rule exceptions can be negotiated informally at the table in a physical card game. However, for the digital card game, we built "soft rules" (warnings rather than hard blocks) so that CHWs can explore "wrong" moves and then receive feedback, which is common in educational game apps.

While designing, we faced a dilemma about the potential inclusion of a 'joker' card, familiar from rummy games, which substitutes for any card. Although feasible mechanically, this feature posed a significant risk of creating a misconception among CHWs that vital immunization steps, like the measles vaccine at 9 months or the tetanus toxoid booster for pregnant women, were optional or skippable. While real-world practices occasionally bypass checks (e.g., the four antenatal care visits or multiple pentavalent/OPV doses), formal health protocols prohibit this. After brief deliberation, the Joker card idea was abandoned to avoid undermining procedural rigor. This episode highlights the need for caution when adapting traditional games for health training, as symbolic flexibility may inadvertently compromise adherence to evidence-based knowledge or practices.

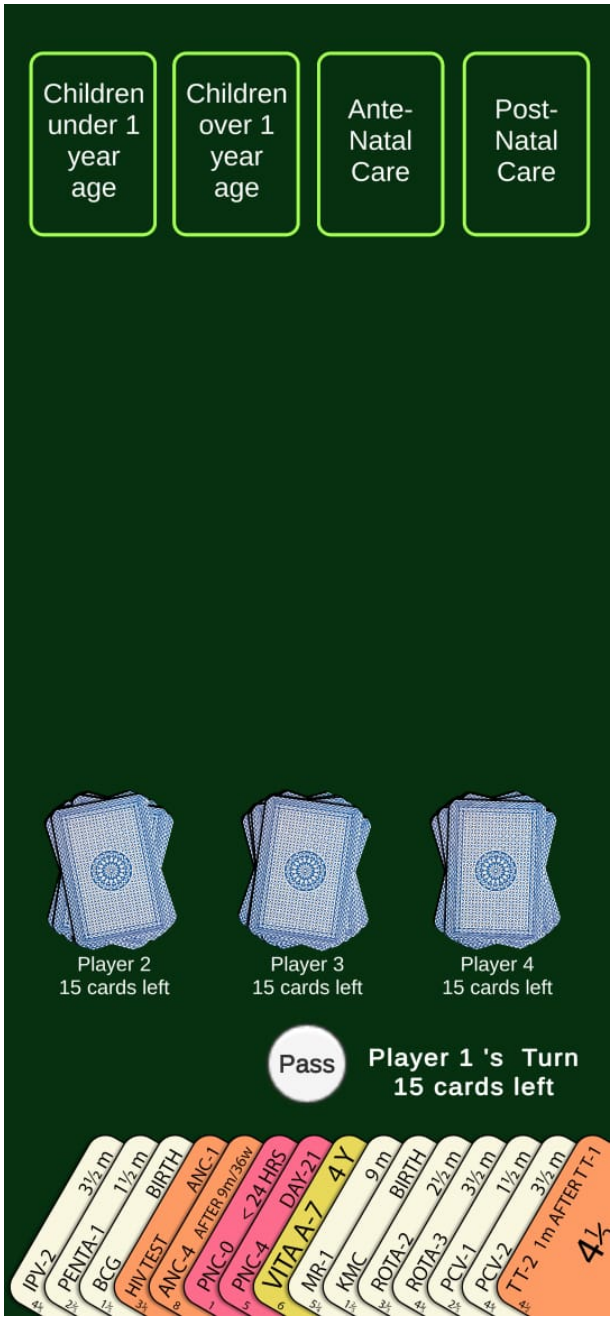


Figure 7.8: Game start Screenshot

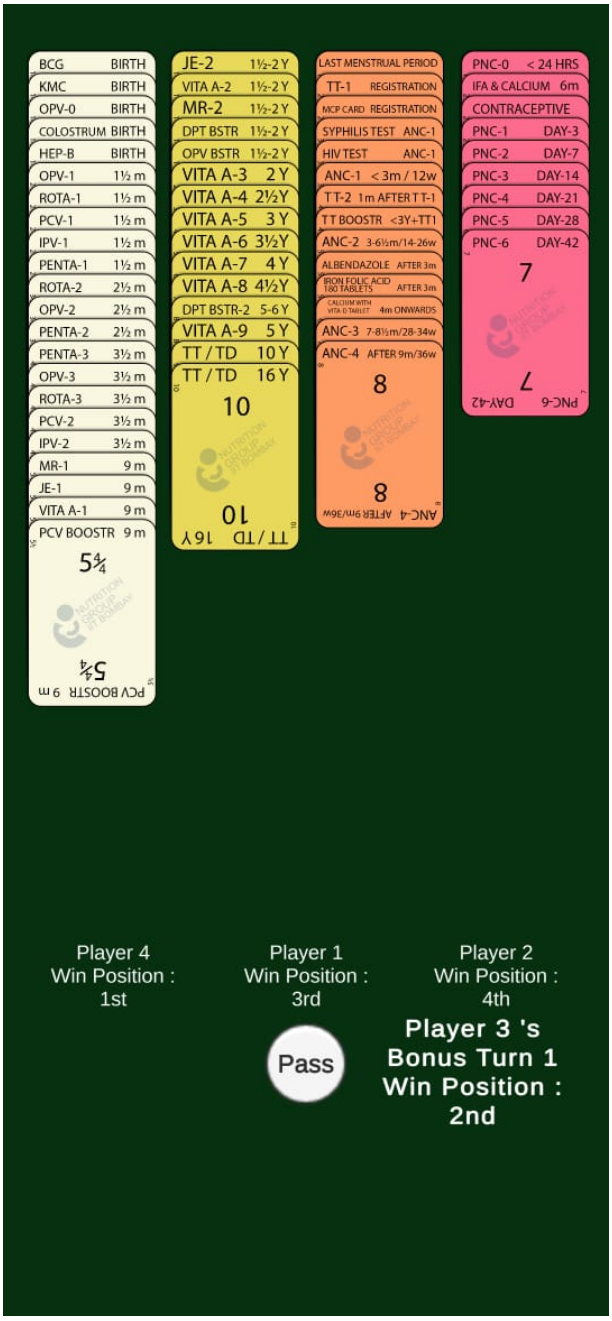


Figure 7.9: Game Over Screenshot

7.4.2 Design of the Card Deck

The design of the cards evolved through many iterations. We began with a minimal card design, featuring only the name of the vaccine and the time of administration. We realised that there was not enough information to communicate. Also, it is not convenient to turn cards upside down after shuffling. In the second iteration, we added relevant images of dosage, diseases that the vaccines prevent, and other relevant information. We attempted to cater to local people by using the local language.

Then we realised that in some time periods, multiple vaccines are administered simultaneously. We added a mixed fraction to denote the time of intervention. In some cases, there were two mixed fractions in the card, one for time and one for order. CHWs found it more complex and were unable to figure it out on their own. So we dropped the idea of mixed fractions for time period display. Instead, we placed a set of small white boxes at the card corner below time period to depict multiple doses of vaccines at a time. We made two versions of the card, focusing on digital app usage and a physical card deck. The design is iterated based on what information can be shown better while stacked physically or digitally.

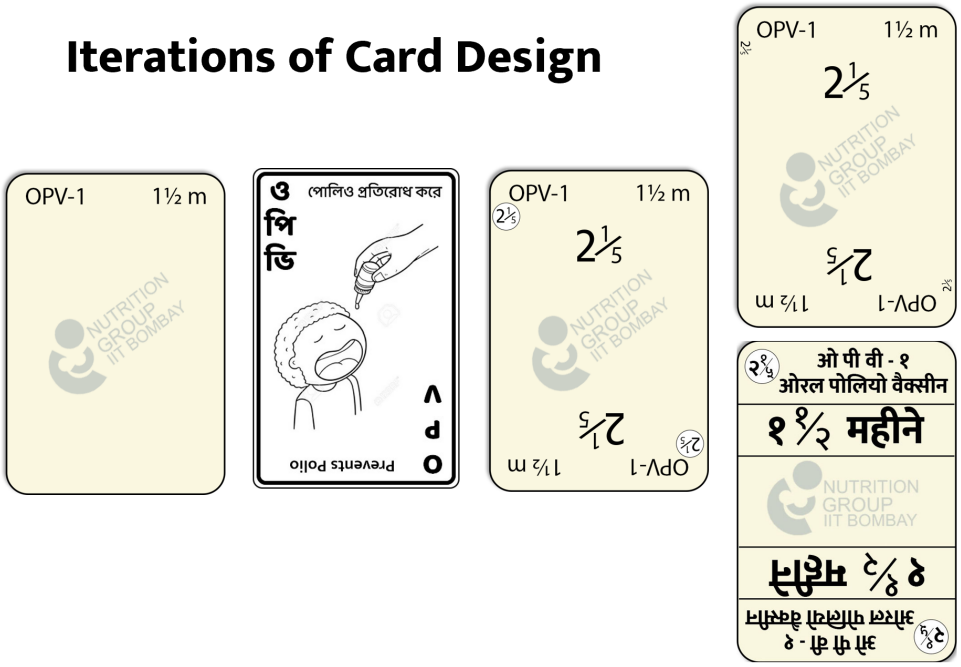


Figure 7.10: Iterations of Card Design

Figure 7.11: New Card Design

Figure 7.12: Card design finalized

Figure 7.13: Immunization cards for children below 1 year (continued on next page)

Figure 7.14: Immunization cards for children below 1 year (continued from previous page)

Figure 7.15: Immunization cards for children above 1 year

Figure 7.16: Ante-Natal-Care (ANC)

Figure 7.17: Post-Natal-Care (PNC)

7.4.3 Coding of the game

Out of all the scripts, we will discuss about a few important ones which makes the major decisions in the game design. The link to the repository is in the footnote [3].

7.4.3.1 Game Manager Script

The Game Manager script executes upon loading the game scene and initializes cards across four foundations. Four string arrays define the cards using prefixes: "c" for children under one year, "C" for children over one year, "A" for antenatal care (ANC), and "P" for postnatal care (PNC), with unequal card counts per category. The second character denotes intervention timing (e.g., "c1" for birth, "c2" for $1\frac{1}{2}$ months, "c3" for $2\frac{1}{2}$ months, "c4" for $3\frac{1}{2}$ months, "c5" for 9 months), while a third character sequentially identifies multiple vaccines or interventions within each period (e.g., "c10" for exclusive breastfeeding, "c11" for OPV-0, "c12" for hepatitis B, "c13" for BCG).

```
1 public static string[] child0_1 = new string[] { "c10", "c11", "c12", "
    c13", "c14", "c20", "c21", "c22", "c23", "c24", "c30", "c31", "c32",
    "c40", "c41", "c42", "c43", "c44", "c50", "c51", "c52", "c53" };
2 public static string[] child1_16 = new string[] { "C00", "C01", "C02", "
    C03", "C04", "C10", "C20", "C30", "C40", "C50", "C60", "C70", "C71",
    "C80", "C90" };
3 public static string[] anc = new string[] { "A10", "A20", "A21", "A30",
    "A31", "A32", "A40", "A41", "A50", "A51", "A52", "A60", "A70", "A80"
    };
4 public static string[] pnc = new string[] { "P10", "P20", "P21", "P22",
    "P30", "P40", "P50", "P60", "P70" };
```

Listing 7.2: Cards Initialization

The four foundations are then initialized to store cards in a Badam Satti-style structure, with each foundation requiring three distinct stacks: top ("T"), center ("C"), and bottom ("B"). The foundations correspond to "c", "C", "A", and "P", resulting in $4 \times 3 = 12$ total stacks for organizing the vaccine cards.

[3] Link to the repository: `https://github.com/arkamajhi/Card-Game-for-ASHAs`

```
Foundations = new List<string>[] {
    FoundationcT, FoundationcC, FoundationcB,
    FoundationCT, FoundationCC, FoundationCB,
    FoundationAT, FoundationAC, FoundationAB,
    FoundationPT, FoundationPC, FoundationPB
};
```

Listing 7.3: Foundation and Stacks Initialization

A new deck is then generated by sequentially adding cards from each of the 12 stacks across the four foundations.

```
public static List<string> GenerateDeck()
{
    List<string> newDeck = new List<string>();
    foreach (string a in child0_1)
    {
        newDeck.Add(a);
    }
    foreach (string a in child1_16)
    {
        newDeck.Add(a);
    }
    foreach (string a in anc)
    {
        newDeck.Add(a);
    }
    foreach (string a in pnc)
    {
        newDeck.Add(a);
    }
    return newDeck;
}
```

Listing 7.4: Generate Deck Function

The deck is then shuffled to randomize card order by iterating 60 times, swapping each of the 60 cards with a randomly selected position using a temporary variable. This Fisher-Yates-inspired algorithm ensures thorough randomization for the sequencing task. The process prepares the playable deck with vaccine cards in an unpredictable order to challenge players'

recall of the immunization schedule.

```
void Shuffle<T>(List<T> list)
{
    System.Random random = new System.Random();
    int n = list.Count;
    while (n > 1)
    {
        int k = random.Next(n);
        n--;
        T temp = list[k];
        list[k] = list[n];
        list[n] = temp;
    }
}
```

Listing 7.5: Shuffling of Deck Function

The deck is then dealt among the players, with HandTransform defining each player's initial hand position. These hand positions are synchronized across all players' devices using PhotonNetwork for consistent multiplayer gameplay. This distribution enables collaborative or competitive sequencing of vaccine cards.

```
IEnumerator Deal()
{
    foreach (string card in deck)
    {
        yield return new WaitForSeconds(0.01f);

        Vector3 position = handTransform.position + new Vector3(cardposx
    * gapx, 0f, -1 * (cardposx) * gapz);

        GameObject cardObj = PhotonNetwork.Instantiate(Path.Combine("
   PhotonPrefabs", "cardPrefab"), position, Quaternion.Euler(0, 0, 60));

        PhotonView cardPhotonView = cardObj.GetComponent<PhotonView>();

        cardObj.GetComponent<DynamicObjectName>().UpdateObjectName(card)
   ;
        cardObj.GetComponent<Selectable>().faceUp = true;
```

```
15          cardPhotonView.TransferOwnership(PhotonNetwork.PlayerList[
    playernumber]);
16
17          cardIndex++;
18
19          playernumber++;
20          if (playernumber == playerCount)
21          {
22              playernumber = 0;
23              cardposx++;
24          }
25      }
26
27      Vector3 PBposition = Vector3.zero + new Vector3(2, -2, 0);
28
29      GameObject PButton = PhotonNetwork.Instantiate(Path.Combine("
    PhotonPrefabs", "PassButton"), PBposition, Quaternion.Euler(0, 0, 0))
    ;
30      Debug.Log("Instantiated PassButton");
31
32  }
```

Listing 7.6: Dealing of Deck Function

7.4.3.2 User Input Script

After dealing cards to players' hands, the game commences with Player 1 taking the first turn. This script manages all user inputs by first verifying Photon Server connectivity and the player's host/client status before initializing turns. Correct moves grant two bonus turns; after these, or upon passing, the turn advances to the next player.

```
1  void UpdateText()
2  {
3      if (Photon.Pun.PhotonNetwork.IsConnected)
4      {
5          p_no.text = "Player Number: " + playerNumber.ToString();
6      }
7      else
```

```
    {
        p_no.text = "Player Number: N/A";
    }

    if (PhotonNetwork.IsMasterClient)
    {
        m_c.text = "This is Master";
    }
    else
    {
        m_c.text = "This is Client";
    }

    foreach (GameObject g in GameObject.FindGameObjectsWithTag("PassButton"))
    {
        int ownernumber = g.GetComponent<PhotonView>().Owner.ActorNumber;
        owner.text = "Owner : " + ownernumber;
        if (ownernumber == playerNumber)
        {
            if (countstack == 0)
            {
                displaytext.text = ("Your Turn now");
            }
            else if(countstack==1)
            {
                displaytext.text = ("Your First Bonus Turn");
            }
            else if (countstack == 2)
            {
                displaytext.text = ("Your Second Bonus Turn");
            }
        }
        else
        {
            displaytext.text = (PhotonNetwork.PlayerList[ownernumber - 1].NickName + "'s turn now \n Wait for your Turn");
        }
```

```
        break;
    }
}
```

Listing 7.7: User Input Script

This code segment validates whether a card can be stacked onto the foundations and identifies the precise position (top, center, or bottom stack) where it fits. It enforces the Badam Satti-style stacking rules aligned with the immunization sequence across the four foundations.

```
    void Stackable(GameObject selected)
{
    //Check if can be stacked
    string symbol = selected.name.Substring(0, 1);

    for(int i = 0;i<FoundationsCharArray.Length;i++)
    {
        if (symbol == FoundationsCharArray[i])
        {
            /*
            //if center has zero cards
                //Stack (by default it happens at top)
            //else
                //(if center has cards)
                //if second letter is one less than the first
                    //Stack (by default it happens at top)
                //else if second letter is one more than the last
                    //Stack (by default it happens at below)
            */

            if (Foundations[(i*3)+1].Count==0)
            {
                if (Foundations[(i * 3)].Count == 0)//if no top cards
                {
                    Stack(selected);
                    Debug.Log("Stacked at center (initialization) of
  foundation : " + symbol);
                }
                else
                {
```

```
30                      if (selected.name.Substring(1, 1) == Foundations[(i
 * 3)][0].Substring(1, 1)) //Ifcenter is blank but cards are at top
initially
31                      {
32                          Stack(selected);
33                          Debug.Log("Center empty and Stacked at top of
foundation : " + symbol);
34                      }
35                  }
36
37              }
38              else
39              {
40                  if(int.Parse(selected.name.Substring(1, 1))== int.Parse
(Foundations[(i * 3) + 1][0].Substring(1, 1))-1)
41                  {
42                      Stack(selected);
43                      Debug.Log("Stacked at top of foundation : " +
symbol);
44                  }
45                  else if (int.Parse(selected.name.Substring(1, 1)) ==
int.Parse(Foundations[(i * 3) + 1][Foundations[(i * 3) + 1].Count-1].
Substring(1, 1)) + 1)
46                  {
47                      Stack(selected);
48                      Debug.Log("Stacked at bottom of foundation : " +
symbol);
49                  }
50                  else
51                  {
52                      Debug.Log("Can't be Stacked on foundation : " +
symbol);
53                      source.PlayOneShot(cantstack);
54                  }
55              }
56          }
57      }
```

```
58 }
```

Listing 7.8: User Input Script - Check of Stackable

7.4.3.3 Update Sprite

This script manages card visibility states—displaying faces, turning backs, or hiding cards based on ownership and context. Cards in the player's hand or foundations appear face up, while those in other players' hands remain hidden for multiplayer fairness. During a player's turn, clicking a hand card raises it above the deck to indicate selection. Subsequent clicks lower the prior card and elevate the new one.

```
1  public class UpdateSprite : MonoBehaviourPun
2  {
3      void Start()
4      {
5          int i = 0;
6          foreach (string card in deck)
7          {
8              if (this.name == card)
9              {
10                 cardFace = gameManager.cardFaces[i];
11                 break;
12             }
13             i++;
14         }
15         state = "inactive";
16     }
17
18
19     // Update is called once per frame
20     void Update()
21     {
22         if(selectable.faceUp==true)
23         {
24             spriteRenderer.sprite = cardFace;
25         }
26         else
```

```
27          {
28              spriteRenderer.sprite = cardBack;
29          }
30
31          if (name == userInput.slot1.name && tag != "FoundationCard")
32          {
33              Vector3 pos = transform.position;
34              pos.y = -3.15f;
35
36              if (cardrotated == 0)
37              {
38                  cardx = pos.x;
39                  pos.x -= 0.6f;
40                  cardrotated = 1;
41
42                  transform.rotation = Quaternion.Euler(0, 0, 3);
43              }
44
45              transform.position = pos;
46
47              //Debug.Log(name+" is active");
48
49              state = "active";
50          }
51          else
52          {
53              //spriteRenderer.color = Color.white;
54              if (state == "active")
55              {
56                  state = "inactive";
57                  //Debug.Log(name + " is inactive");
58
59                  if (tag != "FoundationCard")
60                  {
61                      Vector3 pos = transform.position;
62                      pos.y = -4.7f;
63
64                      if (cardrotated == 1)
65                      {
```

```
                        pos.x = cardx;
                        cardrotated = 0;
                    }

                    transform.position = pos;
                    transform.rotation = Quaternion.Euler(0, 0, 60);
                }
            }

        }



        if (PV.IsMine && this.CompareTag("Card"))
        {
            spriteRenderer.enabled = true;
        }
        else
        {
            spriteRenderer.enabled = false;
        }

        if (this.CompareTag("Foundations")|| this.CompareTag("
    FoundationCard"))
        {
            spriteRenderer.enabled = true;
        }


        int i = 0;
        foreach (string card in deck)
        {
            if (this.name == card)
            {
                cardFace = gameManager.cardFaces[i];
                break;
            }
            i++;
        }
```

```
104         }
105 }
```

Listing 7.9: Update Sprite Script

Chapter 8

Discussion and Conclusion

8.1 Connecting the Dots: Synthesizing Insights from the Three Studies

1. **Absorption and Retention pattern in Knowledge**

   We identified several patterns that were repeated across the studies. Participants in the intervention groups showed statistically significant improvement in scores from the pre-test phase to the immediate post-test phase (comparing just before and after the gameplay sessions). This rise was followed by a gradual yet significant decline in knowledge during the long-term post-test phase, performed several weeks after the intervention. This raises concerns about the retention of learning outcomes achieved through short-term interventions. The findings suggest the need for continued engagement and reinforcement mechanisms in game design. This can be brought through timely content updates, narrative progression, social competition, and real-world incentives.

2. **Complementary Affordances Model** Synthesizing these findings, we propose viewing the pattern not as contradictions, but as an enabler (or *complementary affordance*) of playful learning modalities. The digital card game mode provides rapid, embodied feedback that engages intuition and supports the acquisition of immediate competence, particularly benefiting learners who are less literate or less technologically experienced. In contrast, physical cards afford deliberative reasoning, peer negotiation, and social scaffolding, sustaining knowledge over time and amplifying relatedness. Rather than conflicting, these modalities complement each other. This strongly suggests a hybrid learning ecosystem that supports different learner profiles and educational goals through designed integration. Conceptually, this model combines playful interventions along two axes: *cognitive pathway* (System 1 intuition vs. System 2 deliberation) and *social pathway* (individual feedback vs. collaborative negotiation). Therefore, practitioners can design learning experiences that leverage the unique strengths of both modes, creating synergistic opportunities for durable, engaging, and context-sensitive knowledge acquisition.

3. **Co-Design**

   Another key conclusion across the studies is the importance of co-design in ensuring relevance, usability, and adoption. CHWs were involved in all phases, including ideation, design, and testing of the games. This shaped the interventions to reflect the ground realities and user preferences of CHWs. The co-design approach ensures that interventions are methodologically sound and socially acceptable. This was particularly relevant when designing the physical cards, as the use of scripts and game rules would then become region-agnostic and would be intuitively adopted, much like regular playing cards.

4. **Difference in Control Groups across studies**

   In all these studies, the baseline knowledge or skills were not significantly different between groups, ensuring baseline equivalence among groups and methodological consistency across all four studies. This condition allowed each study to isolate the effects of the intervention itself, whether digital or physical games, classroom-based, or conventional apps, based on learning outcomes or work efficiency. However, in some studies, the Control group was assigned to a classroom teaching. At a few locations, we faced problems with the unavailability of trainers who could physically attend and conduct regular classroom training. The trainers often had different teaching styles and approaches

to engaging with trainees, which made a noticeable difference even within a single study group. Therefore, in some cases, it was difficult to compare the results of one control group to another.

5. **Effective Training Method**

   The studies show that interventions through games are more effective than traditional methods, as they motivate and encourage participation, and provide immediate learning outcomes. Gamification and game-based research support these findings in learning. It says that direct feedback and active participation play a significant role as they help enhance cognitive processing and memory formation. Digital interventions, especially those on smartphone platforms, offer rich and interactive affordances for acquiring abstract information and making it relatable to CHWs.

   The physical card-based games, while less interactive, provided surprisingly equivalent long-term retention rates. This finding suggests that physical play's haptic and social learning elements entail more elaborate information encoding despite lower short-term gains. Physical gameplay supports collaborative thinking and enhances oral repetition, which are the most effective methods for knowledge reinforcement, especially in collectivist cultural settings like India. These findings challenge the assumption of digital solutions' superiority and argue in favor of multimodal solutions, which cater to diverse learner preferences.

   Subsequent game versions could address this limitation by incorporating design enhancements, such as periodic reinforcement mechanisms, increased difficulty levels, narrative progression, or built-in periodic tests. These enhancements can help maintain learner motivation over the long term and mitigate the loss of knowledge retention indicated in long-term measurements.

   The multi-arm comparative study across digital and physical card game modalities provides significant insights. Both physical and digital game-based learning were found to be better than classical classroom training and placebo or control conditions. Although digital games consistently yielded significantly higher immediate post-test gains, long-term retention scores for digital and physical interventions were not significantly different in most studies. The results suggest that both learning modes, game-based learning, can be equally effective over time, provided learners are sufficiently engaged.

6. **Gamification Eases Digital Comfort, Boosting CHW Performance**
   Gamified digital training apps may enhance the performance of CHWs by reducing discomfort with digital tasks, which is a common issue for CHWs with limited tech experience who often feel anxious or uncertain about digital apps and platforms. Fun elements, such as points, badges, step-by-step challenges, and quick feedback, enable low-risk practice, gradually building comfort, reducing tech fears, and enhancing tool use. This shift enables workers to focus more on learning content than on fiddling with the interface, thereby increasing engagement, completion rates, and results. This has been observed across the four studies. Pre-post tests on technical skills, as well as learning scores, need to be compared to separately evaluate the ease of the tool from the knowledge gains.

8.2 Reflections on Experiments

The experimental design had multiple formats, like physical card games, digital games, traditional classroom sessions, and control groups, to assess the impact of gamified learning. We found the immunization cards game to be significantly effective for refresher training of CHWs, with significant knowledge gain and retention. We compared the results by repeating the experiment with similar groups. We also found that the digital card mode was significantly better than the physical card game in most cases in terms of engagement and knowledge gains, but had similar results in knowledge retention. Different intervention arms were used to compare learning outcomes across various groups. The digital card game on immunization offered immediate feedback to players. Gradual difficulty levels and interactive game elements improved engagement and motivation among CHWs. This finding aligns with Self-Determination Theory (SDT) (Miller et al., 1985; Ryan and Deci, 2000), which emphasizes autonomy, competence, and relatedness as essential for effective learning. The physical card game lacked digital interactivity; however, it improved peer collaboration and discussions among the CHWs. The findings of this study align with Vygotsky's social constructivist theory (Vygotsky, 1978), which posits that social interactions are essential for effective learning.

However, as classroom teaching lacked the interactive and immersive elements of the gamified tool, it resulted in comparatively lower engagement and knowledge retention than the

gamified tool. During delayed retention tests conducted a few weeks after the intervention, the marginal reduction in post-test scores across groups shows the limitations of one-time interventions and reduced motivation to practice.

8.3 Reflections on Learning by Doing through Gameful Activities

After a few months of fieldwork and experimentation with research prototypes involving CHWs, we observed that the ethos of learning by doing appears to be gradually diminishing among CHWs, being replaced by demands for instant, convenient technological solutions. They expect technology to make things increasingly easier, bypassing the laborious processes of exploration, iteration, and skill-building, while still achieving knowledge gains and retention that appear satisfactory on the surface. However, the adoption of such shortcuts diminishes deep engagement and reflective connections. When the learning process is sacrificed for efficiency, the resulting knowledge and engagement often become depersonalized and lack the depth that emerges through routine practice. game

8.4 Effects of Ageing on Learning

In all the field experiments, we observed a significant trend: as CHWs aged, the change in knowledge gain between pre- and post-tests decreased. In other words, it was becoming increasingly difficult for aged CHWs to learn new content and adapt to digital tools. Some CHWs were highly motivated and enthusiastic learners, even at higher ages. Research on learning in other contexts also finds similar results in adult education (SALTHOUSE, 1995; Zwick, 2012). Years of formal education also had an insignificant effect on baseline test scores, as the knowledge and experience gained mainly depend on the years of experience working in the field as a CHW and having hands-on experience dealing with prevalent issues. This finding aligns with other research in India (Sharma et al., 2014) and other contexts (Rogers et al., 2023; Kok et al., 2015).

8.5 Reflections on Social Factors

Social factors, including cultural norms, language diversity, socio-economic conditions, and group dynamics, influenced how learners interacted with the training materials and each other, thereby affecting participants' in-game engagement and knowledge performance. India has 22 official languages and more non-scripted ones. CHWs operate in regions with different language intersections. Standardized but inflexible training materials pose challenges for the CHWs. The designed game offered content in the local language, which might have contributed to improved comprehension and engagement. The CHWs' social and economic context can have influenced their access to digital devices and the internet. Those from economically disadvantaged groups might have utilized the physical card game as a more accessible medium, especially for less digitally literate or resource-poor individuals. Peer-to-peer dynamics and group interactions also played a key role. Collaborative learning was particularly evident in the physical card game sessions, where participants often discussed answers and strategies collectively. This social learning environment likely enhanced engagement and supported deeper understanding through dialogue and peer learning.

8.6 Expanding the Content of the Card Game to other Maternal and Child Health Topics

Future researchers, designers, and content creators can expand the game's content beyond the current coverage of ANC, PNC, and the child immunization module to cover other maternal and child health topics, such as adolescent care, maternal nutrition, breastfeeding practices, complementary feeding, and disease prevention. Adding hands-on skills to the game could also help CHWs develop the competencies necessary for effective healthcare delivery. This will nevertheless not replace the initial hands-on training of CHWs.

8.6.1 Expanding Content Coverage for Holistic Health Education

Healthcare services provided by CHWs cover various health issues concerning mothers and children. The game should include adolescent care modules covering topics such as preventing anemia, managing menstrual hygiene, understanding contraceptive methods, and coping with mental health issues. Maternal and child nutrition should also emphasize the importance of balanced diets before and during pregnancy, exclusive breastfeeding for the first six months, and the introduction of complementary, nutrient-dense foods after six months.

The game's expansion should maintain the game's endogenous mechanics, which should evolve from the content. The game module on nutrition could introduce cards representing food groups and types of nutrients. In the game, players can create balanced meals without repetition for a week, catering to varying age groups, which aligns with the advocacy provided by the CHWs. The adolescent care module may challenge players to identify signs of malnutrition or offer proper counseling for effective menstrual health management. Such mechanics help translate health concepts into concrete, familiar actions, enabling more effective communication of information. Expansion would work best with content that has clear protocols and unambiguous instructions, which is why we took on the content from the ASHA's refresher training in the first place.

8.6.2 Collaborative Learning through Games

The work duties of CHWs require contact and coordination among them, such as organizing health camps, responding to disease outbreaks, or implementing community nutrition programs. Cooperative learning can teach them how to work effectively in teams to achieve common goals by replicating real-world dynamics in their jobs.

Group players can develop a comprehensive care plan for a malnourished child in collaborative modules. Each player would have different information (imperfect information gameplay), like dietary habits, clinical signs, or family history. They must contribute their insights to arrive at an accurate diagnosis and treatment plan. This collaborative mechanism would reinforce the importance of information sharing, communication, and teamwork in maternal and

child healthcare, as well as community healthcare.

Competitive game elements, such as leaderboards, can also enhance engagement without compromising the cooperative nature of healthcare provision in the game. It can motivate CHWs to improve their performance while supporting competition and providing consistent learning.

8.6.3 Addition of new content

Involving CHWs in developing the content will help tailor it to be compatible with local traditions and norms, as part of efforts to ensure the game is familiar and suitable for the intended target group. Incorporating elements of games like Badam Satti and Rummy was helpful in our experimental study. Satti (also called Badam Satti or Sevens) is a popular game in India, featuring a strategic sequence-building aspect where players extend suits from 7s upward (8-K) or downward (6-A). It is characterized by its strategic approaches, including blocking opponents, prioritizing cards, and anticipating sequences. This game has a broad appeal to diverse age groups. Adopting the familiar form and principles from the game helped make the training game more suitable for healthcare providers. Playing cards evolved from recreational games to symbolic representations of status and custom in India. This historical background can be applied by developing card games in a manner that educates and serves as a social activity for bonding among healthcare professionals, fostering teamwork and communication.

8.7 Methodology of Play

This game translates the content of the child immunization schedule, medications, antenatal care (ANC), and postnatal care (PNC) into interactive and tangible mechanics through cards (game elements) representing age-specific vaccine milestones and resource management challenges (game mechanics). The game simulated challenges such as vaccine shortages and disease outbreaks. Players need to make quick, informed decisions throughout the game. The game provided experiential learning through hands-on activities, such as card stacking in the

proper sequence of vaccine administration, and transformed abstract concepts into concrete actions. These interactions reinforced memory pathways, enabling CHWs to internalize and apply what they have learned in daily practical settings. Repeated exposure to these concepts through gameplay scenarios and ongoing processes of learning, application, and reinforcement facilitates long-term knowledge retention.

8.7.1 Game Mechanics: Promoting Emotionally Engaging Learning

The game mechanics, adequate challenges, and constructive feedback invoked curiosity, anticipation, and a sense of achievement. It helped to sustain motivation in the game. The game mechanics were endogenous, meaning they were derived directly from the content. In the game, players stacked cards representing vaccines in the correct chronological order based on a child's age. This game mechanic was similar to the decision-making process that CHWs undergo when scheduling and administering vaccines. The real-world tasks of CHWs are aligned with the gameplay mechanics. They were more likely to comprehend and internalize the sequencing rules, increasing learning and retention. Opposition mechanics, such as artificial vaccine shortages or outbreaks, added complexity and realism to the game. These dynamic, challenging opposition mechanics prompted players to adjust their strategies, thereby further enhancing their critical thinking and problem-solving skills. The unpredictability of opposition mechanics kept players engaged. Overcoming these challenges helped reinforce the protocol in the minds of the CHWs, reinforcing the knowledge gained during gameplay. The rewards and feedback elements induce emotional engagement. Overcoming in-game challenges generated a sense of achievement, which drove continued participation. The emotional satisfaction derived from mastering complex tasks (in-game mastery) deepened the player's connection to the content.

8.7.2 Rules of Play: Structuring Learning Through Interaction

The game's rules were designed to structure learning in incremental, interactive, and engaging ways. The game's sequence was structured with an incremental difficulty scale. Initially, players were introduced to a simple stacking game mechanic to help CHWs become familiar with

the game experience. The game later revealed advanced topics such as resource constraints and age-specific critical immunization as players progressed. This technique is known as 'scaffolding' in educational theory (Gaffney and Rodgers, 2018). The basic concepts are taught and tested first before teaching more advanced concepts. By ensuring a smooth and steady increase in the complexity of content type, CHWs remained motivated while they mastered more advanced topics. As their competence grows, these scaffolds or learning supports can be gradually removed (a process often referred to as 'fading'), enabling CHWs to assume greater responsibility and eventually become more self-directed learners.

Collaboration played a crucial role in the game's rule structure, enabling CHWs to work together or compete to complete vaccine sequences and dosages. This dual collaborative-competitive dynamic enhanced engagement through peer interaction, where players shared strategies and explanations that reinforced their understanding of the content. Social experiences among CHWs made the game more memorable, contributing to improved knowledge retention.

CHWs receive feedback for their actions in the game through bonus points for correct moves or penalties for wrong moves. A real-time feedback loop enables players to identify and correct misconceptions, leading to a practical learning experience. Positive reinforcement also supports retention since correct actions are associated with accomplishment.

8.7.3 Translating Content into Multiple Channels of Learning

The game's design employed various methods to teach the content, enabling people to learn more effectively. It included various methods to accommodate different learners' preferred ways of thinking and learning. This approach made sure everyone could understand and remember the information more easily. We created the integrated learning experience through game elements that are audio-visual, action-based, informational, and decision-making. Visual elements had a substantial impact on improving knowledge recall. Color-coded cards and disease-specific information allowed players to identify vaccines and matching age groups at a glance. These visual cues create mental associations that facilitate long-term recall. Action elements in the game sped up learning by leveraging procedural memory. Card stacking or vaccine

sequencing activities use physical actions that enable the steps of the immunization schedule to be committed to memory. Repeated physical interactions and recall help retain information. The cards' content provided factual information, such as detailed explanations of the purpose and timing of each vaccine. Re-reviewing and re-reading these facts repeatedly reinforced verbal memory, allowing the players to recall key information. These game elements supported the learning process through higher-order cognitive engagement. Players made strategic choices during gameplay, such as deciding which vaccine cards to retain in hand for later reveal to win or support their team. This required applying knowledge under pressure, extending beyond simple memorization to build critical thinking and problem-solving skills.

8.8 Implications for Policy and Practice

1. **Integrating Gamified Learning into Curricula**

   The studies show the effectiveness of digital and physical card games as a refresher training tool, which can be integrated into regular training programs for CHWs. Such tools can be implemented in healthcare training curricula at the state and national levels by adopting hybrid learning models. Traditional classroom instruction can be supplemented with gamified applications as a refresher to improve engagement and retention of knowledge by CHWs. CHWs with high smartphone proficiency can use digital apps as supplementary refresher tools during coursework, while the others can benefit from physical card games in low-resource areas where digital access is a challenge.

2. **Addressing Digital and Resource Barriers**

   In rural and tribal areas in India, ownership of smartphones, access to the internet, and smartphone proficiency among CHWs may still hinder the adoption of smartphone-based digital games. Policy advocacy for improving digital infrastructure is crucial to ensure access to digital gamified refresher tools among CHWs. The mobile application for CHWs should prioritize working offline to accommodate users with low or no internet access. The availability of physical versions of the training games as an alternative can cater to CHWs in resource-constrained settings and ensure that they are not left behind.

3. **Language and Cultural Adaptations**

Training materials should be made accessible in multiple regional languages across India to address linguistic diversity and cultural differences. Policymakers and educational institutions should engage CHWs, their supervisors, and master trainers in co-designing training materials and translating them into local dialects for better acceptance and effectiveness.

4. **Policy-Level Support for Continuous Training and Reinforcement**
   The reduction of knowledge over time observed in all study arms highlights the need for continuous reinforcement learning. Policymakers should advocate for periodic refresher training for CHWs to ensure sustained knowledge retention. Gamified digital refresher sessions, conducted periodically, can provide cost-effective solutions.

5. **Data-Driven Policy and Decision-Making**
   Digital gamified applications enable close monitoring of learner performance through in-app analytics. Training institutions identify knowledge gaps and customize learning pathways for future training improvements, and policymakers can make data-driven decisions by leveraging the collected longitudinal data. Such data can also help assess the long-term impact of gamified interventions on healthcare service delivery and community health outcomes.

8.9 Limitations of the study

- **Hierarchical Settings:** The experiments were conducted in a formal institutional environment at a healthcare center, which differs from the typical field conditions faced by CHWs such as ASHAs and AWWs. These voluntary field functionaries operate as street-level bureaucrats within rigid, top-down bureaucratic frameworks, potentially limiting the translation of experimental findings to their real-world, autonomous decision-making contexts. Future iterations should test the game in actual field settings to capture hierarchical influences on CHW engagement and performance.

- **Sample Size and Diversity:** Although conducted with relatively small sample sizes, the studies demonstrated promising results for the card game's effectiveness across diverse locations and settings. The physical format proved particularly engaging and accessible,

suggesting it merits broader adoption across various CHW training modules. Future research should explore optimal combinations of physical and digital versions—directing younger, more educated CHWs toward digital adaptations while prioritizing physical games for older workers with lower digital skills—to maximize scalability and impact

- **Technological Literacy:** Some CHWs faced challenges with the use of digital and AR-based games due to varying levels of technological literacy. Future interventions should include comprehensive training on the use of digital tools to ensure equitable access and participation.

- **Retention Period:** While the studies assessed knowledge retention over a few weeks, longer follow-up periods are needed to evaluate the impact of game-based interventions on CHWs' knowledge and performance. Longitudinal studies can provide deeper insights into how game-based learning influences long-term behavior changes and healthcare practices. To support longer retention, periodic and structured opportunities for replaying the game may need to be incorporated into training programs, ensuring that knowledge is reinforced consistently over time.

- **Cultural and Contextual Factors:** The studies were conducted in specific cultural and geographical contexts, which may influence the applicability of the findings to other settings. Future research should explore the adaptation of game-based interventions to different cultural and linguistic contexts. This will ensure that the interventions are relevant, engaging, and effective for diverse populations.

8.10 Future Research Directions

Building on the findings and limitations of this dissertation, several directions for future research open up:

- **Comparative Studies:** Future research should conduct large-scale comparative studies to evaluate the effectiveness of different game modalities across diverse populations and healthcare settings. Comparative research can help identify the specific conditions under

which each game modality is most effective, guiding the development of targeted training programs.

Research should explore the integration of game-based interventions with existing health information systems to enhance data collection, monitoring, and decision-making processes. For example, location-based games could be linked to national health databases to provide real-time updates on vaccination coverage and malnutrition rates.

- **Personalization and Adaptation:** Future interventions should focus on developing personalized learning pathways that adapt to the individual needs, preferences, and learning styles of CHWs. Adaptive learning technologies can use data analytics to offer customized content and feedback, ensuring that each learner progresses at their own pace.

- **Gamification of Additional Topics:** While this dissertation focused on immunization and data collection, future research should explore the gamification of other healthcare topics, such as maternal care, chronic disease management, and mental health. Expanding the scope of gamified learning can further enhance CHWs' competencies and improve the overall quality of healthcare services.

- **Behavioral and Motivational Factors:** Further research is needed to understand the behavioral and motivational factors that influence CHWs' engagement with game-based interventions. This will inform the design of more effective and user-centered learning experiences. Understanding intrinsic and extrinsic motivators can help create more meaningful and sustained engagement with learning activities.

8.11 Summary

This research compared the efficacy of gamified refresher training of CHWs in improving knowledge acquisition and retention by comparing the performance of CHWs, across physical and digital game formats, traditional classroom training, and control groups, in three phases, pre (before intervention), post (after intervention) and long-post (weeks after intervention). The study shows the significant effectiveness of game-based learning in different socio-cultural. Digital and physical card game interventions demonstrate significant improvements in knowl-

edge acquisition and retention among CHWs, compared to the classroom training and control groups (who didn't receive training during the experiment but were later compensated), highlighting the effectiveness of gamified training tools across different CHW groups.

In this research, insights from self-determination theory (SDT), flow theory, constructivist learning theory, and other gamification theories informed the framing of the relationship between game mechanics, learner engagement, and knowledge outcomes in CHWs. The findings of this research closely resemble those of previous game-based studies that focus on autonomy, competence, relatedness, and immersion in creating meaningful and effective learning experiences. By combining theoretical reflections with empirical evidence, this research contributes to the growing body of literature on the role of games and gamification in education, particularly in resource-constrained environments.

In some cases, the absence of significant differences between physical and digital-based card game formats indicates that the medium is less important than the quality of the learning experience. The acquired knowledge depreciates over time, underscoring the importance of continuous learning and periodic refresher training as a means to reinforce and ensure sustained knowledge retention.

8.12 Conclusions

This study contributes to the academic discourse on game-based learning through its theoretical and practical insights, demonstrating that gamified interventions can be customized to meet the diverse needs of learners, various educational levels, and diverse social contexts, thereby enhancing motivation, engagement, and learning outcomes. The findings show these tools offer a scalable, low-cost option for CHW refresher training—as a complement to initial hands-on training, not a replacement. The study also contributes to theoretical discussions on the applicability of gamification in real-world learning scenarios, particularly in low-resource settings.

Future game-based research may explore the intersection of game design, motivation, and learning outcomes in community healthcare, as well as across other domains such as environmental education, financial literacy, and public health campaigns. Emerging technologies, such as augmented reality (AR), virtual reality (VR), and artificial intelligence (AI), may help en-

hance the personalization and interactivity of learning experiences for CHWs.

Future research should also explore the long-term impact of gamified learning on behavioral outcomes, service delivery, and societal change driven by CHWs. Game-based analytics should be used to create personalized learning pathways. Co-designing games with CHWs, their supervisors, and their trainers would bring more cultural relevance and inclusivity.

This research shows the transformative potential of game-based learning as a tool for empowerment and lifelong learning. By building on the foundations laid by this study, educators, policymakers, and researchers can harness gamification to create engaging, inclusive, and impactful learning environments for CHWs.

8.13 Personal Reflections

As a researcher with a Masters in Design (M.Des.) background, venturing into the domain of maternal and child healthcare, has been both challenging and rewarding at the same time. This interdisciplinary journey has led me to navigate through uncharted territories, from understanding the complexities of public healthcare content and systems to designing, developing, testing, and evaluating gamified training solutions for Community Health Workers. Despite the challenges, the experience has been deeply satisfying, offering valuable insights into the potential of gamification in the training of Community Health Workers. Although this work is still evolving, I believe it has made a meaningful contribution to the broader effort to improve maternal and child health outcomes in India.

Despite the selfless efforts put in by the CHWs, they were rarely appreciated by the state and sometimes by the community. Recently, the WHO Director-General announced ASHAs of India as one of the six awardees of the Global Health Leaders Awards for their crucial role in linking the community with the health system and ensuring those living in rural poverty can access primary health care services during the crucial times of the COVID-19 pandemic. The intention here is not to romanticize and idealize the lives and struggles of the ASHAs, but to highlight the immense amount of dedication they show towards their job in spite of often having physical fatigue and sometimes carrying emotional breakdowns as a woman and often as a mother.

Appendix A

Designing the Refresher Training Activity

A.1 Content for designing Refresher Training Activity

Conducting a Focused Group Discussion (FGD) among a group of AWWs and ASHAs, it was found that most CHWs lacked complete knowledge of the immunization schedule, growth monitoring chart, and other aspects of the Mother and Child Protection card (MCP card). The findings of other researchers from different parts of India were also similar to our (Bag and Datta, 2017). A study found that the issuing of MCP cards to pregnant mothers was strongly associated with partial immunization in children (Kizhatil et al., 2019). The child immunization table from the MCP card was chosen as content for the game. This schedule also follows the Indian Academy of Paediatrics (IAP) guidelines (Kasi and et.al., 2021). A reference to the immunization schedule is in Table A.1 in the Appendix.

Vaccine Name	Birth	$1\frac{1}{2} months$	$2\frac{1}{2} months$	$3\frac{1}{2} months$	9 months	$1\frac{1}{2} - 2 years$
BCG	✓					
Hepatitis B	✓					
OPV	✓	✓	✓	✓		✓
IPV		✓		✓		
Penta		✓	✓	✓		
PCV		✓		✓	✓	
Rota		✓	✓	✓		
MR					✓	✓
JE					✓	✓
DPT						✓

Table A.1: Child Immunization schedule

A.2 Physical Card Play as a Learning Tool

Initial thoughts of game design were that the information about vaccines is displayed to players through blocks of information. Then the player has to make a decision based on the information and the current scenario. Out of many physical tokens and objects of play, card gameplay was chosen as the game artifact. The possibilities of playing through card games were explored. Physical card games are ubiquitous. The primary reasons for this are its cost-effectiveness and handling ease. A deck of cards can be played in numerous different ways just by altering the rules. Different groups of cards can be added to the existing deck to enable new ways of playing with it. Multiplayer card game researchers (Bochennek et al., 2007) have shown significant learning advantages through engagement and collaboration. However, published research papers on using card games for training community healthcare workers were sparse.

The core learning objective of the game is that the community health workers memorize the sequence and dosage of immunizations and supplements required by the child and mother. Initially, there were thoughts of modding the classic physical card games like Rummy, Solitaire, Bridge, Teen Patti, etc. It was easier to think along the lines of combining the rules and mechanics of a popular card game with educational content. However, in several instances of

design iterations, the associated traditional rules of play restricted the fulfillment of our learning objectives. As a result, cherry-picking rules and mechanics were performed from several popular card games, which would address and fulfill the learning objectives of the ASHAs.

A.3 Designing the cards

The content can be categorized into 4 silos depending on the time of delivering the care to the stakeholders. 4 silos of cards were made each symbolizing an immunization, medicine supplement or a care activity or event. They are categorized as follows: Children below 1yr (n=22), Children above 1yr (n=15), ANC (n=14) and PNC (n=9) combined to form a deck of 60 cards. The deck of card design is displayed through Figures : 7.13, 7.14, 7.15, 7.16 and 7.17 in the Appendix section.

These playing cards are made the size of a standard playing card or credit card (5.5 X 8.5 cm), made of card paper (200 GSM), and have printed information and graphics on them.

A.4 Combination of cards

These 60 playing cards can be equally divided between 2, 3, 4, 5, or 6 players, creating multiplayer team combinations. The most common of them is a 4-player match. Theoretically, all the 60 cards could be arranged in a total of 60! ways. But for each of those arrangements, there are many duplicate arrangements in which cards 1 to 15 are the same, but in a different order. Similarly for cards 16 to 30, and so on. For each group of 15, there are 15! ways to rearrange them and not change the cards in each player's hands. That's why four factors of 15! would be in the denominator, considering a 4-player round. The mathematical expression of the number of permutations possible for each player's hand, considering a 4-player round, is as follows: 60! / (15! X 15! X 15! X 15!) or simplified $\frac{60!}{(15!)^4}$. Google Sheet Expression =FACT(60)/POW(FACT(15),4) which calculates to an astronomical value of 2,845,616,726,065,970,000,000,000,000,000,000

A.5 Designing Challenges, Mechanics and Rules of Play

In the process of game design, after brainstorming the basic game idea of stacking cards as per the schedule, we focused on methods to translate content elements of immunization into game elements. We can call this as translation phase. This helped towards achieving endogenousness in design, where game mechanics and resources emerged directly from the immunization content itself.

We started with reimagining content elements as game elements. In the designed game, the immunization content was represented by the vaccines, which act as resources in the game. Each vaccine (such as polio, measles, or tetanus) becomes a game resource that players must acquire and administer at the correct time, simulating the real-world immunization schedule.

The designed game involved players stacking "vaccine cards" in chronological order based on when each vaccine needs to be administered according to a child's age. Players would have to strategically decide which vaccines to prioritize, mirroring real-world challenges faced by CHWs in ensuring timely immunizations. These vaccine cards were color-coded or labeled to correspond to specific diseases, time of intervention (in months), and players received points or rewards for correctly stacking the cards in line with a virtual child's immunization needs.

Our strategy of choosing Game Mechanics, firstly involved identifying player actions that can be performed with the game's resources, while the other strategy mapped established mechanics onto content elements. By linking well-known mechanics with content-derived resources on immunization, we tried to develop a system that feels more naturally connected to immunization or the game's subject matter.

We also focused on opposition mechanics, obstructions, or opposing forces within the context of immunization. In the moderately advanced level of the game, the mechanics were included in the form of unforeseen challenges like a vaccine shortage, requiring players to trade or manage limited resources, or a simulated disease outbreak that necessitates quick decision-making to administer booster shots. This created dynamic decision-making and action elements that reflect the content of immunization, transforming real-world health information into an engaging and educational game format. Challenges often arise when opposition mechanics are not apparent in the content, leading to exogenous mechanics. However, these challenges can

also inspire creative solutions and novel opposition mechanics.

Endogenous design arises when resources or mechanics are organically derived from the content. However, when we were unable to synthesize mechanics or resources directly from content, we resorted to exogenous mechanics, using designs like races or other standard systems. These choices might lead to a more disjointed relationship between content and gameplay.

Once a core or opposition mechanic (blocking cards to restrict completing the sequence) has been identified, we followed the Mechanics-Dynamics-Aesthetics (MDA) framework by (Fabricatore Carlo, 2007). The translation of content to game elements, particularly through mechanics and resources, steered whether the design remains endogenous or shifts towards exogenous approaches.

In game-based learning, effective translation of content into game elements enables the delivery of learning through a variety of channels. These include visual elements (4 foundation layouts demonstrating 4 silos of mother and child care, mainly focusing on immunization), action elements (game mechanics like stacking of cards and ordering), information elements (Card details like name of immunization and time of intervention), and decision elements (player choices during the gameplay). Each of these elements contributes to how players interact with the game, allowing learning to emerge organically from engagement with the game system.

However, knowledge acquisition through game-based learning presents significant challenges, especially when introducing novel concepts. Research, including our previous studies (Majhi et al., 2021, 2022, 2024c,a,b), shows that activating prior knowledge is meaningful and sometimes easier than facilitating the acquisition of new knowledge. Designing games for novel knowledge is particularly difficult in complex subjects like immunization concepts.

Thus, while games can be highly effective for knowledge reinforcement and application, they are less frequently successful in facilitating the acquisition of entirely new knowledge, particularly in cases where the content is abstract or requires sensory engagement for full comprehension. However, in our designed game, we restricted our content to only factual information, and learning through simple game mechanics was an appropriate choice.

A.6 Rules of Play (Based on the rules of Badam Satti)

- The game starts by shuffling the deck of 60 cards, and distributing it equally to 4 players, 15 cards each (2 players 30 each or 3 players 20 each).

- Each color belongs to one silo or foundation. There are 4 foundations. Children below 1 year, above 1 year, antenatal care, and postnatal care. Make 4 spaces for 4 foundations (similar to solitaire)

- Then, at each player's turn, they need to put a card of one type onto the foundation of the same silo on the board.

- Let's say they put the cards of 2 years (eg. $2^1/_5$). This means there are 5 cards or immunizations in year 2. This might also be shown in 5 white rectangles in the new version to avoid confusion. They have to complete the 5 cards of 2 years in their turns. After completing the 2 years, they can either move up or down $1^1/_2$

- The cards are numbered in order of the months and year. If there are multiple immunizations/cards in an age group, they are numbered $2^1/_5$, $2^2/_5$, $2^3/_5$ and so on. This acts as a cue for the players to know how many more cards are left to complete the current sequence so that they don't miss a card and jump to the next month or sequence.

- If a player successfully places a card in the placeholder silo, the player gets 2 more chances or play rounds as a bonus.

- If there is no possibility of cards being placed or 2 bonus rounds are complete, then the turn goes to the next/following player.

- After one age group is completed, the following players, in their turn, can place cards in one age group above the upper set or below the lower set on the board.

- The rounds of play continue until all cards are exhausted from a player's hand. The first player to exhaust the holding cards wins the round and is awarded the first rank, followed by the next players until all cards are exhausted by all players.

A.7 Limitations of playing with physical cards

- The rules of the play need to be followed in order to achieve the intended learning. However, players need to constantly monitor the activities in order to catch deviations from the rules. This problem can be solved digitally through rules written in the form of algorithms. Every wrong move would be announced through voice cues.
- In India, there are 22 official languages. Different groups have different number literacy levels. Some are comfortable with the script of Devnagri numbers or regional ones, while some are with English numbers. Making customized decks cater to each group won't be scalable.
- The content or the knowledge to be delivered changes with new findings in medical science, which needs to be implemented over the state or country. This translates to adding new cards or scraping off the distributed cards, printing and resupplying new decks to all ASHAs, which is not a scalable solution.
- Mundane chores of play, like shuffling cards, equally distributing cards between the number of players, and packing them up, take up a lot of time for players to do manually. These can be done digitally in almost no time. It can also be done in an animated sequence to mimic the physical action sequence of shuffling, distributing, and packing up, which are very satisfying to watch in an interface.

A.8 Designing an app-based interface of the same activity

In the digital app, there are 4 foundations on the top which represent 4 silos. The players click on the desired card to pick it and click on the empty foundation to drop it. If there are multiple cards to that sequence (like $2^1/_5$, $2^2/_5$, $2^3/_5 \ldots$), The cards keep stacking over. As soon as the sequence is complete, they collapse into the foundation. It opens up the possibility to stack cards over or under, the bigger and smaller cards in sequence, respectively. A reference to the app's User Interface is shown in Figure **??** in the Appendix section.

A.9 Multiplayer Playing Variation

During previous field studies, it was observed that more than half of the ASHAs had access to smartphones but did not own them. Usually, their husbands own it and share it with their wives, the ASHA. The resource constraints of owning a phone individually could be potentially solved if one smartphone could be used by 4 players, turn by turn, while 4 players can physically arrange themselves sitting in a circle. A pass button in the User Interface to skip the turn to the next player was introduced. After each player plays her move in her turn and ends successfully stacking a card, the next player gets the turn. But if there are no moves possible, then the player can click on the Pass button to pass the turn to the next player. The player, after completing her turn, orients the phone by turning it towards the next player for her turn to play. Also, an assistive voice would be played announcing the current player's name, to reduce the confusion of which player's turn it is currently.

During previous field studies, it was also observed that the Community Healthcare Workers are comparatively busy during their daytime serving the community, going door to door. It becomes hard for them to manage time to physically get together, sit in a circle, and play the game either digitally or physically. A multiplayer remotely connected play would potentially solve this problem. In this case, one of the players creates a server generally called a "Room" having a meaningful name representing the team. The rest of the players connect to that room and the creator of the room starts the game. Similar mechanics are followed. The players get an opportunity to put cards in their turn. When the turn is passed, the moves get frozen until the turn comes around again, completing all the other players.

During trials, with remotely connected players, we observed that there were a lot of misunderstandings due to the lack of communication between players. They felt frustrated if they couldn't perform a task due to the restrictions enforced by the rules of play, for example, when the turn was missed or they couldn't place a card in their turn, etc. So we introduced audio communication similar to a radio which broadcasts audio of one player to all other players connected in the room or with the other player forming a group with the player. We observed peer learning happening through the discussions between the players while explaining the rules to one another and in general team conversations.

Following the same game mechanics, rules, and deck of cards, two variations of the Android game app were developed. The gameplay analytics of each player were saved offline and asynchronously stored in an online database whenever the internet was available, for further analysis. ASHAs having an active data connection would update the scores instantly. It was observed that a number of ASHAs didn't have an active data pack in their smartphone. For them, there would be asynchronous syncing multiple times within a week, whenever she would come to the state institution for reporting and would connect her smartphone to the Wi-Fi network of the institution (which is usually available).

Appendix B

Survey Questionnaire

मैं अपनी हस्ताक्षर के माध्यम से स्वीकार करती हूं कि इस शोध अध्ययन में मेरी भागीदारी स्वैच्छिक है। मैं इस तथ्य से अवगत हूं कि एकत्र की गई व्यक्तिगत जानकारी का उपयोग केवल शैक्षिक उपयोग के लिए किया जाएगा।

_____ _____ / ____ / २०२२

हस्ताक्षर **तिथि**

नीचे दी हुई जगह पर अपना पूरा नाम लिखिए :

_____

अपनी उम्र लिखिए : _____ **साल**

आपकी स्कूली शिक्षा कौनसी श्रेणी तक हुई है :

☐ ८ वि ☐ ९ वि ☐ १० वि ☐ ११वी
☐ १२ वि ☐ ग्रेजुएट ☐ पोस्ट-ग्रेजुएट

आप कितने वर्षो से आशा कार्यकर्ता हैं :

_____ **साल**

कृपया निम्नलिखित प्रश्नों के उत्तर खाली बक्सों में ☑ निशान लगाकर दीजिये । प्रत्येक प्रश्न का केवल एक ही सही उत्तर है ।

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१. ओ पी वी - बूस्टर शिशु को किस आयु मै दिया जता है?

☐ ९ महीने ☐ ११/२ साल
☐ २ साल ☐ ३ १/२ साल

२. पेंटावैलेंट - १ टीका शिशु को किस आयु मै दिया जता है?

☐ ११/२ महीने ☐ २ १/२ महीने
☐ ३ १/२ महीने ☐ ९ महीने

३. एम आर - १ (खसरे का पहला टीका) शिशु को किस आयु मै दिया जता है?

☐ ९ महीने ☐ ११/२ साल
☐ २ साल ☐ ३ १/२ साल

४. पहला डीपीटी बूस्टर शिशु को किस आयु मै दिया जता है?

☐ ९ महीने ☐ ११/२ साल
☐ २ साल ☐ ३ १/२ साल

५. दूसरा डीपीटी बूस्टर शिशु को किस आयु मै दिया जता है?

☐ ९ महीने ☐ ११/२ साल
☐ ५-६ साल ☐ ३ १/२ साल

६. टी टी / टी डी (टेटनेस / डिप्थोरिया) का पहला टीका शिशु को कौनसी आयु मे दिया जाता है?

☐ १० साल ☐ १६ साल
☐ ६ साल ☐ ५ साल

७. टी टी / टी डी (टेटनेस / डिप्थोरिया) का दूसरा टीका शिशु को कौनसी आयु मे दिया जाता है?

☐ १० साल ☐ १६ साल
☐ ६ साल ☐ ५ साल

८. पेंटावैलेंट - २ टीका शिशु को किस आयु मै दिया जता है?

☐ ११/२ महीने ☐ २ १/२ महीने
☐ ३ १/२ महीने ☐ ९ महीने

Figure B.1: Questionnaire Page-1

९. विटामिन ए सबसे पहले बच्चों को कब दिया जाता है?

⬜ ९ महीने ⬜ १ १/२ साल

⬜ २ साल ⬜ ३ १/२ साल

१०. विटामिन ए दूसरी बार बच्चों को कब दिया जाता है?

⬜ ९ महीने ⬜ १ १/२ साल

⬜ २ साल ⬜ ३ १/२ साल

११. दूसरी बार के बाद विटामिन ए बच्चों को किस अंतराल में दिया जाता है?

⬜ ३ महीने ⬜ ४ महीने

⬜ ५ महीने ⬜ ६ महीने

१२. एम आर - २ (खसरे का दूसरा टीका) शिशु को कब दिया जता है?

⬜ ९ महीने ⬜ १ १/२ साल

⬜ २ साल ⬜ ३ १/२ साल

१३. जे ई - १ (जापानीज इन्सेफेलाइटिस) का पहला टीका शिशु को कब दिया जाता है?

⬜ ९ महीने ⬜ १ १/२ साल

⬜ २ साल ⬜ ३ १/२ साल

१४. जे ई - २ (जापानीज इन्सेफेलाइटिस) का दूसरा टीका शिशु को कब दिया जाता है?

⬜ ९ महीने ⬜ १ १/२ साल

⬜ २ साल ⬜ ३ १/२ साल

१५. पी सी वी - १ (न्यूमोकोकल कंजुगेट) का पहला टीका शिशु को कब दिया जाता है?

⬜ १ १/२ महीने ⬜ २ १/२ महीने

⬜ ३ १/२ महीने ⬜ ९ महीने

१६. पी सी वी - २ (न्यूमोकोकल कंजुगेट) का दूसरा टीका शिशु को कब दिया जाता है?

⬜ १ १/२ महीने ⬜ २ १/२ महीने

⬜ ३ १/२ महीने ⬜ ९ महीने

१७. पी सी वी - बूस्टर शिशु को कब दिया जता है?

⬜ १ १/२ महीने ⬜ २ १/२ महीने

⬜ ३ १/२ महीने ⬜ ९ महीने

१८. पेंटावैलेंट - ३ टीका शिशु को किस आयु मै दिया जता है?

⬜ १ १/२ महीने ⬜ २ १/२ महीने

⬜ ३ १/२ महीने ⬜ ९ महीने

१९. हेपेटाईटिस बी टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

२०. रोटा - १ (रोटावायरस) का पहला टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

२१. बी सी जी का टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

२२. रोटा - २ (रोटावायरस) का दूसरा टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

२३. ओ पी वी - ० (ओरल पोलियो वैक्सीन) का पहला टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

२४. रोटा - ३ (रोटावायरस) का तीसरा टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

२५. जन्म के तुरंत बाद शिशु को क्या दिया जाना चाहिए?

⬜ कोलोस्ट्रम ⬜ ओ पी वी - ०

⬜ हेपेटाईटिस बी ⬜ बी सी जी का टीका

Figure B.2: Questionnaire Page-2

२६. ओ पी वी - १ (ओरल पोलियो वैक्सीन) का दूसरा टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

२७. आई पी वी - १ (इनएक्टिवेटेड पोलियो वैक्सीन) का पहला टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

२८. ओ पी वी - २ (ओरल पोलियो वैक्सीन) का दूसरा टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

२९. पेंटावैलेंट - २ टीका शिशु को किस आयु मै दिया जता है?

⬜ ९ महीने ⬜ १ १/२ साल

⬜ २ साल ⬜ ३ १/२ साल

३०. ओ पी वी - ३ (ओरल पोलियो वैक्सीन) का दूसरा टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

३१. आई पी वी - २ (इनएक्टिवेटेड पोलियो वैक्सीन) का पहला टीका शिशु को कब दिया जता है?

⬜ जन्म ⬜ १ १/२ महीने

⬜ २ महीने ⬜ ३ १/२ महीने

३२. एक गर्भवती माँ को अपनी पूरी गर्भावस्था में कितने न्यूनतम प्रसवपूर्व देखभाल सत्रों की आवश्यकता होती है ?

⬜ १ ⬜ २

⬜ ३ ⬜ ४

३३. किस प्रसवपूर्व देखभाल पर गर्भवती माँ को टी टी - १ (टिटनेस टॉक्साइड) का पहला इंजेक्शन दिया जाता है ?

⬜ पहला ⬜ दूसरा

⬜ तीसरा ⬜ चौथा

३४. किस प्रसवपूर्व देखभाल पर गर्भवती माँ को टी टी - २ (टिटनेस टॉक्साइड) का दूसरा इंजेक्शन दिया जाता है ?

⬜ पहला ⬜ दूसरा

⬜ तीसरा ⬜ चौथा

३५. टीटी बूस्टर उन गर्भवती माताओ को दी जाती है जिन्होने पिछले कितनी सालो में टीटी के २ टीके लगवा चुकी है ?

⬜ १ ⬜ २

⬜ ३ ⬜ ४

३६. किस प्रसवपूर्व देखभाल पर गर्भवती माँ को आयरन फोलिक एसिड (आई एफ ए) की गोलियां सेवन चालू करनी चाहिए ?

⬜ पहला ⬜ दूसरा

⬜ तीसरा ⬜ चौथा

३७. गर्भवती महिलाओं को कौन सी कृमिनाशक गोली दी जाती है ?

⬜ कैल्शियम ⬜ एल्बेंडाजोल

⬜ आई एफ ए ⬜ विटामिन डी

३८. आई एफ ए गोलि के साथ किस गोलि का सेवन नहीं करना चाहिए ?

⬜ कैल्शियम ⬜ एल्बेंडाजोल

⬜ आई एफ ए ⬜ विटामिन डी

३९. सिफलिस टेस्ट और एचआईवी टेस्ट किस प्रसवपूर्व देखभाल पर गर्भवती माँ को करवाना चाहिए ?

⬜ पहला ⬜ दूसरा

⬜ तीसरा ⬜ चौथा

४०. पहला प्रसव पश्चात देखभाल कितने घंटों के भीतर करना चाहिए?

⬜ ४ ⬜ ६

⬜ १२ ⬜ २४

Figure B.3: Questionnaire Page-3

	A	B	D	E	F	G
1	Sl	English	IndiaTyping	Google	Corrected Human	Reverse Translated
2	1	I felt content	मला समाधान वाटले	मला सामग्री वाटली	मला समाधान वाटलं	I was satisfied
3	2	I felt skillful	मला कुशल वाटले	मला कुशल वाटले	मला खेळामुळे खूप कुशल वाटलं	I felt very skilled because of the game
4	3	I was interested in the game's story	मला खेळाच्या कथेत रस होता	मला खेळाच्या कथेत रस होता	या खेळामधील गोष्टी साठी मी उत्सुक होते	I was looking forward to the things in this game
5	4	I thought it was fun	मला वाटलं गंमत आहे	मला वाटले की ते मजेदार आहे	मला हा खेळ गम्मत वाटली	I had this game of fun
6	5	I was fully occupied with the game	मी खेळात पूर्णपणे गुंतले होते	मी खेळाने पूर्णपणे ताब्यात घेत होतो	मी खेळामध्ये पूर्ण दंग झालेले	I was full of timber in the game
7	6	I felt happy	मला आनंद वाटला	मला आनंद झाला	मला आनंद वाटला	I was happy
8	7	It gave me a bad mood	मला वाईट मूड दिला	यामुळे मला एक वाईट मूड दिला	याच्यामुळे मला वाईट अनुभव आला	I had a bad experience because of this
9	8	I thought about other things	मी इतर गोष्टींचा विचार केला	मी इतर गोष्टींबद्दल विचार केला	मी खेळताना दुसऱ्या गोष्टींचा विचार करता होते	I was thinking of other things while playing
10	9	I found it tiresome	मला ते कंटाळवाणे वाटले	मला ते कंटाळवाणे वाटले	मला हा खेळ थकवणारा वाटला	I found this game tiring
11	10	I felt competent	मला सक्षम वाटले	मला सक्षम वाटले	मला सक्षम असल्यासारख वाटलं	I felt like I was able to
12	11	I thought it was hard	मला वाटले ते कठीण आहे	मला वाटले ते कठीण आहे	मला वाटले ते कठीण आहे	I thought it was hard
13	12	It was aesthetically pleasing	ते सौंदर्यानि सुखावणारे होते	हे सौंदर्यात्मकदृष्ट्या आनंददायक होते	खेळ दिसायला सुंदर आणि आकर्षक होता	The game was pretty and attractive to see
14	13	I forgot everything around me	मी माझ्या आजूबाजूचे सर्व काही विसरलो	मी माझ्या सभोवतालचे सर्व काही विसरलो	मी सगळं विसरून खेळात आकंठ बुडाले	I forgot all about the game
15	14	I felt good	मला बरं वाटलं	मला बरे वाटले	मला छान वाटल	I would like to feel nice
16	15	I was good at it	मी त्यात चांगला होतो	मी त्यात चांगला होतो	मला खेळात गती होती	I had a speed in the game
17	16	I felt bored	मला कंटाळा आला	मला कंटाळा आला	मला कंटाळा आला	I got bored
18	17	I felt successful	मला यशस्वी वाटले	मला यशस्वी वाटले	मला यशस्वी वाटलं	I felt successful
19	18	I felt imaginative	मला कल्पनारम्य वाटले	मला काल्पनिक वाटले	मझ्या कल्पनाशक्ती ला चालना मिळाली	My imagination got a walk
20	19	I felt that I could explore things	मला वाटले की मी गोष्टी शोधू शकतो	मला वाटले की मी गोष्टी शोधू शकतो	मला असा वाटलं कि मी पण गोष्टी शिकू शकते	I think I can learn things but
21	20	I enjoyed it	मला मजा आली	मी त्याचा आनंद घेतला	मी इंजॉय केला	I did the engoy
22	21	I was fast at reaching the game's targets	मी खेळाचे लक्ष्य गाठण्यात जलद होते	मी खेळाच्या लक्ष्यात पोहोचण्यात वेगवान होतो	मी खूप वेगाने एक की टप्पा पार करत होते	I was crossing a key phase very fast
23	22	I felt annoyed	मला चीड वाटली	मला राग आला	मला खूप त्रासदायक वाटलं	I feel very annoying
24	23	I felt pressured	मला दडपण जाणवले	मला दबाव वाटला	मला दबाव जाणवला	I felt the pressure
25	24	I felt irritable	मला चिडचिड वाटली	मला चिडचिडे वाटले	खेळताना माझी चिडचिड झाली	I got irritated while playing
26	25	I lost track of time	मी वेळेचा मागोवा गमावला	मी वेळेचा मागोवा गमावला	मी खेळताना वेळेचा भान विसरले	I forgot about the idea of time while playing
27	26	I felt challenged	मला आव्हान वाटले	मला आव्हान वाटले	मला खेळ आव्हानात्मक वाटला	I found the game challenging
28	27	I found it impressive	मला ते प्रभावी वाटले	मला ते प्रभावी वाटले	मला खेळ आकर्षक वाटला	I found the game attractive
29	28	I was deeply concentrated in the game	मी खेळात मनापासून एकाग्र होतो	मी गेममध्ये खोलवर लक्ष केंद्रित केले होते	मी खेळ खेळताना खूप एकाग्र झालेले	I
30	29	I felt frustrated	मला निराश वाटले	मी निराश झालो	मला खूप निराश वाटले	I felt very frustrated
31	30	It felt like a rich experience	एक समृद्ध अनुभव वाटला	हा एक समृद्ध अनुभव वाटला	हा खूप आल्हाददायक अनुभव होता	It was a very pleasant experience
32	31	I lost connection with the outside world	माझा बाहेरच्या जगाशी संपर्क तुटला	मी बाह्य जगाशी संबंध गमावला	मी बाकीच्या जगाचा भान विसरले	I forgot about the rest of the world
33	32	I felt time pressure	मला वेळेचा दबाव जाणवला	मला वेळ दबाव वाटला	मला वेळेत पूर्ण करण्याचा दबाव जाणवला	I felt the pressure to complete in time
34	33	I had to put a lot of effort into it	त्यासाठी मला खूप मेहनत घ्यावी लागली	मला त्यात बरेच प्रयत्न करावे लागले	मला खूप मेहनत घ्यावी लागली	I had to work a lot of hard work
35						
36	34	not at all	अजिबात नाही	अजिबात नाही	अजिबात नाही	Not at all
37	35	slightly	किंचित	किंचित	किंचित प्रमाणात	Quantitatively
38	36	moderately	माफक प्रमाणात	माफक प्रमाणात	मध्यम प्रमाणात	To a moderate amount
39	37	fairly	प्रामाणिकपणे	प्रामाणिकपणे	चांगल्या प्रमाणात	Well
40	38	extremely	अत्यंत	अत्यंत	खूप जास्त	Too much

Figure B.4: GEQ Translation Processs

Appendix C

Institute Review Board Approval

IRCC, Office of the Dean R&D, IIT Bombay

INDIAN INSTITUTE OF TECHNOLOGY BOMBAY

Institutional Review Board (IRB)

January 17, 2023

To
Prof. Satish Agnihotri
CTARA
IIT Bombay

Ref: Proposal No. IITB-IRB/2022/051

Sub: Review of the above-mentioned project proposal

Dear Professor,

Thank you for submitting your proposal to the IITB Institutional Review Board (IRB) for review. The IRB has reviewed the proposal submitted by you and the following proposal is approved:

Proposal number	: **IITB-IRB/2022/051**
Title	: **Refresher Training through Games for Frontline Healthcare Workers on Maternal and Newborn care**

The IRB approval is for the ethical conduct of the study. The study is approved for the entire duration and a closure report should be submitted within 2 months of the completion of the study.

Further, it is also confirmed that neither you, nor any of the study team members have participated in the decision-making process of the committee.

In case there are any changes in the proposed work (which is not limited to scope, dates, participants and methodology etc.), please communicate to IRB within 15 days of such a change.

Thank you.
With Best wishes from IRB for your study,

D Parthasarathy
Chairperson IITB-IRB

Figure C.1: IRB Approval

References

Abt, C., 1970. Abt, Clark C. Serious Games. New York: Viking, 1970, 176 pp., $5.95, L.C. 79-83234. American Behavioral Scientist 14 (1), 129–129.

Agarwal, S., Curtis, S. L., Angeles, G., Speizer, I. S., Singh, K., Thomas, J. C., 2019. The impact of India's accredited social health activist (ASHA) program on the utilization of maternity services: A nationally representative longitudinal modelling study. Human Resources for Health 17 (1).

Akl, E. A., Pretorius, R. W., Sackett, K., Erdley, W. S., Bhoopathi, P. S., Alfarah, Z., Schünemann, H. J., 2010. The effect of educational games on medical students' learning outcomes: A systematic review: BEME Guide No 14.

Andre, F. E., Booy, R., Bock, H. L., Clemens, J., Datta, S. K., John, T. J., Lee, B. W., Lolekha, S., Peltola, H., Ruff, T. A., Santosham, M., Schmitt, H. J., 2008. Vaccination greatly reduces disease, disability, death and inequity worldwide.

Angarita, M., Bernal, R., Ortiz, E., Pinzón, B., Esguerra, A., Van Der Merwe, H., Brown, T., 2005. EDUGAMING PCs AND QUICKPDA –eBOOKS: A new model for surpassing fragmentation in mobile learning on e-health. In: MLearn 2005. Vol. Cape Town,.
URL http://www.mlearn.org.za/CD/papers/Angarita.pdf

Arnab, S., Lim, T., Carvalho, M. B., Bellotti, F., De Freitas, S., Louchart, S., Suttie, N., Berta, R., De Gloria, A., 2015. Mapping learning and game mechanics for serious games analysis. British Journal of Educational Technology 46 (2).

Azuma, R. T., 1997. A Survey of Augmented Reality, Presence: Teleoperators and Virtual Environments. Foundations and Trends in Human-Computer Interaction 6 (4).

B, S., Kumar, A., 2017. Analysis of language effectiveness in ASHA training programme: A Study in Karnataka. International Journal of Commerce and Management Research.
URL www.managejournal.com

Bag, S., Datta, M., 2017. Evaluation of mother and child protection card entries in a rural area of West Bengal. International Journal Of Community Medicine And Public Health 4 (7), 2604.

Bandura, A., 1977. Albert Bandura- Social Learning Theory. Simply Psychology.

Bashingwa, J. J. H., et.al., 8 2021. Examining the reach and exposure of a mobile phone-based training programme for frontline health workers (ASHAs) in 13 states across India.

Bassi, A., John, O., Praveen, D., Maulik, P. K., Panda, R., Jha, V., 2018. Current status and future directions of mhealth interventions for health system strengthening in India: Systematic review.

Billinghurst, M., Dünser, A., 2012. Augmented reality in the classroom. Computer 45 (7).

Bochennek, K., Wittekindt, B., Zimmermann, S. Y., Klingebiel, T., 2007. More than mere games: A review of card and board games for medical education.

Braun, V., Clarke, V., 2006. Using thematic analysis in psychology. Qualitative Research in Psychology 3 (2).

Braun, V., Clarke, V., 2019. Reflecting on reflexive thematic analysis.

Breuer, J., Bente, G., 2010. Why so serious? On the relation of serious games and learning. Eludamos. Journal for Computer Game Culture 4 (1).

Charmaz, K., 2014. Constructing Grounded Theory - Kathy Charmaz - Google Books.

de Ribaupierre, S., Kapralos, B., Haji, F., Stroulia, E., Dubrowski, A., Eagleson, R., 2014. Healthcare training enhancement through virtual reality and serious games. Intelligent Systems Reference Library 68, 9–27.

De Wit-Zuurendonk, L. D., Oei, S., 2011. Serious gaming in women's health care.

Deterding, S., 2011. Meaningful Play: Getting Gamification Right.

Deterding, S., 2012. Gamification: designing for motivation. Interactions 19 (4).

Deterding, S., Dixon, D., Khaled, R., Nacke, L., 2011a. From game design elements to gamefulness.

Deterding, S., Khaled, R., Nacke, L., Dixon, D., 2011b. Gamification: Toward a Definition. In Proceedings,. CHI 2011 Gamification Workshop.

Dimitrov, D. M., Rumrill, P. D., 2003. Pretest-posttest designs and measurement of change.

Dunleavy, M., Dede, C., Mitchell, R., 2009. Affordances and limitations of immersive participatory augmented reality simulations for teaching and learning. Journal of Science Education and Technology 18 (1).

Fabricatore Carlo, 2007. University of Huddersfield Repository GAMEPLAY AND GAME MECHANICS DESIGN : A KEY TO. In: ENLACES (MINEDUC Chile) -OECD Expert Meeting on Videogames and Education.

Gaffney, J. S., Rodgers, E., 2018. Scaffolding research: Taking stock at the four-decade mark.

Ganapathy, K., Ravindra, A., 2008. mHealth: A potential tool for health care delivery in India. In: Making the eHealth Connection. Vol. 68. pp. 1–9.
URL http://dx.doi.org/10.1016/j.ijhcs.2010.06.005

Garg, C. C., Karan, A. K., 2009. Reducing out-of-pocket expenditures to reduce poverty: A disaggregated analysis at rural-urban and state level in India. Health Policy and Planning 24 (2), 116–128.

Gee, J. P., 2012. Situated language and learning: A critique of traditional schooling.

Gentry, S. V., Gauthier, A., Ehrstrom, B. L., Wortley, D., Lilienthal, A., Car, L. T., Dauwels-Okutsu, S., Nikolaou, C. K., Zary, N., Campbell, J., Car, J., 3 2019. Serious gaming and gamification education in health professions: systematic review.

Giunti, G., Baum, A., Giunta, D., Plazzotta, F., Benitez, S., Gómez, A., Luna, D., González Bernaldo De Quiros, F., 2015. Serious Games: A Concise Overview on What They Are and Their Potential Applications to Healthcare. In: Studies in Health Technology and Informatics. Vol. 216. pp. 386–390.

Graafland, M., Dankbaar, M., Mert, A., Lagro, J., De Wit-Zuurendonk, L., Schuit, S., Schaafstal, A., Schijven, M., 2014. How to Systematically Assess Serious Games Applied to Health Care. JMIR Serious Games 2 (2), e11.

Gujral, S., Abbi, R., Mujoo, R., Gopaldas, T., 1991. Determinants of community health workers' performance in India. Food & Nutrition Bulletin 13 (4), 277–281.

Guleri, D. S. k., Dixit, D. S., Sakalle, D. S., Bhagwat, D. A. K., Yesikar, D. V., Pandey, D. D., 2014. A rapid appraisal of training issues of ASHAs (Accredited Social Health Activist) in Madhya Pradesh. Public Health Review: International Journal of Public Health Research 1 (1).

Gunter, G. A., Kenny, R. F., Vick, E. H., 2008. Taking educational games seriously: Using the RETAIN model to design endogenous fantasy into standalone educational games.

Gupta, S., Jain, D. K., 2014. MOOCs for medical education: An Indian perspective. In: Proceedings of the 2014 IEEE International Conference on MOOCs, Innovation and Technology in Education, IEEE MITE 2014.

Hamari, J., Koivisto, J., 2015. Why do people use gamification services? International Journal of Information Management 35 (4).

Hamari, J., Koivisto, J., Sarsa, H., 2014. Does gamification work? - A literature review of empirical studies on gamification. In: Proceedings of the Annual Hawaii International Conference on System Sciences.

Hannafin, M. J., Hannafin, K. M., Land, S. M., Oliver, K., 1997. Grounded practice and the design of constructivist learning environments. Educational Technology Research and Development 45 (3), 101–117.

Hattie, J., 2008. Visible learning: A synthesis of over 800 meta-analyses relating to achievement.

Hutchins, E., 1995. Cognition in the Wild.

Ijsselsteijn, W., Van Den Hoogen, W., Klimmt, C., De Kort, Y., Lindley, C., Mathiak, K., Poels, K., Ravaja, N., Turpeinen, M., Vorderer, P., 2008. Measuring the Experience of Digital Game Enjoyment. Tech. rep.

Ijsselsteijn, W. A., Kort, D., Poels, Y. A. W. ., 2013. The Game Experience Questionnaire. Tech. rep., Eindhoven Institute of Technology.

URL https://pure.tue.nl/ws/files/21666907/Game_Experience_Questionnaire_English.pdf

International Institute for Population Sciences (IIPS) and Ministry of Health and Family Welfare, 2016. National Family Health Survey-4 India Fact Sheet.
URL http://rchiips.org/NFHS/NFHS-4Reports/India.pdf

International Institute for Population Sciences (IIPS) and Ministry of Health and Family Welfare, 2021. National Family Health Survey-5 India Fact Sheet. International Institute for Population Sciences (IIPS) and Ministry of Health and Family Welfare.
URL http://rchiips.org/nfhs/factsheet_NFHS-5.shtml

Jensen, L. X., Buhl, A., Hussain, S., Karan, A., Konradsen, F., Bearman, M., 2023. Digital education for health professionals in India: a scoping review of the research. BMC Medical Education 23 (1).

Joshi, V., Joshi, N. K., Bhardwaj, P., Singh, K., Ojha, D., Jain, Y. K., 2023. The Health Impact of mHealth Interventions in India: Systematic Review and Meta-Analysis.

Kahneman, D., 2011. Thinking fast, thinking slow.

Kasi, S. G., et.al., 2021. Indian Academy of Pediatrics (IAP) Advisory Committee on Vaccines and Immunization Practices (ACVIP): Recommended Immunization Schedule (2020–21) and Update on Immunization for Children Aged 0 Through 18 Years. Indian Pediatrics 58 (1).

Kirkpatrick, D. L., 1950. The Kirkpatrick Model.
URL https://www.kirkpatrickpartners.com/the-kirkpatrick-model/

Kizhatil, A., ., R., Hariharan, H. C., John, A., Thomas, A. M., Padmanabhan, G., 2019. Assessment of immunization coverage and associated factors among children in Paravur Taluk of Ernakulam district, Kerala. International Journal Of Community Medicine And Public Health 6 (8), 3594.

Knowles, M. S., 1980. THE MODERN PRACTICE OF ADULT EDUCATION , From Pedagogy to Andragogy What Is Andragogy ? Business, 400.
URL http://www.amazon.co.uk/dp/0695814729

Kok, M. C., Dieleman, M., Taegtmeyer, M., Broerse, J. E., Kane, S. S., Ormel, H., Tijm, M. M., De Koning, K. A., 2015. Which intervention design factors influence performance of community health workers in low- and middle-income countries? A systematic review. Health Policy and Planning 30 (9), 1207–1227.

Kolb, D. A., 1984. Experiential Learning: Experience as The Source of Learning and Development. Prentice Hall, Inc. (1984).

Koster, R., 2010. Theory of Fun for Game Design [Kindle Edition]. Paraglyph, 256.
URL http://www.amazon.com/Theory-Fun-Game-Design/dp/1932111972

Kumar, N., Anderson, R., 2015. Mobile phones for maternal health in rural India. Conference on Human Factors in Computing Systems - Proceedings 2015-April, 427–436.

Laamarti, F., Eid, M., El Saddik, A., 2014. An overview of serious games.

Lave, J., Wenger, E., 1991. Situated Learning Legitimate Peripheral Participation. Cambridge University Press.

Lenihan, D., 2012. Health Games: A Key Component for the Evolution of Wellness Programs. Games for Health Journal 1 (3).

Li, M., Ma, S., Shi, Y., 2023. Examining the effectiveness of gamification as a tool promoting teaching and learning in educational settings: a meta-analysis.

Madanian, S., Parry, D. T., Airehrour, D., Cherrington, M., 2019. MHealth and big-data integration: Promises for healthcare system in India.

Majhi, A., Agnihotri, S., Mondal, A., 4 2022. Physical and Augmented Reality based Playful Activities for Refresher Training of ASHA Workers in India. In: Asian HCI Symposium '22. ACM, New York, NY, USA, pp. 16–22.
URL https://dl.acm.org/doi/10.1145/3516492.3558788

Majhi, A., Joshi, A., Agnihotri, S. B., Mondal, A., 5 2021. Refresher Training through Quiz App for capacity building of Community Healthcare Workers or Anganwadi Workers in India. In: Asian CHI Symposium 2021. ACM, New York, NY, USA, pp. 80–88.
URL https://dl.acm.org/doi/10.1145/3429360.3468186

Majhi, A., Mondal, A., Agnihotri, S. B., 9 2024a. Mapping Child Malnutrition and Measuring Efficiency of Community Healthcare Workers through Location Based Games in India. In: Proceedings of the 2024 International Conference on Information Technology for Social Good. ACM, New York, NY, USA, pp. 377–385.
URL https://dl.acm.org/doi/10.1145/3677525.3678685

Majhi, A., Mondal, A., Agnihotri, S. B., 10 2024b. Refresher Training through Digital and Physical, Card-Based Game for Accredited Social Health Activists (ASHAs) and Anganwadi Workers (AWWs) in India. In: Companion Proceedings of the 2024 Annual Symposium on Computer-Human Interaction in Play. Vol. 1. ACM, New York, NY, USA, pp. 175–180.
URL https://dl.acm.org/doi/10.1145/3665463.3678819

Majhi, A., Mondal, A., Agnihotri, S. B., 2024c. Replay, Revise, and Refresh: Smartphone-Based Refresher Training for Community Healthcare Workers in India. pp. 310–320.
URL https://link.springer.com/10.1007/978-3-031-61966-3_34

Mani, M. R., Johnson, A. R., Joseph, J., Jyothis, S., Joseph, L., Cleetus, R. P., Sulekha, T., 2020. Knowledge Regarding Pregnancy and Child Care Among Mothers in Possession of Mother and Child Protection Card in a Rural Maternity Hospital in Karnataka. Journal of Medical Sciences and Health 6 (2), 36–42.

Mayer, R. E., 1997. Multimedia learning: Are we asking the right questions? Educational Psychologist 32 (1), 1–19.

Mey, I., 2005. The Cultural Nature of Human Development (Barbara Rogoff). Vol. 15.

Miller, K. A., Deci, E. L., Ryan, R. M., 1985. Intrinsic Motivation and Self-Determination in Human Behavior. Contemporary Sociology 17 (2).

MoHFW, 2016. Mission Indradhanush Operational Guidelines. Tech. rep.
URL https://mohfw.gov.in/sites/default/files/216846291201489665182.pdf

MoSPI, 2015. NSS 75th Round (July 2017-June 2018) : Household Social Consumption on Education and Health. Tech. rep.
URL http://www.mospi.nic.in/sites/default/files/publication_reports/Report_585_75th_round_Education_final_1507_0.pdf

Mulcahy, H., Philpott, L. F., O'Driscoll, M., Bradley, R., Leahy-Warren, P., 2022. Breastfeeding skills training for health care professionals: A systematic review. Heliyon 8 (11).

Nieder, J., Nayna Schwerdtle, P., Sauerborn, R., Barteit, S., 2022. Massive Open Online Courses for Health Worker Education in Low- and Middle-Income Countries: A Scoping Review.

Palmer, D., Denelsbeck, A., 2024. The Mathematics and Design of a New Deck of Playing Cards. Tech. rep.
URL https://archive.bridgesmathart.org/2024/bridges2024-439.pdf

Panda, B. K., Kumar, G., Mishra, S., 6 2020. Understanding the full-immunization gap in districts of India: A geospatial approach. Clinical Epidemiology and Global Health 8 (2), 536–543.

Peek, M. E., 2017. Can mHealth Interventions Reduce Health Disparities among Vulnerable Populations? Diversity & Equality in Health and Care 14 (2).

Pérez, M. C., Singh, R., Chandra, D., Ridde, V., Seth, A., Johri, M., 2020. Development of an mHealth Behavior Change Communication Strategy: A case-study from rural Uttar Pradesh in India. COMPASS 2020 - Proceedings of the 2020 3rd ACM SIGCAS Conference on Computing and Sustainable Societies, 274–278.

Ramachandran, D., Canny, J., Das, P. D., Cutrell, E., 2010. Mobile-izing health workers in rural India. Conference on Human Factors in Computing Systems - Proceedings 3, 1889–1898.

Ramjee, P., Chhokar, M., Sachdeva, B., Meena, M., Abdullah, H., Vashistha, A., Nagar, R., Jain, M., 5 2025. ASHABot: An LLM-Powered Chatbot to Support the Informational Needs of Community Health Workers.
URL http://arxiv.org/abs/2409.10913http://dx.doi.org/10.1145/3706598.3713680

Ritterfeld, U., Cody, M., Vorderer, P., 2009. Serious games: Mechanisms and effects. In: Serious Games: Mechanisms and Effects. pp. 1–530.

Ritterfeld, U., Weber, R., 2006. Video games for entertainment and education. In: Playing Video Games: Motives, Responses, and Consequences.

Rogers, A., Goore, L. L., Wamae, J., Starnes, J. R., Okong'o, S. O., Okoth, V., Mudhune, S., 2023. Training and experience outperform literacy and formal education as predictors of community health worker knowledge and performance, results from Rongo sub-county, Kenya. Frontiers in Public Health 11.

Rupal Dalal, 2019. HST Training Modules.
URL https://health.spoken-tutorial.org/hstTrainingModuleView/147/

Ryan, R. M., Deci, E. L., 2000. Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. American Psychologist 55 (1).

SALTHOUSE, T. A., 1995. Influence of Processing Speed On Adult Age Differences
}in Learning. SWISS JOURNAL OF PSYCHOLOGY 54 (2).

Sangrà, A., Vlachopoulos, D., Cabrera, N., 2012. Building an inclusive definition of e-learning: An approach to the conceptual framework. International Review of Research in Open and Distance Learning 13 (2), 145–159.

Santos, M. E. C., Chen, A., Taketomi, T., Yamamoto, G., Miyazaki, J., Kato, H., 2014. Augmented reality learning experiences: Survey of prototype design and evaluation. IEEE Transactions on Learning Technologies 7 (1).

Sarkar, P., Kadam, K., Pillai, J. S., 2020. Learners' approaches, motivation and patterns of problem-solving on lines and angles in geometry using augmented reality. Smart Learning Environments 7 (1).

Sawyer, B., Smith, P., 2008. Serious games taxonomy. Slides from the Serious Games Summit at the Game . . . , 1–54.
URL http://scholar.google.com/scholar?hl=en&btnG=Search&q=intitle:Serious+Games+Taxonomy#0

Shah, M. P., Kamble, P. A., Agnihotri, S. B., 2017. Tackling child malnutrition: An innovative approach for training health workers using ICT: A pilot study. In: IEEE Region 10 Humanitarian Technology Conference 2016, R10-HTC 2016 - Proceedings.

Sharma, R., Webster, P., Bhattacharyya, S., 2014. Factors affecting the performance of community health workers in India: A multi-stakeholder perspective. Global Health Action 7 (1).

Siemens, G., 2005. Connectivism: Learning as network-creation. ASTD Learning News 10 (1).

Sivasubramanian, S., Raval, A., 4 2025. Artificial Intelligence–augmented public health interventions in India. Health Affairs Scholar 3 (5).

Soyoof, A., Reynolds, B. L., Shadiev, R., Vazquez-Calvo, B., 2024. A mixed-methods study of the incidental acquisition of foreign language vocabulary and healthcare knowledge through serious game play. Computer Assisted Language Learning 37 (1-2).

Stefany, S., Cong, L. M., Helmi, J., Roedavan, R., 2024. Integrating Learning and Serious Game Design: The Development and Application of the Learning and Game Integration (LGI) Framework Using the 'Blackout' Serious Game. In: Proceedings of the European Conference on Games-based Learning. Vol. 18.

Sultan, M. A., Miller, E., Tikkanen, R. S., Singh, S., Kullu, A., Cometto, G., Fitzpatrick, S., Ajuebor, O., Gillon, N., Edward, A., Moleman, Y. P., Pandya, S., Park, I., Shen, J. Y., Yu, Y., Perry, H., Scott, K., Closser, S., 12 2025. Competency-based education and training for Community Health Workers: a scoping review. BMC Health Services Research 25 (1).

Sweller, J., 2022. Cognitive load theory. In: International Encyclopedia of Education: Fourth Edition.

Telecom Regulatory Authority of India, 2021. Press Release No. 22/2019. Tech. rep., Secretary of Telecom Regulatory Authority of India, New Delhi.
URL https://www.trai.gov.in/sites/default/files/PR_No.06of2021_0.pdf

Thomas, V., Jenkins, J., George, J., 1 2024. A systematic review and qualitative evidence synthesis of factors affecting mHealth adoption in India. Oxford Open Digital Health 2.

Tulaskar, R., 2020. Study of Instructional Illustrations on ICTs: Considering persona of low-literate users from India. ACM International Conference Proceeding Series, 53–56.

UNESCO, 2025. AI and the future of education: Disruptions, dilemmas and directions. UNESCO.
URL https://www.unesco.org/en/articles/ai-and-future-education-disruptions-dilemmas-a

van Doorslaer, E., O'Donnell, O., Rannan-Eliya, R. P., Somanathan, A., Adhikari, S. R., Garg, C. C., Harbianto, D., Herrin, A. N., Huq, M. N., Ibragimova, S., Karan, A., Ng, C. W., Pande,

B. R., Racelis, R., Tao, S., Tin, K., Tisayaticom, K., Trisnantoro, L., Vasavid, C., Zhao, Y., 2006. Effect of payments for health care on poverty estimates in 11 countries in Asia: an analysis of household survey data. Lancet 368 (9544), 1357–1364.

Van Rozen, R., Bouwer, A., Millenaar, K., 2023. Towards a Unified Language for Card Game Design. In: ACM International Conference Proceeding Series.

Vicente, O., Vicente, S., Martin, D., Rodriguez-Florido, M. A., Maynar, M., 2014. Health gamification. In: Simulation Series. Vol. 46.

Vygotsky, L. S., 1978. Mind and Society: The Development of Higher Psychological Processes.

WHO, 2011. mHealth: New horizons for health through mobile technologies. Tech. rep., World Health Organization, Geneva.
URL http://www.who.int/goe/publications/goe_mhealth_web.pdf%5Cnhttp://www.who.int/goe/publications/ehealth_series_vol3/en/index.html

WHO, 2020. Global strategy on human resources for health: Workforce 2030. Who, 64.
URL https://www.who.int/publications/i/item/9789241511131

WHO, 2024. Immunization coverage fact sheet.
URL https://www.who.int/news-room/fact-sheets/detail/immunization-coverage

Wilson, M., 2002. Six views of embodied cognition.

WMA, 2013. World Medical Association Declaration of Helsinki: Ethical principles for medical research involving human subjects.

Wu, H. K., Lee, S. W. Y., Chang, H. Y., Liang, J. C., 2013. Current status, opportunities and challenges of augmented reality in education. Computers and Education 62.

Yadav, D., Bhandari, A., Singh, P., 2019. LEAP: Scaffolding collaborative learning of community health workers in India. Proceedings of the ACM on Human-Computer Interaction 3 (CSCW).

Yadav, D., et.al., 2017. Sangoshthi: Empowering community health workers through peer learning in rural India. In: 26th International World Wide Web Conference, WWW 2017.

Yadav, D., et.al., 2019. FeedPal: Understanding opportunities for chatbots in breastfeeding education of women in India. Proceedings of the ACM on Human-Computer Interaction 3 (CSCW).

Yadav, D., Malik, P., Dabas, K., Singh, P., 5 2021. Illustrating the gaps and needs in the training support of community healthworkers in india. In: Conference on Human Factors in Computing Systems - Proceedings. Association for Computing Machinery.

Yan, X., Diaconis, P., Rusmevichientong, P., Van Roy, B., 2005. Solitaire: Man versus machine. In: Advances in Neural Information Processing Systems.

Zwick, T., 2012. Why Training Older Employees is Less Effective. SSRN Electronic Journal.

List of Publications

1. **Majhi, A.**, Mondal, A., Joshi, A., Agnihotri, S. B., 2021. Refresher Training through Quiz App for capacity building of Community Healthcare Workers or Anganwadi Workers in India. *In Proceedings of the Asian CHI Symposium 2021 (Asian CHI '21). Association for Computing Machinery, New York, NY, USA, 80–88.*
   (Best Paper award in Long Paper category)
   https://dl.acm.org/doi/10.1145/3429360.3468186

2. **Majhi, A.**, Agnihotri, S. B., Mondal, A., 2022. Physical and Augmented Reality based Playful Activities for Refresher Training of ASHA Workers in India. *In Proceedings of the Asian HCI Symposium 2022 (Asian HCI '22). Association for Computing Machinery, New York, NY, USA, 16–22.*
   (Best Paper and Best Presentation award in Long Paper category)
   https://dl.acm.org/doi/10.1145/3516492.3558788

3. **Majhi, A.**, Agnihotri, S. B., Mondal, A., 2024. Replay, Revise, and Refresh: Smartphone-Based Refresher Training for Community Healthcare Workers in India. *HCI International 2024, Washington DC, USA*
   https://link.springer.com/chapter/10.1007/978-3-031-61966-3_34

4. **Majhi, A.**, Agnihotri, S. B., Mondal, A., 2024. Mapping Child Malnutrition and Measuring Efficiency of Community Healthcare Workers through Location Based Games in India. International Conference on Information Technology for Social Good (GoodIT '24, Bremen, Germany)
   https://dl.acm.org/doi/10.1145/3677525.3678685

5. **Majhi, A.**, Agnihotri, S. B., Mondal, A., 2024. Refresher Training through Digital and Physical, Card-Based Game for Accredited Social Health Activists (ASHAs) and Anganwadi Workers (AWWs) in India. CHI PLAY Companion '24, October 14–17, 2024, Tampere, Finland
   https://dl.acm.org/doi/10.1145/3665463.3678819

6. **Majhi, A.**, Agnihotri, S. B., Mondal, A., 2026. Design Guidelines for Game-Based Refresher Training of Community Health Workers in Low-Resource Contexts. Interactive Health Conference (IH '26), July 05–08, 2026, Porto, Portugal. https://doi.org/10.1145/3786579.3804932

7. **Majhi, A.**, Agnihotri, S. B., Mondal, A., 2026. Healthcare App Design in Low-Resource Contexts: Challenges, Practices, and Opportunities. Interactive Health Conference (IH '26), July 05–08, 2026, Porto, Portugal. https://doi.org/10.1145/3786579.3803814

Author's Contributions

The following list presents the research contributions of the included publications' authors using the Contributor Roles Taxonomy (CRediT).

1-7: The Publication Numbers

AM1: Arka Majhi

AM2: Aparajita Mondal (Her contributions to research are not part of her academic pursuits)

SBA: Satish B. Agnihotri

AJ: Anirudha Joshi

	AM1	AM2	SBA	AJ
Conceptualization	1-7		1-7	1
Data curation	1-7	1-7		
Formal analysis	1-7			
Funding acquisition	1-7		1-7	1
Investigation	1-7	1-7	1-7	1
Field Work	1-7	1-7		
Methodology	1-7		1-7	1
Project administration	1-7	1-7	1-7	1
Resources	1-7			
Software	1-7			
Supervision			1-7	1
Validation	1-7			
Visualization	1-7			
Writing – original draft	1-7			
Writing – review & editing			2-7	1

Table 8.1: Author's Contributions

Reference of definition and details of contributor roles (https://credit.niso.org/)

1. **Conceptualization** – Ideas; formulation or evolution of overarching research goals and aims.

2. **Data curation** – Management activities to annotate (produce metadata), scrub data and maintain research data (including software code, where it is necessary for interpreting the data itself) for initial use and later re-use.

3. **Formal analysis** – Application of statistical, mathematical, computational, or other formal techniques to analyze or synthesize study data.

4. **Funding acquisition** - Acquisition of the financial support for the project leading to this publication.

5. **Investigation** – Conducting a research and investigation process, specifically performing the experiments, or data/evidence collection.

6. **Methodology** – Development or design of methodology; creation of models.

7. **Project administration** – Management and coordination responsibility for the research activity planning and execution.

8. **Resources** – Provision of study materials, reagents, materials, patients, laboratory samples, animals, instrumentation, computing resources, or other analysis tools.

9. **Software** – Programming, software development; designing computer programs; implementation of the computer code and supporting algorithms; testing of existing code components.

10. **Supervision** – Oversight and leadership responsibility for the research activity planning and execution, including mentorship external to the core team.

11. **Validation** – Verification, whether as a part of the activity or separate, of the overall replication/reproducibility of results/experiments and other research outputs.

12. **Visualization** – Preparation, creation and/or presentation of the published work, specifically visualization/data presentation.

13. **Writing – original draft** – Preparation, creation and/or presentation of the published work, specifically writing the initial draft (including substantive translation).

14. **Writing – review & editing** – Preparation, creation and/or presentation of the published work by those from the original research group, specifically critical review, commentary or revision – including pre- or post-publication stages.

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